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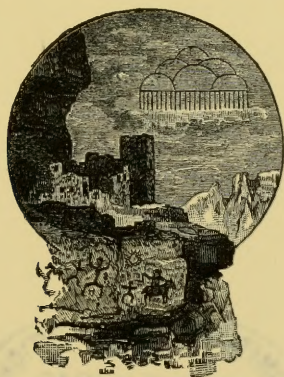
**HANDBOOK
OF
SOUTH AMERICAN INDIANS**

JULIAN H. STEWARD, *Editor*

Volume 5

**THE COMPARATIVE ETHNOLOGY
OF SOUTH AMERICAN INDIANS**

Prepared in Cooperation With the United States Department of State as a Project
of the Interdepartmental Committee on Scientific and Cultural Cooperation



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LETTER OF TRANSMITTAL

SMITHSONIAN INSTITUTION,
BUREAU OF AMERICAN ETHNOLOGY,
Washington, D. C., March 1, 1947.

SIR: I have the honor to transmit herewith a manuscript entitled "Handbook of South American Indians. Volume 5. The Comparative Ethnology of South American Indians," edited by Julian H. Steward, and to recommend that it be published as a bulletin of the Bureau of American Ethnology.

Very respectfully yours,

M. W. STIRLING, *Director.*

DR. ALEXANDER WETMORE,
Secretary of the Smithsonian Institution.

SEP 14 1949

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PART 3. THE NATIVE POPULATION OF SOUTH AMERICA

By JULIAN H. STEWARD

The population density per unit of area is a rough measure of the success of subsistence activities in the area, and it is correlated to some degree with cultural development. It does not throw much direct light on cultural problems, however, because mere densities show populations only in an average or statistical sense as if they were spread evenly over a region. Actually, large areas were temporarily or permanently unpopulated, while people clustered in bands or communities. It is the size, composition, and permanency of these communities which provide the setting for sociopolitical patterns and cultural behavior. The first section will give population densities. A subsequent section will relate settlement composition to sociopolitical types by presenting data on community size.

METHODOLOGICAL PROBLEMS

The methodological difficulties inherent in studies of aboriginal American population are evident from the discrepancies in the results of reliable scientists. The Hemisphere totals vary from Kroeber's (1939) 8,400,000 (a figure less than Ricketson's (1937) 13,000,000 for Yucatán alone) to Rivet's (1924 a) and Sapper's (1924) 40,000,000–50,000,000 and Spinden's (1928) 50,000,000–75,000,000. South American totals range from Kroeber's 4,300,000 to Rivet's 25,000,000; Means (1931) arrives at 16,000,000–32,000,000 for the Andes alone. Table 1 shows the breakdown of these estimates. It is obvious that either the data or the methods were at fault.

Previous population studies have dealt mainly with North and Middle America. All South American estimates, except that of Rosenblat (1945), who has carefully culled original sources, and a few estimates of limited areas, are made by analogy with North or Central America or by sheer guess.

Rosenblat reckons from the modern population of 1940 backward to 1825, 1650, 1570, and 1492. He accepts the more conservative contemporary estimates of each period, but 1570 yields data on only a few areas and 1492 is estimated by a hypothetical backward extension of the population curve. His method is shown in detail only for Haiti and Santo Domingo.

TABLE 1.—*Estimates of native American population*¹

Area	Population estimate according to—			
	Sapper (1924)	Rosenblat (1945)	Kroeber (1939, pp. 131-181)	Steward ²
North of México-----	2,000,000-3,500,000	1,000,000	1,000,880	1,000,880
México-----	12,000,000-15,000,000	4,500,000	3,000,000	4,500,000
West Indies-----	3,000,000-4,000,000	300,000	200,000	225,000
Central America-----	5,000,000-6,000,000	800,000	(³)	736,000
Total-----	20,000,000-25,000,000	5,600,000	-----	5,461,880
South America:				
Andes ⁴ -----	12,000,000-15,000,000	4,750,000	3,000,000	6,131,000
Remainder of South America-----	3,000,000-5,000,000	2,035,000	3,334,000	2,898,000
Total-----	15,000,000-20,000,000	6,785,000	4,300,000	9,129,000
Hemisphere total-----	37,000,000-48,500,000	13,385,000	8,400,000	15,590,880

¹ All are for approximately 1492.² The North American estimate follows Kroeber, 1939; México is from Rosenblat, 1945; other estimates are given in detail below. A preliminary estimate was given by Steward, 1945.³ Central America is included with South America.⁴ From Colombia to Chile.

Rivet (1924 a, pp. 599-602) notes that in North America the aboriginal figure of 1,148,000 has been reduced by one-third to the present population of 403,000. Assuming a similar reduction throughout the Hemisphere, he multiplies the present population by three to calculate the aboriginal figure, taking no account of local differences.

Sapper (1924), a geographer, bases his calculations on the assumed productiveness of different types of land use: in areas of hunting, fishing, and collecting the population will be sparse; in areas of cultivation, especially in the warm open highlands of México, Middle America, and South America, it will be dense. For example, modern and aboriginal Guatemala are assumed to have had about the same populations because their types of farming were about the same.

Kroeber's estimates for North America are based mainly on Mooney's dead reckoning; for other areas, they are made by comparison with North America, taking into account the cultural and natural areas. Kroeber assumes that: (1) most contemporary estimates, particularly by the early Spanish missionaries and administrators, were too high; (2) a competent ethologist may correct such estimates for an area he knows well; (3) modern populations give some indication of native populations, but the ratio of increase is not everywhere the same; (4) a rich ecology usually means a greater native density, but such factors as iron tools and friable soils must be taken into account in comparing modern and native densities; and (5) a rich culture is usually an index of a high density.

The data on South American demography are such that estimates reached by any methods can only approximate accuracy, and the margin of error will always be very great, perhaps as much as 50 per-

cent. Contemporary estimates for the early periods come largely from soldiers and missionaries, but sometimes from administrators. Kroeber's suspicion of such estimates is certainly justifiable in the case of soldiers, who obviously exaggerated the number of their enemies. Missionary guesses of independent tribes were also liable to exaggeration. The careful counts of the Indians at mission stations seem reliable, and often they are our only source; but they do not take into account the Indians still at large, and they tend to lump Indians of different tribes. The value of administrators' estimates varies; in the Central Andes they seem to have been based on fairly careful census counts, but the ratio of tribute payers to the total population varied from 5 to 1 to 2 to 21 (Rosenblat, 1945).

For the Tropical Forest and Circum-Caribbean areas, we have few contemporary estimates. The earliest available figure may be anywhere from 50 to 200 years after the tribe was first contacted by Whites; disease, warfare, slaving, cultural and racial assimilation, and other factors already had reduced the population by half or more and in some cases virtually wiped them out. The *Omagua* were reduced from 15,000 to 7,000 between 1641 and 1681, and the Indians of Hispaniola were nearly extinct within a generation.

The modern Indian population is an unreliable index of the aboriginal, not only because each area has a distinctive population curve but because the modern censuses encounter great difficulties in counting Indians in remote areas and because in Hispanic America there is a tendency to classify Indians on a cultural rather than racial basis. An Indian is considered to be a person who lives like an Indian, especially one who speaks an Indian language; when he has acquired sufficient European culture he passes into the category of Mestizo, criollo, ladino, or caboclo, as the partially assimilated Indians are variously called, although racially he may be wholly or largely Indian. The method of projecting the population curve backward to aboriginal times would be valid only if the curve were first established on the basis of reliable counts at all periods. Even then, its applicability to tribes other than that for which it was constructed would be questionable, because too many factors are involved: whether or not the Indians remained in missions; the effect of disease; and special factors of their culture, environment, and contact with the White man. Thus, while the *Omagua* were halved in 40 years in the 17th century, the neighboring *Cocama* retained nearly their native numbers to the present day.

The present method utilizes the earliest data that appear to be reliable—in some cases missionary or administrator's reports, in others, travelers' accounts—and extends the density calculated for the tribe in question to others who had similar cultures and lived in similar areas. As the tribes seem generally to have declined somewhat

by the time these estimates were made, the densities calculated from them are, even allowing for exaggerations, likely to be too small rather than too large.

NATIVE DENSITIES

The present estimates are mere approximations and should be considered preliminary. Some areas, which are completely unknown, are estimated by analogy with neighboring areas for which reasonable samples exist. On the whole, this gives a fairly coherent picture so that the margin of error for certain areas is probably not over 10 to 20 percent. In some areas the error may reach 100 percent, but even to narrow the error this much is a considerable achievement in view of previous estimates. The obvious need is full utilization of source material in a tribe by tribe and area by area count like that of Kubler's and Rowe's for the Andes (Handbook, vol. 2, pp. 334-341, 184).

Map 16 follows Kroeber's map for North America (Kroeber, 1939) in showing the number of persons per 100 km.².

The estimates for the southern Marginal tribes are probably fairly accurate, although they were made well after the Contact. If wrong, they are too low. Early counts give the *Guaraní* and Chaco numbers which seem reasonable, but the *Abipón* would have a density of 5, which is revised to 15.

The *Araucanian* and Chilotan densities, the latter greater than the Central Andes, are surprisingly high. Even a reduction of the *Araucanian* estimate by half would leave a figure exceeding the densities of neighboring tribes; but the modern *Araucanians* number 300,000. If a native *Araucanian* population of 1,000,000 is not a gross error (see Rosenblat, 1945, pp. 77-78), Central Chile under Indian farming was far more productive than has been recognized. By contrast, the *Atacameño* figure, based on a small sample, is surprisingly low. Few portions of the North Chilean deserts are inhabitable, however, and the *Atacameño* total could not have exceeded 40,000 by much. The *Diaguita* must have had a density between 13, which is estimated for the *Atacameño*, and 15, which is fairly well established for the *Comechingón*.

The Marginal tribes of the plateaus of eastern Brazil are largely unknown. The average density of 10 per 100 km.², based on several seemingly reasonable samples of the upper Xingú-Tapajós area, is extended to the *Ge*. The slightly greater density of 15 for the *Botocudo* and their neighbors near the coast is a slight upward revision of several fairly recent samples of these peoples. By contrast, the coastal *Tupí*, to judge by certain of the *Tupinamba*, had a native density of 60.

In the Tropical Forest tribes of the Amazon Basin, several independent samples run fairly consistently between 17 and 25. There



MAP 16.—Population densities of native South America. Numbers indicate persons per 100 km.² of territory.

are some Tropical Forest areas, however, which do not fit the general picture. The territory south of the upper Guaporé River in eastern Bolivia had 36 persons per 100 km.², according to what seem to be fairly good figures. Unless these are too high, the Yungas density of 20 may be too low. The Yungas also seem too low by comparison with the Montaña to the north; possibly the population of the Yungas, which came into contact with Whites at a very early time, was halved by the time estimates were made. The Montaña is high by comparison with the Amazon lowlands. Unless the missionaries erred in a very consistent manner in several independent Montaña estimates, it may be suspected that some of the lowland estimates, which generally are later, are too low. The *Witotoans* are an island of great density among comparatively sparse peoples. The figures taken from such reliable observers as Preuss and Whiffen may mean that estimates for neighboring peoples are too low; for the present, however, the inconsistency is allowed to stand.

Guiana is almost certainly too low. Figures are needed on the Contact Period. Recent samples give a density of about 10 per 100 km.²; natively, it must have been half again as great, perhaps more. Whether the swampy coast had an unusually high density is an open question.

The best source for Venezuela is Humboldt (1862), whose observations made about 1800 give sample densities of 38 per 100 km.². The native population was greater, but it is impossible to say how much greater without utilization of Contact Period sources. It is allowed a density of 45.

Colombia, too, needs to be reckoned for early sources. The present estimate accepts 300,000 for the *Chibcha* and 700,000—a density of 184—for the remainder of Colombia, a total of a 1,000,000, though some estimates allow the *Chibcha* alone 1,000,000. A reliable estimate of 1586 gives Colombia 715,000 (Rosenblat, 1945, pp. 77-78), and 1,000,000 for all native Colombia seems reasonable by comparison with Perú and probably not too large. Some chroniclers estimate the Popayán Valley alone at 100,000 to 500,000 persons, a density of 500 to 2,500 per 100 km.²

Highland Ecuador is reckoned entirely by analogy. A half million native population, which is close to the figure for the modern Indians, gives a density of 300 as compared with Perú's 390 persons per 100 km.². Certainly this figure is not too small; if, in Ecuador as in Perú, there are more Indians now than at the Conquest, it is too large.

Central America also needs estimates based on the early chroniclers. The present Indian population, which is extinct or greatly reduced in most areas, throws little light on native numbers. The few early samples suggest that the population was very dense in eastern Panamá,

300 persons per 100 km.², but thinned toward the west. Costa Rica, El Salvador, and the southern Highlands of Guatemala had greater density (200) than Nicaragua or northern Honduras (40).

The Antilles had a very large population, though its size frequently has been exaggerated. I have followed Rosenblat's conservative estimates even though his Hispaniola figure gives a density of 767, about twice that of any other area in America. The lesser Antilles, however, reach a density of over 500, to judge by certain samples. By contrast, Cuba, which the *Arawak* shared with the primitive *Giboney*, had a density of only 69.

In the table 2, the data are taken from Handbook articles except where the source is indicated.

TABLE 2.—*Native population numbers and densities of South American tribes*

Area	Population	Size in units of 100 km. ²	Persons per 100 km. ²	Source and comments
Marginal Tribes:				
Southern Marginal:				
Archipelago.....	9,000	1,051	9	<i>Chono</i> , 1,000; <i>Alacaluf</i> , 400 after 1900; <i>Yahgan</i> , 3,000, 1875; <i>Ona</i> , 2,000, 1875-1900.
Pampas-Patagonia.....	36,125	14,450	2.5	Various 19th-century estimates.
<i>Querandi</i>	4,000	1,000	4	
<i>Comechingón-Huarpe</i> , etc.....	52,550	3,570	15	<i>Comechingón</i> native estimate 30,000 (Serrano, 1945).
<i>Charrua-Caracara</i>	9,000	3,000	3	By analogy with Pampas.
Total, Southern Marginal..	110,675	23,071		
Western Chaco.....	186,400	4,320	29	Earliest totals for main tribes (Métraux, Handbook, vol. 1).
Eastern Brazil:				
<i>Bororo</i>	16,000	1,600	10	By analogy; recent estimates seem too low.
<i>Southern Cayapó-Guató</i>	59,000	5,900	10	By analogy; no data.
<i>Caingang</i>	17,500	2,500	7	Recent sample; may be too low.
<i>Bacairi</i>	6,000	600	10	By analogy; no data.
<i>Nambicuara</i>	22,000	1,100	20	Good estimates (Lévi-Strauss, Handbook, vol. 2).
Upper Xingú.....	10,000	1,000	10	Analogy with <i>Nambicuara</i> , conservative reduction; Von den Steinen's 3,000 too low.
Northwest and Central Ge.....	98,000	9,800	10	Recent samples give 6 to 8 per 100 km. ² .
<i>Creye-Timbira</i>	1,440	160	9	By analogy, other <i>Ge</i> .
<i>Timbira</i>	22,000	2,200	10	No data; by analogy.
<i>Jeico</i>	10,000	1,000	10	No data; by analogy.
<i>Botocudo</i> , etc.....	43,500	2,900	15	Estimates of last century give density of 11 to 14 per kilometer. ²
<i>Tapirapé</i>	4,000	160	25	1910 estimate, 1,000; <i>Carajá</i> reduction by 1910 estimated at one-fourth original.
<i>Carajá</i>	57,000	2,000	28	Krause's estimate of 100,000 for 1845 reduced by one-half.
<i>Teremembe</i>	21,000	700	30	No data; density was probably half that of coastal <i>Tupí</i> .
Total, eastern Brazil.....	387,440	31,620		
Amazon Marginals:				
<i>Mura</i>	30,000	1,400	21	Reduction by half of early estimate of 60,000 native population.
Province of Mainas.....	42,500	1,900	22	Missionary estimates.
<i>Carijona</i>	19,400	970	20	Analogy; no data.
Others.....				Included with neighboring tribes for lack of data.
Total, Amazon Marginals..	91,900	4,270		

TABLE 2.—*Native population numbers and densities of South American tribes—Continued*

Area	Population	Size in units of 100 km. ²	Persons per 100 km. ²	Source and comments
Tropical Forest Tribes:				
Eastern Chaco:				
<i>Abipón</i> and neighbors.....	50,250	3,350	15	Earliest totals of main tribes (Métreaux, Handbook, vol. 1).
<i>Payaguá, Chané, Mbayá</i>	30,000	900	33	Earliest totals (Métreaux, Handbook, vol. 1).
Total eastern Chaco.....	80,250	4,250		
Tupí-Guaraní:				
Paraguay-Brazil.....	200,000	7,200	28	Estimate of 300,000 <i>Guaraní</i> killed or captured at Conquest reduced to 200,000; or 30 missions with 6,000 each in 1708.
Paraná Delta.....	24,000	800	30	By comparison coastal <i>Tupí</i> .
Upper São Francisco River region.....	49,000	4,900	10	No data; analogy with <i>Ge</i> area.
<i>Cariri</i> and neighbors.....	65,500	6,550	10	No data; analogy with <i>Ge</i> area.
Coastal <i>Tupí</i> south of Amazon.....	189,000	3,150	60	4,200 km. of coast, 75 km deep. <i>Tupinamba</i> sample had 27,000 persons in 45,000-km. ² Island of Maranhão, density of 906 per 100 km. ² (Métreaux, Handbook, vol. 3).
<i>Tenelehara</i>	60,800	1,520	60	By analogy coastal <i>Tupí</i> ; an unknown number of <i>Amanayé</i> and <i>Turiwara</i> in area not counted.
Lower Amazon <i>Tupí</i>	100,000	4,000	25	Extensions of <i>Maué</i> density; <i>Maué</i> sample in 1939 gave 25 per 100 km. ² Total may be too low.
<i>Apiacá-Cayabí</i>	18,000	1,600	11	Probably reliable estimate of 16,000 <i>Apiacá</i> , 2,000 <i>Cayabí</i> .
<i>Mundurucú</i>	36,200	1,800	20	Martius' estimate about 1880, and double that of Tocantins, 1877.
<i>Parintintin-Cawahib</i>	25,500	1,500	17	By analogy neighboring <i>Tupí</i> .
Total <i>Tupí</i>	768,000	33,020		
Amazon River:				
<i>Cocama</i>	12,000	400	30	These estimates allow 50 km. each side of the river. Missionary estimates. Today, about 10,000.
<i>Omagua</i>	16,000	800	20	Missionary estimates, 15,000, 1641; 7,000, 1681.
Amazon below <i>Omagua</i>	72,000	1,800	40	No data; analogy coastal <i>Tupí</i> and upper Amazon; Carvajal mouth of Iça.
<i>Arua (Marajó)</i>	30,000	500	60	No data; by analogy coastal <i>Tupí</i> .
Total Amazon.....	130,000	3,500		
Southwest Amazon:				
Juruá-Purús.....	139,400	7,200	17	By analogy from neighbors; present White and Indian about 100,000.
North of Guaporé.....	46,000	2,300	20	No data; by analogy with neighbors.
South of Guaporé River.....	72,160	1,950	36	Average of several estimates ranging from 25 to 45 per 100 km. ²
Total Southwest Amazon.....	257,560	11,450		
Eastern Bolivia:				
<i>Tacanas</i>	25,000	1,250	20	Mission estimates 40 to 150 years ago total 18,800; perhaps $\frac{2}{3}$ were in missions.
Province of Chiquitos.....	42,000	2,000	21	Estimate of 23,788 in 1766, 200 years after Contact.
Province of Mojos.....	6,000	400	15	Estimate for 1680.
<i>Paressí</i>	5,000	50	10	
<i>Yungas</i>	31,000	1,550	20	Estimates are much lower; density of 20 per 100 km. ² is by analogy.
<i>Chiriguano</i>	48,000	800	60	Fairly accurate early estimate.
Total Eastern Bolivia.....	157,000	6,500		

TABLE 2.—*Native population numbers and densities of South American tribes—Continued*

Area	Population	Size in units of 100 km. ²	Persons per 100 km. ²	Source and comments
Tropical Forest Tribes—Continued				
Montaña:				
Ucayali-Madeira.....	31,740	1,800	17	1940 census.
Campa.....	20,000	510	38	17th century missionary estimates; 1940 census gives 20,000 also.
Huallaga-Ucayali.....	105,070	2,765	38	Sample missionary estimates; analogy neighbors.
Jivaro.....	26,600	700	38	Stirling, 1938, pp. 28-38.
Cofán, Quijo, Canelo.....	25,000	255	100	Missionary estimates; see Rosenblat, 1945, pp. 77-78.
Total Montaña.....	208,410	6,030		
Northeast Amazon:				
Amazon-Rio Negro.....	71,250	2,500	25	Most conservative native estimate for <i>Yagua</i> used as sample.
Western <i>Tucanoans</i>	16,000	800	20	Estimate of 1635 gives density of 15 per 100 km. ² adjusted to 20.
<i>Witotoans</i> , etc.....	67,000	1,000	67	Cf. Handbook, vol. 3, estimates, which seem too high by comparison with neighbors.
Uaupés-Caquetá.....	45,000	2,250	20	Total <i>Tucanoans</i> in 1900 is 8,700 or 10 persons per 100 km. ² ; native density probably double this.
Eastern Colombia.....	70,500	3,525	20	By analogy.
Total Northwest Amazon.....	269,750	10,075		
Guianas:				
North of Amazon, Guianas, including Marginals.....	213,750	14,250	15	Modern samples give densities around 10 persons per 100 km. ² ; native must have been half again this, perhaps more.
Warrau.....	6,300	525	12	Gumilla, 1791.
Total Guianas.....	220,050	14,775		
Northwestern South America:				
<i>Motilon</i> es and neighbors.....	40,000	1,000	40	Recent <i>Motilon</i> es density is 25 persons per 100 km. ²
<i>Goajiro</i>	31,300	113	277	20th-century estimate.
Colombia Pacific Coast.....	15,000	600	25	Recent <i>Chocó</i> estimate gives density 17 persons per 100 km. ² .
Total Northwestern South America.....	86,300	1,713		
Sub-Andean Tribes:				
Venezuela, north of Orinoco.....	144,000	3,200	45	Humboldt's estimates (1822-27) give 38 persons per 100 km. ² in 1800.
Colombia:				
<i>Chibcha</i>	300,000	240	1,070	Native estimates.
Remainder Highland Colombia.....	700,000	3,800	184	Allowing 1,000,000 for Highland Colombia, minus 300,000 <i>Chibcha</i> .
Total Sub-Andean.....	1,144,000	7,240		
Andean Tribes:				
Ecuador.....	500,000	1,800	300	Handbook, vol. 2.
Southern Andes:				
<i>Atacameño</i>	40,000	3,008	13	Salar de Atacama, 13 persons in 100 km. ² , 1581; total in 1800, 7,400 (Boman, 1908).
<i>Diaguita</i>	41,000	3,160	13	By analogy with Atacama probably estimates too low.
<i>Araucanians</i>	1,000,000	3,650	273	Native estimate (Cooper, Handbook, vol. 2).
Chiloé.....	50,000	90	555	Cooper, Handbook, vol. 2.
Total Andean Tribes.....	1,631,000	11,708		
Central Highlands:				
Highland Perú.....	2,330,000	5,965	390	Handbook, vol. 2.
Highland Bolivia.....	1,170,000	3,000	390	Handbook, vol. 2.
Total Central Andes.....	3,500,000	8,965		

TABLE 2.—*Native population numbers and densities of South American tribes—Continued*

Area	Population	Size in units of 100 km. ²	Persons per 100 km. ²	Source and comments
Total South America.....	9, 228, 735	182, 507	-----	
Central America:				
Eastern Panamá.....	150, 000	500	300	Density probably near that of modern <i>Cuna</i> , 350 per 100 km. ² .
Western Panamá.....	74, 600	373	200	Population of Panamá thinned toward west, with Chanaguena density 125 in 1709.
Costa Rica.....	119, 400	597	200	Probably same density as western Panamá.
East coast Nicaragua and Honduras.....	180, 000	1, 180	100	<i>Mosquito</i> density now 55; native was certainly double.
North coast Honduras.....	25, 000	500	50	<i>Jicaque</i> density, 20 in 1674; <i>Paya</i> density, 50 in 1800.
Honduran Highland and El Salvador.....	187, 500	750	250	Modern <i>Lenca</i> density about 190; Kroeber gives 292 for El Salvador.
Total Central America....	736, 500	3, 900	-----	
Antilles:				
Cuba.....	80, 000	1, 147	69	Rosenblat.
Jamaica.....	40, 000	115	348	Rosenblat.
Bahamas.....	20, 000	114	175	Half native estimate.
Haiti and Santo Domingo.....	100, 000	767	555	Rosenblat.
Puerto Rico.....	50, 000	90	555	Rosenblat; De Hostos (Handbook, vol. 4) gives 200,000. Some estimates were 600,000.
Lesser Antilles.....	35, 000	70	500	2,000 on St. Vincent in 1700; if 3,000 native, St. Vincent density is 750 density for Dominica is 375.
Total Antilles.....	225, 000	2, 303	-----	

DENSITY, CULTURE, AND ECOLOGY

The population densities on map 16 represent approximations to absolute figures that have an error of from 10 to 100 percent. The relative densities are far more important to cultural problems, and the significance of error is greatly reduced if relative densities are considered. For example, by present estimates, the Central Andes had about 150 times the density of Patagonia. If the Patagonian population were doubled and that of the Andes halved, the former would have 40 times the density of the latter, which would still be highly important in terms of the numbers of people who may aggregate in sociopolitical groups. Assuming that the present figures are approximately correct, the densities seem to vary significantly among the cultural groups. Taking the Central Andes as the yardstick, it is found to have 44 times the density of the Chilean Archipelago, 14 to 40 times that of the other Marginal tribes, 20 times that of the Tropical Forest peoples, and about twice that of the Antilles, Highland Colombia, and Central America.

In ecological terms, the greatest densities were in the semiarid Andes, where subsistence was based on intensive farming, which employed both irrigation and fertilization. It has been estimated that in the Andes only one-half acre of cultivated ground was required to sup-

port each person as compared with 1 acre in native México and in the eastern United States (Kroeber, 1939, pp. 146-147, 163) and sevenths acre in Yucatán (Ricketson and Ricketson, 1937, pp. 16-17). The next greatest densities were in the Circum-Caribbean area, where the people relied on ocean resources as much as on farming. The situation is comparable to the Pacific Coast of North America, which, despite the absence of agriculture, had one of the densest populations on the Continent. Around the Caribbean, the additional agricultural resources supported many more people than on the North American coasts. The coastal and fluvial areas of the Guianas and Brazil were more sparsely populated than the Circum-Caribbean area, probably because of the decreased intensity of farming. One has the impression, however, that vast regions of potentially arable land were never utilized and that the population may have been expanding at the time of the Conquest. In the savanna lands scattered around the Amazon and Orinoco Basins and in the brush country of eastern Brazil and the Gran Chaco, where the tropical root crops could not be grown and the tribes were hunters and gatherers, the population was extremely sparse. The lowest density was the plains of Argentina, which though well adapted to animal raising and plough agriculture, did not yield to Indian farm methods.

POPULATION TRENDS

The aboriginal population of South America is estimated at about 9,000,000, of which approximately half were in the Central Andes. (See table 3.) Today nearly 7,000,000 persons are classed, largely for cultural reasons, as Indians. (Some estimates double this figure.) There are several million caboclos, criollos, and Mestizos of various sorts, who, though predominantly Indian in race, are not classed as Indians in the census.¹ Biologically, therefore, there are considerably more than 7,000,000 persons of Indian ancestry. In fact, the Indian race is undoubtedly as numerous as at the time of the Conquest, and it has probably made a substantial net gain. The gain, however, is attributable entirely to the Andes. Elsewhere, the population has declined, though not always to the extent that the census indicates. The loss, which is primarily a function of the intensity and duration of European contact and of the aboriginal population density, varies in each area (map 17).

The greatest decline was along the coasts and large rivers, where the Europeans, water-borne and maintaining contact with the homeland, came with overwhelming force. The Indians of the coasts of Brazil,

¹ Acceptance of census estimates may easily obscure the considerable numbers of Indians and consequently the true population trends in these countries. (See, for example, Shapiro, 1942.)

TABLE 3.—*South American Indian population*

Location	Indian population in—		Percent of national population (1946)
	1500	1940	
Southern Nations:			
Chile.....	1, 000, 000	290, 000	9. 0
Argentina.....	300, 000	120, 000	1. 0
Uruguay.....	20, 000	Extinct	0
Total.....	1, 320, 000	410, 000	
Tropical, subtropical, and Circum-Caribbean nations:			
Paraguay.....	100, 000	60, 000	6. 0
Brazil.....	1, 100, 000	500, 000	11. 0
Guianas.....	90, 000	11, 000	2. 4
Venezuela.....	350, 000	103, 000	3. 7
Antilles.....	300, 000	Extinct	0
Lowland Colombia.....	350, 000	105, 000	5. 7
Highland Colombia.....	800, 000	60, 000	3. 2
Panamás.....	70, 000	42, 000	9. 0
Costa Rica.....	40, 000	3, 000	0. 6
Nicaragua, Honduras, and El Salvador.....	100, 000	85, 000	9. 0
Total.....	3, 300, 000	969, 000	
Highland Nations:			
Ecuador.....	1, 000, 000	960, 000	50. 0
Perú.....	3, 500, 000	2, 800, 000	40. 0
Bolivia.....		1, 800, 000	51. 0
Total.....	4, 500, 000	5, 560, 000	
South American total.....	9, 120, 000	6, 939, 000	

the Guianas, Venezuela, Colombia, Central America, and Ecuador, though very numerous, were quickly submerged; along the Amazon and Orinoco they also suffered a rapid decline. In these areas, their place was partly taken by Negroes from Africa. In most of the coastal and fluvial areas, the very identity of the native tribes is in doubt. In the Antillean islands, which may be regarded as entirely coastal, the Indians were nearly extinct within a century and a half. Las Casas estimated that the native population of Puerto Rico and Jamaica had shrunk from 3,000,000 (a figure that is much too high) in 1509 to 200 in 1542.

As the Europeans penetrated inland along the Amazon and Orinoco, the Indians gradually succumbed, some perishing, others becoming culturally assimilated and adding their blood to that of the Whites and Negroes to form mixed populations. Today the main survival of Indians is in the areas beyond easy navigation, i. e., in the great U which surrounds the Amazon Basin and includes the Amazon-Orinoco watershed, the Northwest Amazon, the Montaña, Mato Grosso, parts of the Chaco, and the eastern Brazilian plateau. This was precisely the area of Marginal and Semi-Marginal tribes, who, in pre-Colombian times had remained comparatively uninfluenced by the Tropical Forest type of culture, which was also water-borne. (Compare the culture types, map 18, and the distribution of modern Indians, map 17.) Elsewhere



MAP 17.—Distribution of Indians in 1940.

in the Tropical Forests and around the Caribbean Sea, the Indians who survive today live mainly in areas that are unsuited for European occupation, such as the swamps of the Orinoco Delta and of the Mosquito Coast in Nicaragua and the mountainous areas of western Venezuela and Central America.

In Argentina and Uruguay, where the native population was sparse, the coastal tribes were also first to succumb. Later, the Europeans swept inland, finding the country suitable for their economy, and the

Indians have all but vanished. In Chile, the story was the same except that the *Araucanians*, driven south but not surrendering, were finally isolated on a reservation.

There were numerous factors in this decline, and their operation in different localities made for very unlike population curves. (See Rosenblat, 1945.) European diseases, which were most devastating where the Indians were concentrated, as in missions or forced settlements, wiped out some tribes within a century or two, whereas immediate neighbors survived in some strength to the present day. Warfare, both with the Whites and with tribes dislocated by the Conquest, took a terrific toll. The Indians also suffered from disrupted economic and cultural life. It is perhaps premature to guess whether better health measures and reservation systems will check this decline.

By contrast to the remainder of South America, the Andes, especially the Central Andes, have more Indians today than at the time of the Conquest. As elsewhere, the Coast suffered most from White contacts, but later the Whites penetrated the interior and the total population was probably halved. Aboriginally numerous, culturally stable, and comparatively unaffected by epidemics, it has recovered and is now larger and increasing.

PART 4. SOUTH AMERICAN CULTURES: AN INTERPRETATIVE SUMMARY ¹

By JULIAN H. STEWARD

INTRODUCTION

It is the purpose of the present article to provide a basis for classifying South American Indian cultures and to present comparative summaries of the principal cultural types in terms of their ecological adaptations and historical development. It endeavors to reduce the bewildering variety of cultural data to categories which have a real and historical meaning. In this respect, it differs from the method of historical particularizing which treats each tribe and culture as unique, emphasizing their peculiarities and stressing the exceptional rather than the general. The latter method is valuable in detailed analyses of individual tribes, but, applied to large areas, it gives the impression that cultural elements and patterns occur in a random and fortuitous manner.

Admittedly, an attempt to subsume large numbers of tribal cultures under general types, as undertaken here, encounters difficulties presented by borderline cases, by insufficient data, and by possible misinterpretation of data. The general character of the main cultural types will doubtless have to be redescribed in the light of new information and more detailed comparisons, and many tribes will quite probably be found to belong to types other than those to which they are here assigned. Authors familiar with certain tribes will find that the generalizations do not do their tribes justice. Science will be best served, however, by correcting faulty generalizations with better generalizations; carping at these constructs because a few special facts do not fit them can only lead to the impression that culture development is utterly capricious and haphazard.

The fourfold classification used herein (map 18) corresponds in general to the four volumes of the Handbook—Marginal tribes, Tropical Forest peoples, Circum-Caribbean and Sub-Andean peoples, and Andean civilizations—except that in retrospect it is evident that many

¹ I am grateful to Drs. A. L. Kroeber, Robert H. Lowie, W. D. Strong, John M. Cooper, and Gordon R. Willey for having read and criticized this summary.



MAP 18.—Distribution of aboriginal culture types. (Solid black, Central Andes; horizontal hachure, Circum-Caribbean; cross-hachure, Tropical Forest and Southern Andes; diagonal hachure, Semi-Marginal; stipple, Marginal.)

tribes were improperly classified. In the preceding volumes of the Handbook, the tribes were classed more or less on the basis of impressions. The classification, like previous ones,² used principally the general element content of the cultures rather than a systematic comparison of the patterns. Special weight was accorded one or another feature in each case. The Marginal peoples were distinguished by their lack of farming and their generally simple cultures; the Tropical Forest peoples were identified by their agriculture and various material traits, which were adapted to the tropical rain forests; the Circum-Caribbean and Sub-Andean peoples were grouped together because of their class system and temple cult; and the Andean peoples, from southern Colombia to Central Chile and Northwest Argentina, were distinguished by their Central Andean technology, material culture, and ritual complex.

The present classification is based primarily on sociopolitical and religious patterns. Culture elements are accorded secondary importance because too many of them are independent variables. Their distributions were dissonant with those of the sociopolitical and religious patterns, and they occurred in quite different patterns. They were the building materials of culture and did not greatly affect the architecture. The bow and atlatl, for example, occurred in widely differing patterns of hunting and warfare. Ritual elements, such as hair cutting, flagellation, or the scratching stick, served a very different purpose in each local context. Deities, such as the sun or stars, were variously mythological characters, shamans' spirits, or tribal gods. Items of adornment, such as face painting, tattooing, and ear-plugs, were badges of tribal membership, sex, society affiliation, or class status, according to the tribal patterns they entered. Even element complexes, that is, stable groups of elements, were found in different settings. For example, the initiation of boys into a secret organization, which used a sacred trumpet to represent the voice of the gods, was part of an ancestor cult in the northwest Amazon and of a priest-temple cult among the *Mojo*.

A classification based on culture elements would not at all correspond to one based on sociopolitical patterns. In terms of culture elements, eastern Bolivia would be classed with the Tropical Forests, and the Northern and Southern Andes would be included with the Central Andes. The Circum-Caribbean peoples would belong with the Tropical Forests if material elements were emphasized and with the Andes if social and ritual elements were given more weight. Culture elements have greatest classificatory significance in the case of the Marginal peoples, who, though very heterogeneous, differed from all other South American tribes in certain important absences. They

² Wissler's five-fold grouping (1922), Cooper's three-fold (1942), and Stout's nine-fold (1938).

characteristically lacked agriculture, loom weaving, twilled and woven basketry, dugout canoes, and a considerable list of lesser items found among their neighbors.

The patterns chosen herein as the basis for classification are those which integrate the institutions of the sociopolitical unit. The unit is the cohesive group whose members live in more or less permanent association with one another, participate in the same economic, social, and religious activities, and submit to the same in-group sanctions and political controls. It is the group of persons whose varied and reciprocal behavior patterns form a self-contained cultural whole. It is sometimes but not always the tribe; the term "tribe" is often applied to a group of units which, though culturally and linguistically similar, are politically independent of one another. The magnitude of the sociopolitical unit varies: In rare cases, it is the conjugal family; more often, it is the extended family or lineage; frequently, it is the multi-family or multilineage community; and, in some areas, it is the multi-community state, federation, or empire.

The pattern or structure of each unit varies not only with its size and composition but with its special cleavages, which may be based variously on kinship, sex, age, and status and on military, religious, and economic activities. The units are thus distinguished by such institutions as clans, lineages, and other kin groups, division of labor, secret societies, special associations, warrior classes, a priesthood, a nobility, and the like. Though every independent sociopolitical unit differed somewhat from all others in the way it patterned these institutions, certain broad configurations occurred over wide areas, and it is possible to group them in four principal types, as follows:

(1) The Marginal peoples had sociopolitical units which consisted either of a single kin group or of several loosely organized kin groups. Members of the unit were differentiated on the basis of age, sex, economic activities, and sometimes associations. Behavior was sanctioned by the informal and often unconscious influences of tribal custom operating through these institutions. These tribes had very similar crisis rites, shamanism, and magic, and their technology and material culture, though not homogeneous, was rudimentary and generally lacked the developed agriculture, building arts, and manufacturing processes found among other South American Indians. The sociopolitical patterns, however, varied with local conditions. Because the Marginal tribes lived in areas of limited resources and had elementary exploitative devices, the size and composition of their groups and many of their institutions had to be adapted to subsistence needs.

(2) The Tropical Forest and Southern Andean peoples also had sociopolitical units consisting principally of kin groups and structured

along lines of age, sex, and associations, but theirs differed from those of the Marginal tribes in that more developed exploitative devices, which included farming, and better transportation afforded by the canoe, permitted larger and more stable units. Social control was informal, except in a few communal activities such as warfare, which often had a special chief. They also had a richer technology and material culture, but their crisis rites and shamanistic patterns were of the same types as those of the Marginal tribes.

(3) The Sub-Andean and Circum-Caribbean peoples, though similar in technology and material culture to the Tropical Forest Tribes, had a more effective subsistence complex which supported a denser population and larger and more permanent villages. The villages were composed of many non-kin groups and were organized on the basis of classes rather than merely of age, sex, and associations. Warfare, carried on by the Marginal peoples mainly for revenge and by the Tropical Forest tribes for revenge and for personal prestige, became, among the Sub-Andean and Circum-Caribbean peoples, the principal means of achieving membership in the upper class, and war captives formed a slave class. Individual behavior was still largely sanctioned by custom, but governmental regulation through state law was foreshadowed by special and delimited powers accorded chiefs, warriors, and shamans in particular contexts and for delimited periods. Among the Tropical Forest and Marginal peoples, the shaman dealt principally with his own spirit helper, and his principal function was to cure and practice magic; among the Sub-Andean and Circum-Caribbean tribes, he served also as the priest in a temple-idol cult dedicated to the tribal gods, though his rites tended to be private oracular sessions rather than public ceremonies in a ritual calendar.

(4) The Central Andean peoples had the most developed agriculture in South America, the most dense population, and very efficient transportation, the combination of which permitted the growth of true urban centers and the extension of social interrelations and political controls over large areas. Patterns implicit in the Circum-Caribbean and Sub-Andean culture were fully developed in the Central Andes: a rigidly hereditary class system; war for conquest rather than for personal advancement through the capture of slaves; a temple cult with a hierarchy of gods and priests; public ceremonies forming a ritual cycle; and regimentation of a considerable portion of the lives of the commoners through political controls enforced by the state more than by the sanctions of tribal custom. These institutions overshadowed and to some degree replaced behavior patterns pertaining essentially to the kin group and the community. Central Andean material culture differs little technologically from other areas, except in

metallurgy and the building arts, but it is outstanding in the quantity, variety, and excellence of its products, and finer goods were produced by special craftsmen for the upper classes.

This fourfold classification has developmental implications in that some institutions and practices were necessarily antecedent to others, but it is not a unilinear scheme. As will be shown in a concluding section, a strong historical tradition carried certain sociopolitical institutions and probably several technologies throughout a considerable portion of South America, but the acceptance and patterning of such institutions was always contingent upon local potentialities. The potentialities were a function of the local ecology, that is, the interaction of environment, exploitative devices, and socioeconomic habits. In each case, the exigencies of making a living in a given environment with a specific set of devices and methods for obtaining, transporting, and preparing food and other essential goods set limits to the dispersal or grouping of the people and to the composition of settlements, and it strongly influenced many of their modes of behavior.

The *Yahgan*, for example, quite obviously had to live in small, widely spaced groups if they were to obtain sufficient shellfish in their coastal habitat. The *Ona* and *Tehuelche* could live in larger, nomadic hunting bands, and those of the *Tehuelche* undoubtedly grew in size after they obtained the horse. Probably as a function of the hunting economy, the bands were made up of patrilineal lineages. Along the rivers and coasts of the Amazon area, a more abundant subsistence and developed canoe travel permitted large population concentrations, and the settlement pattern and sociopolitical institutions had wider latitude for variation. The Circum-Caribbean area probably had still greater latitude, but it differed from the Tropical Forests because of the historical influences which brought it a class-structured society. The Andes had the most efficient subsistence pattern of all South America, and it developed over a long period within a strong historical tradition. The ultimate origin of its patterns has not yet been disclosed, but their influence on non-Andean peoples and their final formulation in the Central Andes by the *Inca* can be traced with some certainty.

In the following pages, population densities are related to community size. South American cultural data are then classified and summarized according to the four sociopolitical patterns. Under each, the subsistence activities and settlement patterns are described; the social, religious, and political institutions are analyzed; and the technology and material culture are briefly sketched. A concluding section traces basic trends in the development of South American cultures from the earliest known periods to the present day.

POPULATION AND COMMUNITY SIZE

Population densities suggest potentialities for community development, but far greater insight into the actual setting of sociopolitical patterns is afforded if they are analyzed in relation to settlement types and community size (map 19).

Population density and community size are by no means correlated, because the number of people who may live together in permanent association depends not only upon the number of persons per unit of area but on the quantity and distribution of resources, the transportation facilities, and the sociological factors, such as warfare and religion, that determine settlement patterns.

The Marginal tribes of Southern Argentina and Chile had a comparatively sparse population, ranging from 2.5 to 9 persons per 100 km.² In addition, their reliance on wild foods, which were widely scattered, and the necessity of transporting their goods and foods on their own backs (in some place they used bark canoes) precluded large, permanent settlements. Even where their population density equaled that of neighboring Tropical Forest tribes, their nomadism prevented community growth, for example among the *Nambicuará*, *Mura*, *Sirionó*, and Western Amazon Semi-Marginal tribes. (Cf. maps 16 and 19.)

A striking illustration of the importance of transportation facilities to community size is afforded by the tribes which adopted the horse after the Conquest. Among the peoples of Patagonia, the Pampas, and the Chaco, the bands increased several fold after the adoption of the horse.

Many of the Marginal tribes had groups consisting of only a single, conjugal family, that is, five or six persons, or slightly more if there were polygamy. Where a greater number of people remained in permanent association, they tended to form an extended family or lineage rather than a group of unrelated conjugal families. The limit of the lineage, however, was usually about 60 persons. When a community or band exceeded this number, they often lost a sense of being related to one another. Larger groups, therefore, generally consisted of several unrelated lineages, as among the somewhat agricultural tribes of the Chaco, who had communities of some 200 persons, and of eastern Brazil, where villages had 400 or more.

The agricultural tribes had permanent food resources near their villages, but farming did not have the same relationship to community size in each area. In the Tropical Forests, where the population density ranged from 10 to 50 persons per 100 km.², cultivation



MAP 19.—Size of native communities. The average size is given for each area.

was by the slash-and-burn method, and the fields, and in some cases the village, had to be moved periodically. The soil was tilled by women, who, having many household duties, could go only a limited distance to the fields. The Tropical Forest villages, which had as many as 1,000 persons among the riparian and littoral tribes, were supported more by the combination of water resources and transportation in efficient, dugout canoes than by farming. Conceivably, communities could have been larger had they been dispersed, but warfare kept them tightly nucleated; several large houses, each sheltering a lineage, were clustered inside a palisade. In some areas, however, as along the upper Amazon, houses were strung along the river without special grouping, though the population was comparatively dense (at least 30 persons per 100 km.²).

In the Circum-Caribbean area, the denser population (several hundred persons per 100 km.²) and the larger communities, which had 1,000 to 3,000 persons each, represent an intensification of the Tropical Forest ecology: somewhat better agriculture, abundant sea foods, and excellent canoe transportation. The large villages, however, were not solely the product of the numerous population. The villages tended to grow because they had become administrative and religious centers and because their population was augmented with captive slaves. The *Araucanians*, also good farmers and with a population density about equal to that of the Circum-Caribbean area, lacked these sociological factors which made for community growth. Instead of fewer and larger communities, they had numerous small ones, each constituting a lineage of 100 to 150 persons. Many other tribes similarly failed to achieve the community size which their ecology would have permitted.

In the Central Andes, the population, which had a density of 720 persons per 100 km.², was so concentrated in the fertile Coastal and Highland valleys that the settlement pattern was one of almost continuous dispersal over areas that were inhabitable. Instead of concentrating in fortified villages (the North Coast was probably an exception), refuge was sought in hilltop forts. Religious and administrative centers, however, provided nucleating points for communities of varying sizes. The ayllu village usually had several hundred inhabitants. The more important centers had several thousand people, and some communities were true cities, for example, Cuzco, which had some 100,000 persons. The basis for these urban communities was the combination of the dense population, which was supported by highly efficient agriculture, coastal transportation by sea-going balsa rafts and inland transportation by pack llamas traveling on roads, and the strong development of religious and administrative centers.

These general, comparative data on population densities and community size will serve as a background to the analyses of sociopolitical and cultural types which follow.

THE MARGINAL TRIBES

The Marginal tribes were essentially hunters and gatherers, and they exploited comparatively unproductive environments by means of a simple technology. Their sociopolitical units were small, somewhat unstable, and frequently nomadic, and the relations of their members to one another were governed by kinship, age, sex, and associations. The size, permanency, and composition of their groups were strongly affected by subsistence patterns, each adapted to a distinctive environment, and by special local developments, such as clans, moieties, and associations. Sociopolitically, therefore, the Marginal peoples differed from one another as much as they differed from other South American Indians.

In terms of culture elements, the Marginal tribes were also extremely varied not only because many of the comparatively archaic features are unevenly distributed among them but because locally they borrowed many items from their Tropical Forest and Andean neighbors.

The following are some of the elements which certain Chaco tribes borrowed from the Andes: Loom weaving, painted cloth, tie dyeing, ceramic styles, dice games, some farming, use of salt and other condiments, fermented drinks, slings, feather fire fans, sandals, body paint applied with stamps, drums, flutes, whistles, eyed needles, some myth themes, and other minor items (Handbook, 1:210-211). From the Tropical Forest tribes they evidently took mortars, scalping, and use of tobacco; a few tribes also borrowed house types, hammocks, urucú, and arrow types.

The tribes of eastern Brazil acquired a considerable list of items from their Tropical Forest neighbors, and the *Northwest* and *Central Ge* might even be considered transitional between the Marginal and Tropical Forest tribes. These items include: Farming, large thatched houses, dugout canoes (*Carajá*, *Guató*), reburial and urn burial, hammocks (some *Ge*, *Cararí*, *Mashacalí*, *Carajá*), rubber balls (*Apinayé*, *Sherente*), various musical instruments, penis sheath (*Bororo*, *Cayapó*, *Camacán*, "*Tapuya*"), penis thread ("*Tapuya*," *Patashó*), feather-work (*Goytacá*, *Botocudo*, *Timbira*, *Central Ge*), earplugs, labrets, use of genipa and urucú as body paint, and others.

The *Charrua* and *Querandí* near the Paraná Delta adopted the taking of trophy heads from the Tropical Forest tribes and, doubtless under the influence of the neighboring *Tupian* peoples, they made canoes and thatched houses and wore ear, nose, and lip plugs.

Stripped of these borrowings, the Marginal cultures would be extremely simple. They would lack agriculture and domesticated animals. They would also lack loom weaving and probably finger-weaving or twining, though some of them would probably know netting techniques. They would have no basketry (the Fuegian coiled baskets are probably Andean-derived). They would lack pottery or make only a very crude ware. Their canoes would be only of bark and their houses one-family, conical, domed, or lean-to type shelters. They would prepare their food directly on the coals, in an earth-oven, or in water made to boil with hot stones, and they would eat it without salt or condiments. They would use no clothing, except skin robes in the colder areas, and they would lack ear, nose, and lip ornaments, adorning their bodies only with necklaces and paint. Their musical instruments would be limited to rhythm beaters, but probably excluding rattles as well as drums, and their only games would be shuttlecock and athletic contests. They would use no narcotics or stimulants.

THE ECOLOGICAL BASIS

Subsistence.—The subsistence potentialities of the different Marginal groups are difficult to estimate quantitatively, but there is no doubt that food could be had only in limited amounts, so that the general population was extremely sparse. In addition, transportation facilities were so poor that, unable to assemble food and other goods at points of concentrated population, people had to disperse so as to be near their resources.

On the whole, the Marginal peoples inhabited plains or savannas and relied upon collecting wild seeds and fruits, hunting small game, and fishing. This required that they split into small groups, frequently of individual families, and move seasonally from place to place as different foods became available. The Patagonians and the *Ona* of Tierra del Fuego, however, were somewhat distinctive for the emphasis on large-game hunting (the rhea and guanaco), the yield of which supported cohesive bands of 50 or 60 people. The maritime peoples of the Chilean Archipelago, on the other hand, were essentially shellfish gatherers who lived scattered in small clusters along the seashore. Though the Marginal tribes generally avoided large streams, many *Guató*, *Mura*, and *Yahgan* virtually lived in canoes, each vessel carrying an encampment of a small, bilateral and independent family.

It would be incorrect to characterize all the Marginal tribes as nonfarmers. Although the peoples of southern Argentina and Chile lived south of the limits of any American Indian crops, many of the tribes farther north cultivated on a limited scale. It is more important that this farming was not very productive. Outside the

Andes, with its potatoes and cereal-crop staples, agriculture was based on tropical root crops, especially manioc and sweet potatoes. These require a rain forest environment, but the Marginal tribes generally occupied plains or savannas, and rarely had access to gallery forests where they could be grown.

Agriculture was wholly lacking in the Archipelago, and among the *Tehuelche*, *Puelche*, *Charrua*, *Purí-Coroado*, Guiana Marginals, and *Ciboney*. Of the Orinoco groups, only the *Yaruro* farmed a little. Agriculture occurred mainly among tribes of eastern Brazil, among some near the Paraná Delta, on a small scale in the more favored portions of the Chaco (*Zamuco*, *Tumerehã*, *Lengua*, *Ashluslay*, *Mataco*, *Kaskihá*, *Sanapaná*, *Pilagá*) and among the *Huarpe* and *Comechingón*, the last two showing Andean influence in their maize, beans, and quinoa, and llama herds. During the historic period, the *Caingang*, *Yaruro* and probably *Guayakí* abandoned farming, whereas the *Botocudo*, *Coroado*, and perhaps the *Poya* seem to have adopted it.

The main crop among most tribes was maize (it was the only *Huarpe* crop), but it was eaten green by the *Guayakí*, *Mashacalí*, *Malalí*, and *Yaruro*, presumably a practice of incipient farmers. The tropical root crops were grown only by some of the *Ge* tribes, who cultivated strips of gallery forest.

Transportation.—Transportational facilities are no less important than food resources in determining the ability of a population to group itself in sociopolitical units. They also delimit the quantities of material goods a nomadic people may possess. In aboriginal times, the Marginal tribes used either human carriers or canoes, though there was a doubtful case of dog packing in Patagonia. Perhaps the oldest device was the skin bag, which was used at the Contact Period in the far south. The netted bag and the carrying net—the former widely distributed north to México and the latter found in the Andes and north to western North America—were probably also old methods. These were used in the Chaco, and among the *Caingang*, *Botocudo*, *Mashacalí*, and *Yaruro*. The woven carrying basket of the Tropical Forests was adopted in eastern Brazil (*Northwest* and *Central Ge*, *Bororo*, and *Guayakí*), and by the *Guahibo*.

The canoe was of major importance to the Tropical Forest peoples, but among the Marginal tribes its use was limited and its construction primitive. Many Marginal tribes lacked canoes simply because they did not live on large streams. At the same time, they often purposely avoided large rivers because stronger, more aggressive, and definitely more aquatic-minded Tropical Forest peoples occupied them.

Canoes were absent among the *Tehuelche* and *Puelche* of the almost streamless Pampas, though in the Pampas and among some Chaco

tribes the bullboat was occasionally used. Other tribes lacking canoes were the *Guayakí*, *Caingang*, *Bororo*, *Purí-Coroado*, *Tapirapé*, *Sirionó*, *Nambicuara*, Guiana Marginals, most of the *Northwest* and *Central Ge*, and many of the Chaco tribes. The *Bororo* and *Botocudo* adopted dugouts in the historic period.

Bark canoes were used by tribes living peripheral to the Amazon Basin on its headwaters, by the *Suya*, *Mura*, and upper Xingú tribes, and by the Archipelagic peoples. The last, however, adopted the dugout early in the historic period, and later the plank canoe. Dugouts were apparently native to the *Guató*, many of whom virtually lived in them, and to the *Guahibo* of the Orinoco, the *Carajá*, some of the Chaco tribes, and the *Charrua*. Under Andean influence, the *Huarpe* made reed balsas.

The importance of transportation facilities to the sociopolitical group is well illustrated by the *Tehuelche*, *Puelche*, and *Charrua*, who adopted the horse during the early historic period. Through increasing the yield of the hunt and facilitating the movement of foods to a central point and of peoples to the sources of food, it enabled the small, native bands of foot Indians to amalgamate into large, multi-lineage hunting bands. Comparably large bands elsewhere were only seasonal and temporary, as in the Chaco during harvests, or in the Archipelago when a whale was stranded.

House types.—The nomadic or seminomadic life and the small social units of these tribes precluded the use of large, permanent, frame, thatched houses of the Tropical Forest types. Some tribes lacked houses altogether (*Nambicuara*, *Guahibo*, *Ciboney*, *Guamo*, *Taparita*, *Gayón*); others made some kind of crude, temporary shelter (*Guaaja*, *Sirionó*, *Mura*, *Yaruro*). Where construction is described, one of four types is indicated: (1) A domed pole hut covered with leaves, brush, or grass (Archipelago, *Mura*, *Botocudo*, *Patashó*, *Mashacalí*, *Macuní*, Chaco); (2) a conical, tipilike pole hut (*Yahgan*, *Guarañoca* in the Chaco); (3) a portable pole toldo covered with mats (*Charrua* and some Chaco tribes, especially the horsemen) or skins (Pampas, Patagonia, *Charrua*); (4) a single or double lean-to (*Ona*, *Guató*, *Purí-Coroado*, recent *Caingang*). The large, frame multi-family house is reported only among the *Bororo* and possibly the *Mashacalí*, and it was used by some *Northern Ge* as a clubhouse. The *Huarpe* and *Comechingón*, under Andean influence, made slight use of masonry houses and possibly of semisubterranean houses.

People slept on the ground, except among a few tribes which had borrowed the hammock from the Tropical forests (*Sirionó*, *Charrua*, *Apinayé*, *Eastern Timbira*, *Sherente*, *Purí-Coroado*, *Guahibo*, *Yaruro*, *Gayón*, *Mashacalí*, and a few upper Xingú and Guiana Marginal

tribes) and some of the *Ge* who used a platform bed. The Pilcomayo and Bermejo River tribes of the Chaco used the hammock as a cradle.

SOCIOPOLITICAL AND RELIGIOUS PATTERNS

Three principal sociopolitical types and several subtypes were found among the Marginal tribes. The principal types were: (1) Groups consisting of a single bilateral or conjugal family; (2) the unilateral band, consisting of a single, extended family, either matrilineal or patrilineal; and (3) the mixed band, consisting of several unrelated families, lineages, or clans. Where the group was composed of a single family or lineage, all its members were kin and it was, therefore, exogamous. This usually meant local as well as group exogamy. The relationship of the members to one another was based on kinship, and the headman was the family elder. Where the band was composed of several unrelated families, lineages, or clans, local exogamy was not necessary and social behavior and political controls extended beyond kinship relations.

(1) **Family units.**—The bilateral or conjugal family, consisting merely of father, mother (or mothers), and children, underlies all sociopolitical types. Among the *Guató*, *Mura*, and *Nambicuará*, it was the only permanent sociopolitical unit, and, though several families might associate with one another seasonally for special activities, they lacked permanent cohesion and had no leader or chief.

Comparatively little is known about these tribes. The *Guató* family was polygynous, and it lived in its own hut, or, during much of the year, its canoe. The *Guató* had a loose band organization in that several families occasionally foregathered for temporary association with one another, but the band had no political significance. Among the *Mura*, some two to five individual family houses sometimes were close enough to form a village, but each family retained its independence.

The population of these tribes was usually dispersed, so that there was little opportunity for social activities, but the *Mura* had puberty ceremonies, in which girls were isolated and boys were whipped, given parica snuff, and subjected to the ant ordeal. The elements of the boys' rite were borrowed from the neighboring Tropical Forest peoples. The *Nambicuará* evidently lacked a true secret cult, but in ceremonies to the thunder god they used sacred flageolets which were taboo to women and children. These tribes had little or no warfare by which individuals could win honors, but among the *Guató*, individual prestige was won by killing a jaguar.

(2) **Unilineal bands.**—The unilineal band consisted of a single extended family or lineage, i. e., a man, his wife, and their descendants through either the male or female line. The band was patrilineal when postmarital residence was consistently partilocal, women going

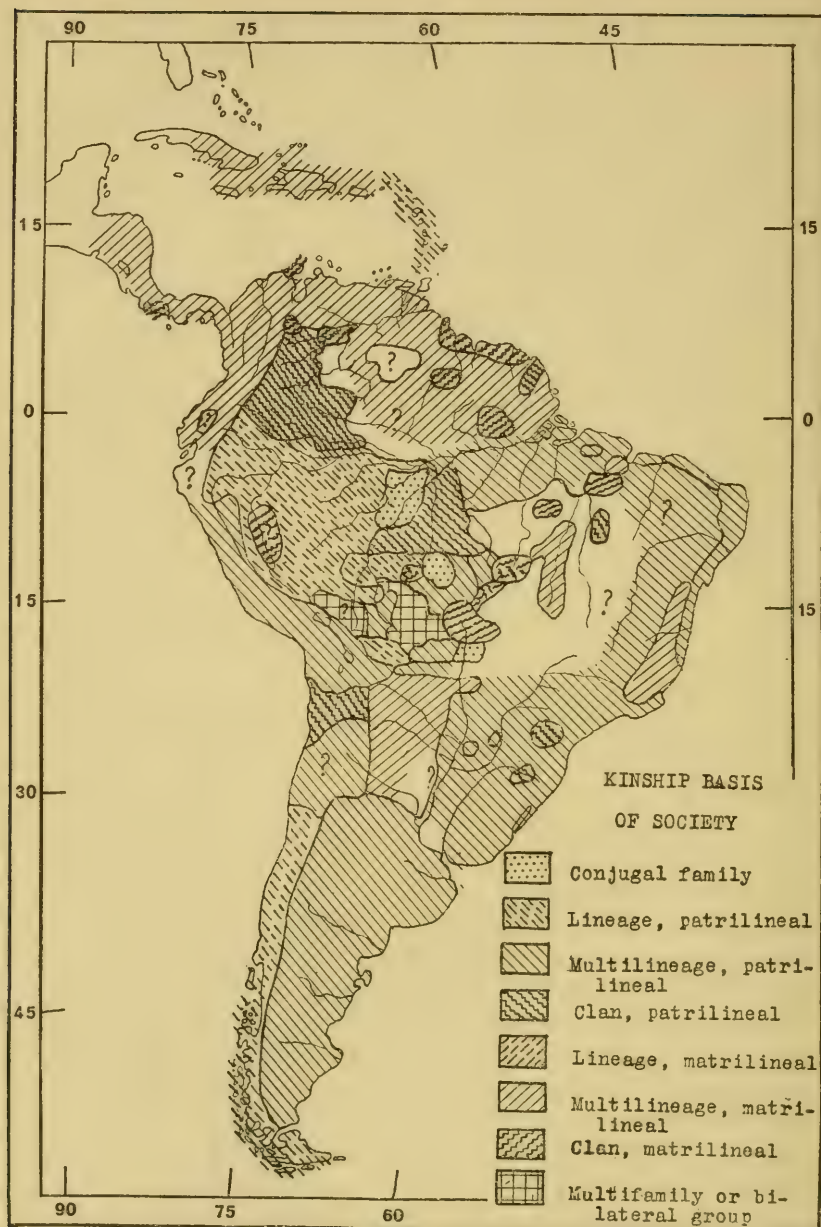
to their husband's family. In this case, it consisted of husband, wife, unmarried daughters, and married sons whose wives came from other bands. When the residence rule was matrilineal, the band was matrilineal and consisted of parents and their unmarried sons and married daughters whose husbands came from elsewhere. Extended families were also common in the Tropical Forest area, where, however, they were much larger than those of the Marginal peoples, often consisting of 100 or more persons occupying a large, communal house (map 20).

(a) *Patrilineal bands*.—Among the Archipelagic tribes—*Alacaluf*, *Chonó*, *Yahgan*, *Ona*, and *Haush* (Handbook, vol. 1)—there were various gradations from the independent conjugal family to the patrilineal band. Among the maritime *Alacaluf* and *Haush*, patrilocal families often lived alone, but among the *Chonó* and *Yahgan*, three or four related families stayed together in shifting villages. Each of these small sociopolitical units claimed a certain territory within which it traveled and which it defended against trespass. Warfare against trespass and in revenge for witchcraft strengthened group cohesion. Political control was synonymous with kinship control by the headman. The only divisions within the local group were those based on sex and age. The former included division of labor, the *Yahgan* secret society (which was probably related to that of the *Ona*), and the various girls' puberty rites. Age gradations were somewhat institutionalized through the initiation of pubescent boys and girls into the tribal society.

The *Ona* socioeconomic group, based on the collective hunting of land game, was larger than that of the Archipelagic peoples, ranging around 60 persons, but it, too, was based on kinship and consisted of about a dozen patrilineally related families. It had the characteristics of the typical patrilineal band (Steward, 1936), being patrilineal, patrilocal, exogamous, and land-owning. Thus, it resembled a localized sib. The lineage head was the headman of the band. The only formal sexual grouping was the secret tribal society to which boys were initiated at puberty. Girls were isolated at puberty. The function of warfare was about the same as among the Archipelagic tribes.

(b) *Matrilineal bands*.—The *Guayakí*, nomadic hunters and fishers, were grouped in bands of some 20 persons each (Handbook, 1: 441). The band had a strict rule of matrilineal residence and so must have tended strongly to consist of a single matrilineal lineage or extended family. There was no formal social differentiation within the band except by sex and age. Pubescent girls were scarified and observed meat taboos; boys had their lips perforated.

The *Siriono* (Handbook, 3: 458) had extended matrilineal families that averaged perhaps 15 persons each. The entire extended family



MAP 20.—Distribution of the types of kinship basis of aboriginal sociopolitical groups.

traveled together and the constituent conjugal families often lived together in a single hut.

(3) **Mixed bands and villages.**—The mixed bands consisted of several unrelated families or lineages. They differed according to whether the component units were individual conjugal families, matrilineal lineages, or patrilineal lineages. Unless fixed by rigid rules of postmarital residence, these types intergraded and overlapped. Nevertheless, a tendency either to matrilocal or patrilocal residence was usually very strong, and most bands appear to have been aggregates of one or the other kind of lineage. In some cases, mixed bands were formed during the historic period, when new economic factors permitted previously independent lineages to band together in permanent association.

(a) *Mixed bands with a matrilineal base.*—The foot Indians of the Chaco—the *Mataco*, *Chorotí*, *Ashluslay*, *Macá*, *Lengua*, *Toba*, and *Lule-Vilela*—had land-owning bands of 50 to 200 persons. The recent *Ashluslay* villages of 1,000 are exceptionally large and may represent a post-Columbian condition. Though the smaller bands or villages may have consisted of a single lineage, most were made up of several extended families, which, being strongly matrilocal, must have been somewhat matrilineal. The band as a whole, however, was endogamous, except among the *Pilagá*.

The occurrence of matrilineal lineages mainly in the Chaco is probably to be explained by the relatively great importance of seed gathering, which placed women in a strong economic and, therefore, social position.

The band chief was usually an elder who attained leadership through personal influence. Some areas had a district chief, an office very possibly resulting in part, at least, from White influence. Hereditary chieftainship was uncommon.

A special bachelors' house or club was not a typical Chaco feature, but occurred among the *Pilagá* and *Ashluslay*. The *Chamacoco* initiated young men in a secret rite. The Chaco held public ceremonies at girl's puberty.

Warfare contributed to social cohesion among these tribes, but did not form the basis for a warrior class. Scalps were the only trophies.

The *Tapirapé* and *Carajá*, also had bilateral bands, each consisting of several extended, matrilineal families.

(b) *Mixed bands with a patrilineal base.*—The *Tehuelche*, *Puelche*, *Pehuenche*, and *Poya*, of the Pampas and Patagonia (Handbook, 1: 127), and the *Huarpe* (1: 169) and *Comechingón* (2: 673) of western Argentina had mixed bands with a patrilineal base. The last two, though little known, had some Andean features (e. g., farming, llama raising, and stone construction) that placed them slightly above a

Marginal level, though sociopolitically they probably resembled their neighbors.

The aboriginal sociopolitical organization of these tribes is not known, but they were hunters rather than seed gatherers and, therefore, were probably predisposed to patrilineal families. Their ecology and social patterns very likely resembled those of the *Ona*. When first described in any detail, they already possessed the horse, which they obtained early in the 18th century from the Spaniards. The horse must have revolutionized their ecological adaptations through its usefulness both in hunting rheas and guanacos on the plains and as a means of transportation. The *Tehuelche*, *Puelche*, and probably the *Poya* and *Pehuenche* became organized in bands of 100 or more persons, which, unlike those of the Archipelagic tribes, consisted of several *tolderías*, or households, each comprising four to five patrilineally related families. A certain fluidity of band membership prevented a band from consisting only of relatives, and there was consequently no rule of band exogamy. Each *Tehuelche* band claimed hunting territories, but the boundaries appeared to have been somewhat flexible and to have depended on the band's ability to defend them.

Warfare was carried on mainly for revenge against witchcraft, murder, and trespass, as in the Archipelago, but a few captives were taken as drudge slaves. These slaves together with chiefs, who had some power though not hereditary status, marked incipient classes. The classes were far less developed, however, than among the horse tribes of the Chaco (*Abipón Mocoví*, *Caduveo*, and *Mbayá*, who may have had an oboriginal predisposition in this direction) or the *Goajiro* of Venezuela. The bands were essentially democratic, and the chief ruled and held a following by persuasion rather than through kinship ties, as in the Archipelago, or through hereditary status.

The Patagonian tribes may have had a men's secret society; otherwise, the main sexual differences were in division of labor.

In religion, the Patagonian and Pampean tribes were on the whole like those of the Archipelago, though group ceremonialism was manifest in the *Tehuelche* girls' puberty ceremony, in which a horse was sacrificed, and in the *Puelche* El-el ceremony, in which an evil spirit was impersonated.

The *Comechingón* (Handbook, 2: 673; Serrano, 1945, pp. 329-332), though Andean-influenced in having permanent settlements based on farming and llama raising, were somewhat Marginal in that each settlement evidently consisted of a patrilineal lineage occupying 10 to 40 houses. New lineages budded off, but what were evidently groups of related lineages formed "ayllus," or larger kinship groups, each owning its land and having a hereditary chief. A horse period is not recorded for the *Comechingón*.

The *Huarpe* had patrilineal, probably extended families, each with its own farm lands and chief. Men had a special drinking house.

The *Querandí* (Handbook, 1: 180), *Charrua*, and some of their neighbors of Uruguay (1:191) probably resemble the tribes of Patagonia and the Pampas in sociopolitical type, but differed from them in possessing certain social traits borrowed from the *Tupian* tribes of the Paraná Delta and certain religious traits—finger mutilations, placing skewers in the flesh, and vision seeking—that are unique in South America and resemble those of the Plains Indians of North America.

These tribes became horsemen, organized in hunting bands. The *Charrua* band consisted of 40 to 60 persons under a weak chief. Each *toldería*, or household, consisted of a single family, but the family composition is not known. Tropical Forest traits, some of them probably of *Tupian* origin, are: war captives killed at their master's death (*Charrua*, *Querandí*), skull trophies (*Charrua*, *Querandí*), and distinctive dress and hammock for the chief (*Yaro*). These suggest incipient class structure, which, however, cannot have been well developed in groups so small.

The *Guahibo* band, averaging 30 members, may have been unilineal, but data on its composition are not available.

(c) *Mixed bands with exogamous clans and moieties*.—Sibs or clans and exogamous moieties, which are really pairs of sibs, differ from lineages or extended families in that their membership is based on the fiction of common descent, through either the male or female line, even though the descent cannot actually be traced, whereas lineages consist only of true relatives. In addition, a particular clan or moiety is typically distributed over several places and is, therefore, exogamous by group rather than by locality, whereas a lineage occurs only in one place and is exogamous by locality. A lineage could become an exogamous sib or moiety if such features as origin myths and totemism reinforced the sense of kinship of its members after their genealogical connection had become untraceable and if it continued to be exogamous without regard to where the members lived.

We do not here attempt a theory of sib origin, but it should be pointed out that the difference between the two is not great and that the lineage or unilineal band organization is very widespread among the Marginal tribes. It is patrilineal among the Southern Hunters, matrilineal among the seed gatherers of the Chaco, and among many farmers. If two lineages were brought together in a single group or community and continued to be exogamous, they would constitute moieties, and if three or more were brought together and continued to be exogamous they would constitute sibs. A change in subsistence, for example from hunting and gathering to farming, might permit the formation of

such multilineage groups, and there is perhaps some significance in the occurrence of clans and moieties mainly in eastern Brazil, where many of the tribes practiced considerable farming and where villages were unusually large for Marginal tribes (the *Apinayé* villages, for example, had 500 to 1,400 persons). Once the pattern of sibs and moieties had been established it could readily spread among neighboring tribes, and a tendency might develop to form nonexogamous tribal or community divisions copied from the exogamous ones (amp 20, p. 684).

Whatever the origin of sibs, moieties, and various associations in eastern Brazil, it is of interest that these sociological refinements developed mainly in the patterns of the Marginal tribes. There was no tendency whatever toward a class system.

Exogamous matrilineal moieties occurred among the *Paru d'Arco*, *Canella*, *Bororo* (each subdivided into sibs), and *Yaruro*. Exogamous patrilineal moieties occurred among the *Sherente*, where each was subdivided into sibs, and among the *Caingang*, where each was subdivided into two groups with reciprocal functions in mortuary ceremonies and where there were also some kind of preferential marriage classes. In spite of their patrilineal moieties, the *Caingang* practiced matrilocal residence.

Exogamous patrilineal sibs and matrilineal sibs were found among various upper Xingú tribes. Possibly the former had some connection with the patrilineal sibs of the neighboring *Tupian* tribes of the Tapajóz-Madeira region.

Eastern Brazil had a considerable proliferation of moieties in addition to those which regulated marriage and of age classes and other groupings. The *Canella* were divided into exogamous matrilineal moieties, three sets of nonexogamous moieties based respectively on divisions of nature, personal names, and sports groups, and four age classes. The *Sherente* had not only patrilineal exogamous moieties with four sibs in each, but seven grades in the bachelor's hut and four other men's associations. The *Apinayé* had four exogamous marriage classes and also nonexogamous matrilineal moieties. The *Tapirapé*, *Tenetebara*, and *Carajá*, though lacking exogamous marriage groups, had nonexogamous ceremonial moieties.

A striking feature of eastern Brazil which links some of the tribes with the Tropical Forest rather than Marginal peoples is a men's club house: *Sherente*, *Northern Cayapó*, *Paru d'Arco*, *Bororo*.

(4) **Sociopolitical units of uncertain composition.**—The *Boto-cudo* bands of 50 to 200 persons consisted of several extended and polygynous families, but descent is not stated. The shaman was band chief.

Puri-Coroado bands had up to 40 persons. Some were single lineages, and thus of the unilineal band type, and others consisted of two

lineages, but the type of descent is not known. The family elder was the chief. Warfare was carried on, and skull and flute trophies were taken.

The *Mashacali* may at one time have had communal houses, perhaps implying lineages. Unusual social features were the warriors' council and the men's house, which served as the center of a cult of the dead and was taboo to women and uninitiated children. The souls of the deceased were supposed to appear from the sky during ceremonies in which the bull-roarer and masks were used.

Information on the Guiana Marginals, most of the Orinoco Marginals, and on the *Ciboney* of the Antilles permits no appraisal of their sociopolitical structure. The same is true of the *Guaharibo*, beyond the fact that they had bands of about 40 persons.

Warfare.—Warfare among the Marginal tribes was rudimentary and contributed to group cohesion rather than to individual status. It was virtually never aggressive and was defensive only where the Marginals adjoined predatory tribes or where trespass on hunting and gathering territories was involved, as in Patagonia, the Chilean Archipelago, and some of the Chaco. The most frequent cause of warfare was revenge for witchcraft against a group member. Typically, it was unorganized, there was no cannibalism, except perhaps among the *Botocudo*, and the taking of trophies or captives was rare. The *Coroado*, *Charrua*, *Querandi*, and neighbors took trophy heads; the Chaco scalped. Exceptional military practices have been mentioned under Sociopolitical Patterns.

Religion.—Religion characteristically gave great prominence to shamanism and magic, and it served individual rather than group objectives. The Marginal tribes lacked a cult religion, a priesthood, and group ceremonies, except those mentioned above in connection with initiation rites, and certain Chaco religious feasts and rites to control sickness and weather. The concepts of the supernatural, the specific practices and beliefs concerning witchcraft, magic, and omens, and the ritual elements varied greatly from tribe to tribe.

Religious concepts.—The Marginal tribes fall into three groups, according to the supernatural beings to which they attached greatest importance.

(1) Belief in a High God prevailed among the Patagonian, Pampean, and Archipelagic tribes, but the god is known to have been an object of prayer only among the last group. There is no proof that belief in a High God was ever universal in America. To the plains tribes, lesser spirits, especially evil ones, were of greater importance in daily life than the High God. The *Huarpe* also worshiped celestial beings and hills.

(2) Among the *Nambicuara*, upper Xingú tribes, and the *Northern*

Ge, the mythological creator was a celestial being, usually the Sun or Moon. The *Northern Ge* had true celestial deities, and the *Nambicuara* held a ceremony to the thunder god, but in both tribes and among the *Sirionó* and *Caingang* spirits, especially of nature, were more concerned with the affairs of men.

(3) The *Bororo* and eastern Brazilian tribes attached great religious importance to ghosts or souls of the dead. The *Bororo* and *Mashacali* impersonated their ancestors in a cult, and the *Caingang* believed that their ancestors, though invisible, attended ceremonies. In the *Canella* boys' initiation, the spirits of the dead entered the neophytes' bodies.

Shamanism.—Shamanism is a near-universal feature of primitive cultures and it is certainly very ancient, but it was absent among some Marginal tribes and very undeveloped among others. Often, both shamanistic and lay curing of disease was connected with ghosts or ancestors.

The *Sirionó* and *Canella* lacked shamans. Among the former, illness was cured by anyone using the bones of the dead, and among the *Canella* by direct appeal to one's deceased kin for aid. Shamans were rare among the *Bacairi*. The *Caingang* shaman merely consulted spirits; a sick person was cured by his own relatives. Among the *Carajá*, the shaman only mediated between the living and ghosts; among the *Purí-Coroado*, they conjured up souls of the dead to ask them questions; and among the *Botocudo*, they conjured sky spirits to a post in the village. These conjuring functions seem to foreshadow the oracular duties of the Circum-Caribbean shaman, but among the Marginal peoples they were exceptional and did not involve a well-formulated tribal cult and deity.

The cases just cited seem to have lacked the concept of the shaman's spirit helper, but the concept was evidently present elsewhere, though the nature of the spirit and the manner in which the shaman obtained it are seldom clearly described. The Archipelagic shaman acquired his spirit helper, often a deceased shaman, in a dream. The *Bororo* shaman's helper was also the soul of a dead shaman; among the *Yahgan*, it was a dwarf or female spirit. In the Chaco, the nature of spirit helper varied.

Shamans performed various services, including prognostication, officiating at a few public rites, finding lost objects, and weather making, but their most important task was to cure illness. Regardless of concepts of their supernatural power, they cured diseases everywhere by blowing, massaging, and sucking out a foreign object believed to have lodged in the body. The only exception is that among some Archipelagic tribes and the *Caingang*, *Sherente*, *Apinayé*, *Northern*

Cayapó, and *Botocudo* shamans recovered a patient's soul, the loss of which was believed to cause the sickness.

The concept that any individual may have a supernatural guardian spirit was doubtless more widespread than recorded. It is attributed only to some of the Archipelagic tribes and to the *Caingang* and *Charrua*; *Charrua* men fasted and scarified themselves to obtain it.

Ritual elements.—A number of ritual elements occur in various religious contexts. Scarification or blood letting, often about the same thing, was very widespread: *Charrua*, *Chaco*, *Purí-Coroado* (who used the venisection bow), *Northern Cayapó*, "*Tapuya*," *Botocudo*, and *Southern Cayapó*. Other elements are of very limited distribution and occur also among some of the more Marginal peoples of North America: the drink tube (Archipelagic shamans, *Yahgan* initiation rites); bull-roarer (*Mashacalí*); scratching stick (*Yahgan* initiations); cutting off a finger joint (*Charrua*, *Minuané*, and neighbors); running a wooden skewer under the skin (*Charrua*, *Minuané*, and neighbors); steam bath for curing (*Purí-Coroado*, *Botocudo*); sacrifices (*Ona*; *Tehuelche*, of horses; *Huarpe*, of chicha and maize); and blowing tobacco to the four directions (*Tehuelche*, historic?). The ant ordeal (*Mura*), flagellation (*Puelche*, *Botocudo*, *Mura*), and taking parica snuff (*Mura*) were probably borrowed from the Tropical Forest tribes.

Disposal of the dead.—In contrast to the tribes with a class structure, the Marginal tribes disposed of all their deceased in the same manner. Simple earth burial was the usual method, but the *Yahgan*, the *Guaharibo*, the *Aweicoma Caingang*, and sometimes the *Toba* cremated (the *Guaharibo* preserved the ashes); the *Purí-Coroado* practiced urn burial, a Tropical Forest trait; the *Charrua*, *Chaná*, *Bororo*, *Camacá*, some *Northern Ge*, and *Mataco* gave reburial (the *Mataco*, after tree burial); and the *Caingang* (*Guayaná*), mound burial. The "*Tapuya*" were accredited with endocannibalism. The *Bororo* and *Caingang* held mortuary festivals.

TECHNOLOGY AND MATERIAL CULTURE

All Marginal tribes had a primitive technology which, in both positive and negative traits, represented a very early—certainly a prehorticultural—American cultural level. More advanced technologies were absent to a surprising degree, even among the tribes who adjoined or formed enclaves within the Tropical Forest peoples and would seem to have had considerable opportunity for borrowing. Some primitive features were contingent upon a nomadic life, e. g., small temporary shelters and the restricted number and size of material objects. Others reflected the environment, e. g., the lack of canoes in areas without large streams or the absence of agriculture in the far south. And a few

primitive features persisted where tribes, such as those of Tierra del Fuego, were too remote from the sources of more advanced things. Many primitive survivals, however, seem to represent a sheer cultural inertia with respect to certain practical aspects of life, even where it would have facilitated daily affairs to have borrowed new processes. For example, simple woven and twilled baskets are easier and faster to make than coiled or twined ones, loom weaving is more efficient than netting or finger twining, and in many localities even dugout canoes would have repaid the labor of constructing them. It is as if these tribes had not yet reached the threshold of paying serious attention to material security; they exhibited a certain recklessness toward the hazards of existence and the food scarcity that kept their population at a low density.

On the other hand, the Marginal tribes laid considerable stress on the more recreational aspects of life. Among the tribes in contact with Tropical Forest or Andean peoples, personal adornment was richly developed among certain groups, musical instruments were present in considerable number, and most tribes had a fairly large number of dances, songs, and games.

Fishing devices.—Hunting weapons are mentioned below (p. 695). Fishing was apparently done by all tribes with nets, harpoons, and spears, and by all but those of the Archipelago with the bow. Fish-hooks and fish drugs, though occurring sporadically, are not characteristic of the Marginal peoples.

Food preparation.—The Marginal tribes are distinctive even in their methods of preparing food. A striking feature is their failure to use salt; it is reported only among the *Tehuelche* and certain Chaco tribes, who obtained it from the Andes. Even the *Purí-Coroado*, who had pepper, used no salt.

The tribes without pottery (see below) had no cooking containers and consequently broiled or roasted their food. The babracot, or grill, so characteristic of the Tropical Forest tribes, was found only among the *Caingang*, and the *Mashacalí*, and some of the latter's neighbors. Other tribes placed food directly on the fire, but some used the earth oven, evidently an old, prepottery trait (the Chaco, *Caingang*, *Ge*, *Purí-Coroado*, *Botocudo*, and some Orinoco Marginals). Stone boiling, which, like the earth oven, has North American parallels, is reported for the *Northwestern Ge*.

Pottery.—Pottery was absent among the Archipelagic tribes and the *Ciboney*, and it apparently reached the *Tehuelche* only recently, perhaps in the historic period. (See map 3.) Its aboriginal presence among the *Northwest* and *Central Ge*, eastern *Nambicuará*, upper Xingú, *Pirahá*, *Mura*, and *Botocudo* is doubtful; even if made by these tribes, it cannot have been of great importance. The remaining tribes

were unquestionably potters, but their ware was plain, incised, stick-impressed, or cord-marked. Painted ceramics are accredited to only the *Huarpe* and the western Chaco, who borrowed it from the Andes, and perhaps the *Guaharibo*, to whom a "beautiful" ware is attributed.

Basketry.—Basketry (map 21) had a limited distribution and was



MAP 21.—Distribution of basket weaves.

probably introduced comparatively late among these tribes. It was absent among the *Puelche* and *Tehuelche*, who used skin and wooden containers, the *Charrua* and most Chaco tribes, who used pots, and the *Botocudo* and *Camacán*.

Coiling would seem to be a fairly old technique, to judge by its occurrence among living tribes at the northern and southern extremes of the hemisphere and its spotty archeological and ethnographic distribution between, but in South America it is the characteristic Andean technique and it has a very restricted distribution elsewhere (map 21). In the Central Andes, it appears to be later than twined basketry; in Northern Chile, coiling is older than twining. Coiling was used by the Archipelagic tribes, the *Comechingón*, the *Pancararú*, some of the *Northwestern* and *Central Ge*, and possibly the Chaco (*Mataco*). Twining, which has a more restricted hemisphere distribution, is reported from Patagonia, the *Huarpe*, the *Carajá*, and the *Shirianá*. Several Chaco tribes used it for mat making.

Baskets and often mats and fans were woven of a single palm frond. These, evidently a primitive form of true woven and twilled baskets, are accredited to the *Mura*, *Guató*, and *Sirionó*. Twilled and woven baskets, borrowed from the Tropical Forests, occurred mainly in eastern Brazil (*Guayakí*, *Caingang*, *Purí-Coroado*, *Northwestern* and *Central Ge*, *Carajá*, *Tapirapé*), and among the *Yaruro*, *Guató*, *Bororo*, *Bacairi*, and *Caingúa*.

Fabrics.—Fabrics were finger-made, either netted or twined, rather than loom-woven. Thread was made of wild bast; cotton was grown only by some of the *Ge* and the *Mashacalí*. Fish nets were made by many of the tribes, and were the only Archipelagic textile product. The Chaco, *Carajá*, *Caingang*, *Patashó*, and *Mashacalí* made netted bags, and the *Charrua*, the Orinoco Marginals, and perhaps the *Mashacalí*, netted hammocks. The *Botocudo*, *Guayakí*, *Bororo*, *Yaruro*, and *Camacán* used also netting for various objects.

Twining was perhaps more common than is known, for textile techniques are seldom described. It is accredited to the Chaco, *Guató*, *Sirionó*, and possibly some of the eastern Brazilian peoples.

Several tribes had adopted the true loom, but, as most Marginal peoples either went nude, or wore skin garments, textiles were limited to woven bands and a few skirts, and loom weaving was a minor craft. The loom and true weaving were probably pre-Columbian among the *Huarpe*, *Caingang*, *Northern Cayapó*, *Camacán*, the *Arawakans* of the upper Xingú, the *Nambicuará*, *Tapirapé*, *Chono*, and *Sirionó*. The *Tehuelche* seemingly adopted it from the *Araucanians* in the historic period. Some western Chaco tribes had learned weaving with cotton or wild bast from the Andes in prehistoric times, but they intensified the craft after receiving sheep.

Weapons.—The Marginal peoples differed from the Tropical Forest tribes in their almost complete lack of the blowgun and in the use among the southern tribes of the bolas and sling, though both groups had bows. Some differences in weapons reflected the environment: blowguns were used in the jungle, not in the more open plains; slings and bolas are plains weapons; arrowpoints were of stone in the plains of the south (*Charrua* to Tierra del Fuego), of wood farther north. But there are nonenvironmental differences: virtually no Marginal tribes used arrow poison (only the *Guahibo* and possibly *Puelche*); their arrows did not have the Tropical Forest variety of specialized points; and clubs were absent among the *Charrua* and Patagonians.

The bow, though found throughout the hemisphere, is late in the Fuegian sequence and it may have spread rapidly at the expense of other weapons, particularly the dart and spear thrower. Outside the Andes, the spear thrower had a sporadic occurrence, representing local survivals. Among the Marginal tribes, it was limited to tribes near the Paraná Delta (*Chaná*, and possibly the *Querandí*), and to the upper Xingú and *Carajá*. The pellet bow, more of a toy and probably of European origin, occurred in the Chaco and among some eastern Brazilian tribes.

A spear or lance was general. Among the horsemen (*Tehuelche*, *Puelche*, *Charrua*, and some Chaco tribes) it was developed to extreme length and became a major weapon, even at the expense of the bow.

The bolas is fairly old in the Fuegian shell mounds, though at the Conquest it extended from the Chaco south only to the Río de la Plata. With the horse, it spread farther south.

Clubs occurred from eastern Brazil to the Chaco, but were not characteristic of the Southern Hunters.

The sling, a characteristic Andean trait, occurred in the Archipelago, among the *Charrua*, and as a toy in the Chaco, but was absent among other Marginal tribes, except the *Ciboney* of the Antilles.

Defensive devices are obviously late acquisitions from the Andean peoples. The *Tehuelche* and *Puelche* used skin-tunic armor and a helmet; the Chaco, a woven shirt or hide armor.

Dress and adornment.—Except in Patagonia, Tierra del Fuego, and among some Chaco tribes, where fur and skin cloaks provided protection during cold winters, the body was adorned rather than clothed. Adornment included painting, featherwork, ear, nose, and lip ornaments, coiffure, headdress, necklaces, arm and leg bands, and tattoo. Ear, nose, and lip ornaments were absent among the Archipelagic peoples, and, therefore, are probably not among the oldest South American culture traits. Elsewhere, these and other ornaments occurred in such variety that their types, distribution, and

history require special study. For the present purpose, it is of greatest interest to observe that when these items were not purely ornamental, they served to distinguish tribes, sexes, societies, or other associations rather than to indicate social status. Chieftains' insignia were rare, and class insignia were probably unknown.

RECREATIONAL ACTIVITIES

Musical instruments.—The Marginal peoples once lacked any musical instruments, except perhaps a stick to beat time, the only instrument accredited to the Archipelagic peoples at the Conquest. No instruments are reported for the *Charrua*. A rhythm beater in the form of a bamboo stamping tube became characteristic of the *Bororo*, *Ge*, and other tribes of eastern Brazil, and rattles, often of bark but of gourd where there was some farming, were widely distributed, except in the Archipelago. The Chaco evidently had only the rattle at one time, but it borrowed a considerable number of other instruments, especially wind instruments, from the Andes. Similarly, eastern Brazil borrowed trumpets, panpipes, flutes, and other instruments from the Tropical Forests. From the same source, the *Guahibo* acquired rattles, flutes, and panpipes.

Intoxicating beverages.—Chicha, a fermented alcoholic drink made of various cultivated plants or wild fruits, was not originally part of the Marginal culture. At the Conquest it was unknown to the Archipelagic and plains tribes, except the *Puelche* and *Huarpe*. Its principal occurrence was eastern Brazil: *Caingang*, *Ge*, *Tenete-hara*, *Nambicuara*, upper Xingú tribes, *Mura*, *Tapirapé*, *Carajá*, probably the *Purí-Coroado*, and other Marginals near the Brazilian coast. Here, it was certainly borrowed from the Tropical Forests. It was also made by the Orinoco River Marginals. The Chaco made chicha of wild fruits, and the *Guató* and *Bororo* are accredited with a palm wine.

Games.—Games had a limited occurrence. Of special interest are the *Ge* log races and the dice games (Andean), hockey, bull-roarer, stilts, "snow-snake," and shuttlecock found among many of the tribes, especially in the Chaco. (See p. 503.)

Tobacco and other narcotics.—Tobacco was comparatively recent among the Marginals. It is reported only for the *Bororo*, who smoked cigars; the Chaco Indians, who chewed it or smoked it in elbow and tubular pipes; and the *Coroado* and southern Amazon Marginals, who usually smoked it in a pipe (map 9). Pipes have been found archeologically in the Pampas, Patagonia, and *Caingang* territory, but none of these tribes used tobacco in historic times.

Other narcotics are equally rare among the Marginals. The Chaco used a narcotic snuff, the *Mashacalí* a vision-producing grub, the *Mura*

and *Guahíbo* parica snuff, and the *Pancararú* a vision-producing drink, all borrowed from Tropical Forest neighbors.

THE TROPICAL FOREST AND SOUTHERN ANDEAN TRIBES

The Tropical Forest and southern Andean peoples differed from the Circum-Caribbean and Central Andean tribes and resembled the Marginal peoples in having sociopolitical groups which were based on kinship and lacked classes. Their groups, however, were characteristically large, semipermanent villages, whereas the Marginal groups were small and more or less nomadic. But each village was independent; there were no federations or empires, and, political units that transcended the community were exceptional.

Both the Tropical Forest and Southern Andean peoples had loom-woven textiles, fairly commodious houses, developed basketry, dugout canoes or balsa rafts, and other material traits and technologies that were characteristically absent or little developed among the Marginal tribes. On the basis of the specific forms of these features and of certain environmental adaptations, however, the Tropical Forest peoples must be distinguished from those of the southern Andes. The former, occupying the tropical rain forests of the Guianas, the Amazon, eastern Brazil, the coast south to the Paraná Delta, and the Paraguay River, were linked with the Circum-Caribbean area. They had tropical root crops, frame-and-thatched houses, dugout canoes, twilled and woven basketry, and the same general tradition of ceramics. The southern Andean peoples were linked with the central Andes in that they had potatoes, cereal crops, and agricultural techniques adapted to deserts and semideserts, coiled baskets, some stone houses, balsa rafts, metallurgy, llama breeding, Andean type clothing, and many minor items derived from Perú and Bolivia.

Thus, two streams of historical influence are evident in the material culture and technology of these tribes: one carrying central Andean traits southward, especially after the *Inca* conquest, to Chile and Northwest Argentina, and representing desert and Highland adaptations; the other, carrying Circum-Caribbean traits down the Atlantic Coast and up great rivers, and representing coastal, riparian, and rain forest adaptations. In the Chaco, the farming *Arawakan Chané* (*Chaná* and *Guaná*) in the north, and the natively riparian and later equestrian *Abipón*, *Mocoví*, and perhaps *Payagua* and *Mbayá* of the south and east were evidently influenced by both streams, and it is possible that considerable Tropical Forest influence even reached the *Araucanians* of Chile.

Because of the great difference in the element content of their cultures, the Tropical Forest and the southern Andean tribes are treated

separately. Their sociopolitical patterns, however, were of the same general type, though the latter approached sub-Andean types.

THE TROPICAL FOREST TRIBES

The Tropical Forest culture, though comparatively uniform, had several subtypes. As volume 3 of the Handbook carried a description of the general culture (pp. 886-895) and of the subtypes (pp. 896-899), the Tropical Forests will be accorded less detail than was given the Marginal groups.

Some of the subtypes, especially those of the western and southern Amazon, represented gradations between the Tropical Forest and Marginal cultures and they are classed as semi-Marginal (map 18). Particularly, they lacked certain characteristic material and technological elements of the former and they had several distinctive social and political features.

THE ECOLOGICAL BASIS

The Tropical Forest tribes had a fairly adequate subsistence, but it was by no means based entirely on farming. These peoples were riparian and maritime, and water resources were no less important to them than crops. In fact, population density was greatest on the coasts and rivers.

After receiving the horse, the riparian *Abipón*, *Mbayá*, and *Mocoví* gave up what farming they had practiced to become looters or overlords of farming tribes (Handbook, 1:250). The *Goajiro* turned from farming to cattle raising.

Farming was by slash-and-burn in the rain forests, a method requiring periodic shifting of the fields if not of the village. The most important crop was bitter manioc (prepared with the tipití or manioc squeezer) which, however, was little grown in eastern Bolivia and farther south. Also staples were: sweet manioc; sweetpotatoes (*Ipomoea batatas*); a species of yam (*Dioscorea*); *Xanthosoma*, the New World equivalent of taro; arrowroot (*Maranta arundinacea*); maize, often in many varieties; pepper (*Capsicum*); beans and squash. The Tropical Forest people also cultivated various palms, gourds (*Lagenaria vulgaris*), calabashes (*Crescentia cujete*), such dyes as bixa (*Bixa orellana*) and genipa (*Genipa americana*), cotton, tobacco (*Nicotiana tabacum*), and in various localities, arrow reeds, cane, palms, and other special plants. Although some opinion tends to place the original home of maize and perhaps other crops in eastern Bolivia or Mato Grosso, the Tropical Forests cannot definitely claim the first domestication of any but the tropical root crops, which are distinctive of them.

The Tropical Forest peoples characteristically lived along the coasts and rivers, and they were skilled boatmen and fishermen. The most

common craft was the dugout canoe, but the *Island Carib*, expert ocean navigators, used planked canoes; some of the tribes on the smaller tributaries of the great rivers, however, made only bark canoes, and the eastern Chaco had bullboats. Fish, a major food resource, were taken with drugs (p. 277), baskets, nets, weirs, multiprong and harpoon arrows, and, frequently, hooks. Important foods along the Amazon and its main tributaries were turtles, turtle eggs, and large water mammals (especially manatee (*Trichechus inunguis*) and river dolphins), the last taken with harpoons. On the coast, fish and shellfish were eaten.

Land resources were also exploited. Wild plants (see Lévi-Strauss, Handbook, vol. 6) were collected with the aid of a great variety of baskets and, in many localities, the climbing ring (a loop held between the feet and placed against a tree trunk to aid climbing). Game was taken in various kinds of traps (p. 265) or it was killed with the spear, the bow, a great variety of special types of arrows (p. 229), and, in the western portion of the area, the blowgun (map 4). Most of these tribes kept Muscovy ducks.

These food resources not only supported a population that was much denser than among the Marginal peoples, but, combined with the transportational advantages of the canoe, they permitted people to live together in larger and more permanent communities. Whereas the Marginal band usually had 10 to 60 persons, the Tropical Forest village consisted of hundreds and even 1,000 or more people welded into a more or less cohesive unit. The village, moreover, could remain in one place several years, but it was rarely permanent. Soil exhaustion periodically required that it be moved to the site of a new plantation, and many tribes customarily destroyed the village at the death of one of its inhabitants (p. 354).

SOCIOPOLITICAL AND RELIGIOUS PATTERNS

The Tropical Forest village, like the Marginal band, usually consisted of an extended family or lineage, i. e., several generations of families related through either the male or the female line. In the case of the Marginal peoples, a lineage ordinarily could not exceed 50 or 60 people before it had to split, a new lineage budding off to become an independent band; only an unusual ecology, such as that based on the horse in Patagonia, permitted several lineages to remain together in a single mixed band. Among the Tropical Forest tribes, the greater resources permitted the lineage to become much larger before a new lineage separated off, and the new one often set up its own house not far from the parent lineage. Where several houses made up a community, the arrangement varied with local circumstances. In some areas, they were tightly clustered around the chief's

house and the men's club, and they were enclosed by a palisade—the coastal *Tupí* village had up to 700 or 800 persons. In other places, as along the Amazon River, they were closely spaced for a distance of many miles. In either case, it would seem that the household typically consisted of a lineage, had its own chief (the family head), and retained considerable independence.

More rarely, as among the *Apiacá*, the maloca, or large house, accommodated several hundred individuals who evidently constituted several kin groups.

Among the Tropical Forest people as among the Marginal tribes, the distinction between the exogamous sib and the lineage or extended family is often difficult to make. By definition, a sib or clan is exogamous regardless of whether the members live together and whether their relationship can be traced. The fiction of relationship of its members is usually supported by legends that trace their descent from a mythical ancestor, and their cohesion is maintained by ceremonies and other observances. Among the Tropical Forest peoples, the most common sociopolitical unit was the extended patrilineal family, occupying a single large house. Where the house stood alone the group was exogamous by locality as well as by kin. In the northwest Amazon, each exogamous household had ceremonies and myths, and thus might be considered a localized sib. In other regions, however, several extended households formed a village, so that there was household exogamy but not local exogamy, nor sibs properly speaking. In short, there seems to have been a strong predisposition to patrilineality, which showed every gradation to true sibs, though lack of genealogical and other data prevents their adequate classification.

It is logically possible that matrilineal, extended households might have occurred as the counterpart of patrilineal ones. Actually, they and matrilineal sibs are restricted to the Guianas, the *Goajiro*, and the *Patángoro*, and they occur doubtfully among the *Ucayali River Panoans* and perhaps some upper *Guaporé River* tribes.

Strong chieftainship and social classes were not characteristic of the Tropical Forest tribes, because duties and obligations based on kinship overshadowed those based on status. Where the village consisted of a single lineage, the chief was merely the family head and interpersonal relations were controlled by factors of age, sex, and relationship. In the strongly nucleated, multilineage villages, there was a village chief, who usually succeeded his father and was often a shaman. In such villages, the chief controlled some of the activities of people who were not his relatives. This represents the beginning of civil government, but the chief had authority only in limited contexts, such as land use, warfare, or hunting. Among the Circum-Caribbean and Andean tribes, the chief's status carried many special privileges; the

Tropical Forest chief, however, merely served the community, and he gained little personal advantage beyond a few extra wives and some support from village members.

Common people frequently acquired some status through special activities, especially warfare and hunting, but this rarely led to the formation of a special class and never to hereditary rank. Where villages were usually large and warfare was well developed, as in the Guianas and among the coastal *Tupí*, the chiefs had unusual power and a semihereditary position, the warriors achieved fame, and the war captives tended to form a loose slave class. But the classes were not fixed: the slaves were fitted into the kin groups, usually intermarrying into their captive's family and being regarded as relatives, and their children became freeman. The serflike status of the *Macú* with respect to the *Tucano* was exceptional. The eastern Chaco, however, evidently had an aboriginal tendency to a class system. Here, certain tribes carried on considerable warfare and accorded warriors membership in a special class. After the Conquest, the acquisition of the horse so intensified warfare and the spirit of conquest that captives, both individuals and whole tribes, came to form a lower class while the warriors became a definite upper class. These Chaco tribes are not classed with the Sub-Andean peoples, however, because the classes were largely of post-Columbian origin and because the tribes lacked other diagnostic Sub-Andean features, such as the priest-temple cult.

Types of kinship basis.—The lineage seems to have been fundamental to most if not all of Tropical Forest society, and even in multilineage villages each lineage probably retained its identity. The Tropical Forests were overwhelmingly patrilineal, even where the sib was the basis of society (map 20). Matrilineal descent is recorded only in the Guianas and parts of Colombia among tribes adjoining or near the Circum-Caribbean area. As the latter seems to have been definitely matrilineal, the peripheral occurrences of matrilineal descent may have been the result of Circum-Caribbean influence which is postulated also to have introduced considerable material culture and technology to the Tropical Forests. (See p. 726 ff.) The native patrilineal basis is perhaps correlated with the importance of hunting and fishing, both of which were masculine activities.

The patrilineal tribes.—The extended, patrilineal household was characteristic of the upper Amazon Tropical Forest (*Panoan*, *Tupian*) and the semi-Marginal peoples (*Záparoans*, *Pebans*, *Western Tucanoans*, and some *Panoans* and *Arawakans* (Handbook, 3, pp. 507–798)). Usually the village consisted of a single communal house, but the *Tupí* on the Amazon and Marañón Rivers had larger, multihouse villages, and they and the *Quijó* kept captive slaves. The *Panoan*

tribes of the lower Ucayali River are accredited with totemic, matrilineal clans, but the data are not conclusive. The *Arawakans* of the neighboring Juruá-Purús region had animal-named patrilineal households. Among the *Tenetehara*, the coastal *Tupí*, and the *Guaraní*, the villages were extremely large and each house held 10 or more patrilineally related families (several hundred persons) under a powerful chief who was usually the shaman. Intensive warfare contributed to *Tupí* village cohesion.

Tribes with patrilineal sibs, except the *Palicour* and *Chacó*, adjoined those with the patrilineal lineage or household (map 20). In the northwest Amazon, the *Tucuna* had patrilineal, nonlocalized sibs despite some matrilineal households. The *Tucanoans* and possibly the *Arawak* and *Carib* of the Uaupés-Caquetá area and the *Witotoans* had patrilineal sibs, each generally localized in a communal house. The *Witotoans* were cannibals, and kept some captives as slaves. An outstanding *Tucanoan* feature, perhaps shared in some degree by the *Witotoan*, *Tucana*, and *Yurimagua*, was the ancestor cult into which boys were initiated with use of sacred trumpets. Each household or localized sib held its own cult ceremonies. To the north of the northwest Amazon, the *Achagua* and *Sáliva* also had patrilineal, totemic sibs, each perhaps occupying a communal house. Often one such house constituted a village. Chieftainship was weak, and warfare was purely defensive, lacking trophies, cannibalism, or slave taking. Sexual cleavage is manifest in the use of a men's clubhouse, where drinking bouts were held.

A second main center of patrilineal sibs was among the *Tupian* tribes of the lower Madeira-Tapajóz region (*Mundurucú*, *Parintintin*, and *Tupí-Carwahib*). The *Maué* and lower Xingú *Tupians*, however, had small villages consisting of a few single-family houses. The *Mundurucú* settlements, though large, were dispersed, and each house had its own chief.

The *Chocó* also had exogamous, patrilineal lineages, which were perhaps the equivalent of sibs. The chief had little power and warfare was not developed.

The matrilineal tribes.—The matrilineal tribes included several in the Guianas and the *Goajiro*, *Cagabá*, *Patángoro*, and Antillean *Carib*. In the Guianas, nonlocalized matrilineal clans were found among the coastal *Arawak*, the *Wapishana*, and *Aparai*. Although society was based on kinship, war captives formed a loose lower class among both the *Arawak* and *Carib*, and Guiana *Carib* chiefs enhanced their prestige by attaching a special following of freemen, usually sons-in-law, to their households. The *Aparai* and *Papishana*, however, appear to have coupled patrilocal and presumably exogamous settlements with their matrilineal clans.

The *Goajiro*, whose aboriginal culture is virtually unknown, had matrilineal sibs. In post-Columbian times, a cattle-horse complex introduced a secondary series of social cleavages based on wealth. The *Chaké* ("Motilones") south of the *Goajiro* had small shifting farm villages. The *Cágaba*, despite archeological evidence in their territory of a more advanced culture, had a simple unstratified, matrilineal society.

The *Patángoro*, like their Sub-Andean neighbors, had villages of 80 to 90 houses and a ceremonial structure, but the basis of their society was matrilineal clans rather than classes. Their warfare was for the purpose of obtaining victims for cannibalism and not slaves.

Other sociopolitical groups.—Of the remaining tribes of the Tropical Forests, many were doubtless predominantly unilineal and others may have been bilateral, but the basis of their descent is not now clear.

In eastern Colombia, the *Guayupé* and *Sae* lived in palisaded villages, each with a plaza and ceremonial building in the center. They had an elective chief, who used a stool, wore feather blankets, and was cremated at death, but other people acquired status with age. Boys were initiated into warrior status, but war was waged to take victims for cannibalism and human trophies, not to procure slaves. The nearby *Otomac* and *Guamo* also had large villages, and the chiefs controlled groups of houses. The *Betoi* village had one or more houses, each sheltering an extended family and ruled by the elder. A unique *Airico* feature was hired laborers.

The *Cayapa* and *Colorado* in western Ecuador had villages consisting of one to several multifamily houses, but whether these sheltered lineages is not clear.

In eastern Brazil, some of the small non-*Tupian* tribes were on a Tropical Forest level, but little is known of their sociopolitical groups. The "*Tapuya*" had palisaded villages. The *Camacán* communal houses sheltered as many as 20 families. The *Tarairiu* were governed by strong shaman-chiefs, who had special announcers; when they died, these chiefs were cremated and their ashes drunk, whereas deceased commoners were cooked and eaten.

Farther south, around the Paraná Delta, the warlike *Guaraní* and *Timbú* lived in permanent, palisaded villages, in contrast to the nomadic, Marginal *Charrua*, *Querandí*, and other neighbors.

In eastern Bolivia, all tribes except the Sub-Andean *Mojo*, *Bauré*, *Manasí*, and *Paressí* (p. 714 f.) were probably Tropical Forest in their general culture, though little is known of the composition of their sociopolitical groups. Some had small communities and perhaps should be classed with the Marginal peoples: each *Atsahuaca* and *Chapacuran* family lived in its own lean-to; the *Chimane* village consisted of a few small houses; the *Canichana* hamlet was fortified and

contained only 60 persons. At the other extreme were tribes with large communities and some Sub-Andean features. The *Cayuvava* village had 1,800 to 2,000 persons, and as many as 7 villages were controlled by a single chief. Each village had a temple where offerings were made. The *Xaray* communities had 1,000 persons; four communities were controlled by one chief. Perhaps the *Cayuvava* and *Xaray* belong with the Sub-Andean tribes. Between these extremes, the *Yuracare* village consisted of one or several large houses, each perhaps sheltering a lineage and governed by the family head. Similar villages are also reported for the *Guarayú*, *Pauserna*, *Mosetene*, *Tacana* (100 to 200 persons per house), *Tiatinagua* (2 to 8 families per house and each house under its own chief), *Southeastern Panoans* (2 to 3 communal huts per village), *Yamiaca*, and *Chiquito* (several small houses and a men's club in each village, and a village chief and elected council). The *Chiquito* and *Canichana* were somewhat warlike. *Chiquito* captives married into the tribe; *Canichana* captives were eaten or else became slaves, but such slavery seems not to have created a true class system.

In the Gran Chaco, the western tribes belonged with the Marginal peoples, but the *Mbayá*, *Payaguá*, *Mocoví*, *Abipón*, and *Guaná* should certainly be classed with the Tropical Forest tribes. The kinship basis of society is obscure, but to judge by postmarital residence, the *Arawakan Guaná* may have been somewhat patrilineal, like the eastern Bolivian tribes, and the others somewhat matrilineal, like the western Chaco Marginal peoples. These tribes had an aboriginal basis for a stratified society in their special warrior's class. After the Conquest, the acquisition of horses intensified their military activities and true classes emerged. The *Mbayá*, who were exceptional for their wars of conquest, had the following social strata: (1) Several kinds of nobles and chiefs with special privileges; (2) a warrior class into which boys were initiated; and (3) serfs, consisting of conquered tribes, especially the *Guaná*, and their descendants. Among the *Mocoví* and *Abipón*, the privileged class of warriors became a class of nobles, distinct from the commoners. The *Payaguá* drew a distinction between chiefs, nobles, and commoners. The *Tereno* and *Guaná*, though victims of their more aggressive and powerful neighbors, had a somewhat endogamous class of chiefs, a warrior group which achieved its status, and a class of commoners, or "camp followers." The *Tereno* also were divided into two endogamous moieties.

Warfare.—Warfare was probably a major reason for the strongly nucleated communities in much of the Tropical Forest area. Houses were closely grouped and surrounded by palisades, additional protection being afforded by pitfalls, poisoned stakes planted in trails, coltrots, and other devices.

Whereas the Marginal tribes fought only in self defense or for revenge, a large number of Tropical Forest peoples carried on warfare to obtain human trophies and victims for cannibalistic feasts. In addition, a few tribes, including several of the western and northwest Amazon, some in the Guianas, and the horse bands of the Chaco, took slaves. But true slaving was mainly a post-Columbian development; tribes which had formerly killed and eaten their victims and many tribes which had never taken captives at all conducted raids to obtain slaves for sale to the Whites.

Cannibalism, though widespread, was not practiced by all tribes. The *Tupí* and *Carib*, who attached great importance to warfare, killed and ate their captives, often after keeping them a long time. Unlike some of the Sub-Andean peoples, they did not sacrifice their victims in religious rites; ceremonial objectives, though not absent, were secondary. Cannibalism seems to have been motivated by deep hatred of enemy peoples, and there were unending reprisals against former hostilities. Cannibalism was also practiced among the *Tupí* and some of the *Zaparoans* of the upper Amazon, the Guiana *Arawak*, the *Witoto*, and other tribes of the northwest Amazon, the *Patángoro* in Colombia, the *Canichana*, in eastern Bolivia, and the *Amniapa* and *Guaratagaja* of the Guaporé River, but elsewhere it was absent.

Human trophies, which were displayed in order to give their takers prestige and to insult the enemy, were of many kinds; shrunk whole heads (Montaña), skulls (often made into cups), flayed skins (*Arara*), scalps, and flutes made of long bones (p. 409). Trophy taking tended to be associated with cannibalism.

Crisis rites.—In societies based essentially on kinship and lacking public ceremonialism, crisis rites had great importance in underlining sex differences, in marking age status, and in providing focal points for ritual activity.

Birth customs involved many magical observances and taboos, and, though they were primarily concerned with the mother and child, the father was often deeply involved. (See The Couvade, p. 369.)

Pubescent girls were always subjected to certain rites, which, though more or less concerned with their physiological condition, also signaled a social transition to womanhood. The girl was isolated and, in different areas, whipped, made to dance, scarified, her hair cut, her clitoris excised (*Sáliva*), given drugs, and the like. In many tribes, boys passed informally into manhood, their status being evidenced not only by dress and behavior but often by their occupation of the men's clubhouse. Frequently, however, their hair was cut, their noses pierced, or some other public rite celebrated, and in some instances, they were initiated into a secret society or cult, as mentioned below. The *Sáliva* boy was circumcised.

In contrast to the Marginal tribes, who almost always buried their dead, the Tropical Forest peoples disposed of the body in many ways, but they rarely accorded chiefs or leading men special treatment. Only the *Guayupé* and *Sae*, both doubtless under Andean influence, gave chiefs burial different from other people.

Direct earth burial was the principal Tropical Forest method, but many tribes subsequently exhumed the bones and reburied them in the earth (*Mundurucú*, *Apiacá*, *Abipón*, *Mocoví*), in a cave (*Otomac*), or in urns (upper Amazon *Tupí*, *Goajiro*, some *Island Carib*, *Guaraní*, coastal *Tupí*, and probably many other tribes), or they preserved the bones in the house or elsewhere (*Juruá-Purús Arawak*). The *Atorai* and *Rucuyen* cremated their dead, and several tribes drank the cremated ashes in chicha (*Montaña*, *Sáliva*, *Island Carib* in the case of chiefs, *Arapium*, *Guayupé*, and *Sae*). The *Montaña*, *Tarairiu*, and *Arapium* also ate the corpse. The *Chaké* exposed the body and later made a bundle of the bones.

The practice of mummification, which in the Andes is associated with strong ancestor worship, is accredited to only the *Piaroa* of the upper Orinoco. Many Tropical Forest tribes, however, had what amounted to a cult of the dead; for example, the northwest Amazon and Guianas, where there were public mourning ceremonies.

Religion and shamanism.—Tropical Forest religion resembled that of the Marginal peoples in that it centered in the shaman, who had his own supernatural helper and whose main functions were to cure, prognosticate, and perform magic. The Circum-Caribbean and Andean pattern was slightly foreshadowed in that the shaman sometimes performed oracular functions, but he probably consulted his own spirit helper rather than a tribal god. In fact, true tribal gods were rare and there were no temples or idols. Celestial beings, which, in the Circum-Caribbean and Andean areas, were frequently tribal deities and the objects of public ceremonialism, were usually myth characters. Several Tropical Forest tribes, however, had public ceremonials, and a few of these involved tribal gods and spirits, though most of them were secret society rites.

There were considerable local differences in religion, owing to the limited distribution of secret societies, cults, ritual elements, public festivals, and particular concepts of gods and spirits.

Cult religion.—Cult religion is exemplified by the sib ancestor cult found in the northwest Amazon (*Handbook*, 3: 885) and perhaps among the *Achagua* and *Sáliva* (*Handbook*, 4: 410). This, however, was as much a secret society as a tribal cult: boys were initiated to membership and to the secret of certain sacred religious paraphernalia that were taboo to women and children. The *Juruá-Purús* feast of nature spirits, though providing public festivals, also involved sex

distinctions in that the sacred musical instruments were taboo to women and children. Even the *Tacanan* priest-temple-idol cult (Handbook, 3: 447), though somewhat Andean in general features, seems really to have been a boys' initiation rite, women being excluded.

Public ceremonies.—The *Tupian* Grandfather cult, with its messianic features and hope of a better life (Handbook, 3: 131), was a true tribal religion. The *Sáliva* ceremonies in honor of the creator (Handbook, 4: 410), though in a wholly different pattern, were also tribal rites. So were the harvest ceremonies of the lower Amazon (*Tupinamba*, *Tenetchara*, *Maué*, *Mundurucú*), the *Guaraní*, the *Tairairiu*, and the *Chaké*, the fertility rites of the *Manao* and the *Juruá-Purús Panoans*, the first fish ceremonies of the *Achagua* and *Sáliva*, the Pleiades festival of the Chaco, the bush spirit ceremonies of the *Tupinamba*, *Mundurucú*, and *Guaraní*, the ghost rituals of the lower Xingú tribes, and the mourning ceremonies of the Guianas and Northwest Amazon.

Supernatural beings.—The principal deities tended to be myth characters rather than objects of religious worship, except that among the *Tupian Guaraní*, *Tupinamba*, *Parintintin*, and *Apiacá* they entered something of a cult. The main gods or mythological beings were most often the sun, moon, stars, and thunder, and it is easy to see in them prototypes of some of the major Andean tribal gods. Gods of mountains, springs, rocks, and ocean, and place spirits generally were much less important than in the Andes. There was, however, belief in a multitude of bush and other nature spirits, and many of them were revered and feared. Ghosts, too, were commonly of religious importance, especially as shamans' helpers.

Shamanism.—Shamanism of the Tropical Forests was in the general pattern of the Marginal peoples. The shaman obtained a special spirit helper. Little pertinent information is available, but it seems usually to have been a nature spirit or human ghost (for example: a tree spirit, *Tucuna*; a jaguar, *Witotoans*, *Tucanoans*; a dead shaman, *Jivaro*). The werejaguar concept was widespread. The shaman's principal function was to cure disease, which he did with the aid of a drug or narcotic, especially tobacco but locally with other plants (this volume, pp. 588-599), and by massaging and sucking out an object-believed to have caused the sickness but sometimes by exorcising evil spirits or by recovering the patient's "lost soul." The shaman also practiced black magic, prognosticated, controlled weather, and, where there were group ceremonials, he generally officiated. In fact, his supernatural power was often so great as to give him special social status, and, in many tribes, great political power. Among the *Tupí*, he was often the chief.

Ritual elements.—Many ritual elements, such as the ant ordeal, snuff taking, gashing and bleeding, circumcision, scarification, flogging, use of sacred trumpets, cutting the hair, use of the scratching stick, and the like were, very widespread, but they entered different local contexts. For example, the ant ordeal was used by the *Rucuyen* and some of the *Tupí* south of the Amazon to test young boys, by the *Andaqui* to try out warriors before battle, by the *Sáliva* to induct new chiefs, by the Guiana *Arawak* as a feature of death rites, and by other Guiana tribes in hunting ritual. Whipping was used by the *Macushí* and Montaña tribes for pubescent girls, by the *Guayupé* and *Sae* to make boys into warriors, by the *Yurimagua* while initiating boys to the tribal cult, by the *Manao* for young boys, and by the *Chaké* (*Motilonés*) for participants in the harvest festival. The scratching stick was employed by *Tucanoan* and *Chocó* girls during their isolation at puberty.

TECHNOLOGY AND MATERIAL CULTURE

The Tropical Forest peoples had a far richer technology and material culture than the Marginal tribes. These features were found also in the Andean civilization, which, however, elaborated them far beyond the Tropical Forest. The Tropical Forests did true loom weaving with cotton, but used no wool and lacked the fine and intricate Andean weaves. They often made good painted or modeled pottery but fell far short of the Andes in number of colors and intricacy and variety of design. The Tropical Forests entirely lacked metallurgy, stone construction, and stone sculpture, the last two precluded by the absence of stone.

On the other hand, the Tropical Forests had a number of traits not found in the Andes: frame houses, hammocks, mosquito shelters, the blowgun, pepper pot, rubber, bark cloth, dugout canoe, climbing ring, hollow-log drum, babracot, and poisoned arrow. Some of these had only a partial distribution in the Tropical Forests, but many extended to the Antilles and Central America.

Basketry.—The characteristic basketry type is woven, usually twilled, but in many places it is also hexagonal and of wicker. Some tribes such as certain of the *Tupí* south of the Amazon (*Parintintin*, *Tupí-Cawahíb*), however, used only a simple twilled basket woven of a single palm leaf. The Guianas had the greatest variety of weaves and basketry types. Twining is found only in the northwest Amazon and in one or two east Bolivian tribes (map 21), and coiling, which is essentially Andean, only among the *Chocó*.

Weaving.—True loom weaving—a simple technique with the weft passed in and out between the warps—was widespread, extending from the Guianas to the upper Amazon and south to the *Tupí-Cawahíb*, *Guaraní*, and *Mbayá*. Many tribes, however, only wove bands, be-

cause large fabrics had little use in a climate in which people generally wore few garments. The main yarn was cotton, usually domesticated though sometimes wild, but some tribes used bast of wild plants. Designs were largely limited to woven-in stripes, but tribes bordering the Andes painted designs on their textiles.

The *Tucanoans* and most of the *Tupí* tribes had a twined or finger-weave without a loom—a technique similar to that of twined basketry in which a pair of wefts enclose each warp element, being given one turn or twist before enclosing the next.

In addition, netting occurred in the Guianas, northwest Venezuela, and eastern Bolivia, and among the *Achagua* and *Sáliva* (who had no true loom).

Ceramics.—Pottery was made by all these tribes, but the wares varied greatly. They were plain on the Juruá-Purús and in northwestern Venezuela. Elaborately modeled wares occurred in the lower Amazon and may be linked with the Cirum-Caribbean area. Painted vessels were found in the same area as the last and also up the tributaries of the Amazon and south among the *Tupí-Guaraní*.

Houses and household goods.—The Tropical Forest house, though varying greatly in detail, was typically a frame, thatch-covered dwelling which accommodated several families and often had considerable size. The eastern Chaco horsemen, however, used portable, mat-covered dwellings. The house was furnished with wooden stools, hammocks (woven, twined, or netted of various materials, according to the locality), mosquito shelters (Montaña, *Otomao*), babracots, mortars and pestles, and storage baskets. The hammock has spread rapidly in the historic period, and the platform bed, a trait of the eastern Andes, has survived only in a few tribes bordering the Andes.

Weapons.—The bow and arrow occurred among practically all Tropical Forest tribes. The bow often had great length, and arrows were characteristically equipped with a large variety of wooden points—multiprungs, barbed rods, lanceolate blades, knobs, harpoon heads, and other forms for special purposes (this volume, p. 235 ff.)—and were frequently poisoned (*Achagua*, *Sáliva*, *Patángoro*, *Lucalia*, *Guiana Carib*, eastern Bolivia). Stone points, though typical of the Marginal hunting tribes of the south, were not found, owing mainly to the scarcity or lack of stone in the forests.

The spear thrower survived in a few areas—Montaña, Juruá-Purús *Arawak* (pp. 244–247)—but was evidently being replaced by the bow, a process that continued into the historic period (Handbook, 3: 526).

The blowgun, an early Andean trait and possibly of trans-Pacific origin, was limited to the western Amazon and extended across the northern Andes to Central America (pp. 248–252; map 4). It con-

tinued to spread during the historic period, a condition of its diffusion probably being increased intertribal contacts and trade of poison without which it is worthless. Its value lay in providing a silent, highly effective means of hunting arboreal game. There seem to be few records of its use in warfare, even by tribes which did not hesitate to kill their enemies with poisoned arrows.

Shields occur among some Tropical Forest tribes (p. 259 f.; map 5).

Bark cloth.—Bark cloth, a trait of possible trans-Pacific origin, had a distribution very similar to that of the blowgun (pp. 67–68). Among many tribes which practiced loom-weaving, it was used primarily for religious masks and as an alternate for cloth. Among others, such as the *Chocó*, it was used instead of cloth.

Metallurgy.—None of the Tropical Forest tribes smelted or cast metal in pre-Columbian times, but objects of gold, silver, and sometimes copper from the Andes had reached a great number of them, many a considerable distance away; for example, the *Xaray* and *Guaraní* of the Paraná and Paraguay Rivers and the tribes of the middle Amazon.

Rubber.—Many but not all the Tropical Forests shared a knowledge of rubber working with Central America and México (p. 227). Despite its present-day importance, rubber was of minor value to the natives, being used only to make balls, syringes, rings, and figurines.

Musical instruments.—The Tropical Forests had a considerable number of musical instruments, some of special interest. Gourd rattles, which were very widespread, were generally shamans' instruments. Trumpets tended to be associated with a secret cult, in which they represented the voice of the deities and were taboo to the uninitiated (p. 576 f.), but the *Otomac* used them in lay festivals and the *Sáliva* in funeral ceremonies. Hollow-log signal drums, possibly of trans-Pacific origin, had a distribution very similar to that of bark cloth and the blowgun (Central America, Colombia, *Witotoans*, western Guiana, the *Betoi* and their neighbors, *Achagua*, *Chocó*). The pan-pipes, perhaps also of trans-Pacific origin, were even more widespread than the hollow-log drum, and they occurred also in all periods of Andean history. Among other instruments were various flutes, jingles, and stamping tubes to mark time.

Narcotics and stimulants.—Chicha was common to all Tropical Forest tribes, who drank it in quantity on festival occasions. An *Nex* was drunk as maté in Paraguay and eastern Brazil, and another *Nex* as guayusa in the western Amazon (map 10). Tobacco was the main narcotic, being used by all tribes except some of the *Tupí* (p. 525 f.; map 9). Other narcotics had a more restricted distribution, and occurred especially in the northwest Amazon (map 15).

THE SOUTHERN ANDEAN TRIBES

The principle southern Andean tribes were the *Kunza*-speaking *Atacameño* (Handbook 2: 41, 599), the *Kakan*-speaking *Diaguita* or *Calchaquí* (Handbook 2: 39–40, 633–637), who occupied the extreme deserts, especially those west of the Andes, and the *Araucanians* (Handbook, 2: 42–44, 687), who lived in the temperate zone of winter rainfall in Central Chile.

Many features of the cultures of these tribes were of unquestioned central Andean origin; the agricultural complex, llama raising, guinea pigs, dogs, wool and cotton weaving, metallurgy, ceramic styles, slings, quipus, garment types, blood sacrifice, stone houses, stone cist graves, and mummification. The *Araucanians*, who experienced somewhat less Andean influence than the *Diaguita* or *Atacameño*, had a number of traits that linked them with the Tropical Forest tribes: frame, thatched houses, mortars, dugout canoes, cannibalism (including drinking cremated ashes with chicha), human trophies, sporting games, and urn burial. In all three tribes, however, the sociopolitical units were in the Tropical Forest rather than Andean pattern.

THE ECOLOGICAL BASIS

These tribes were intensive farmers and their basic crops—maize, quinoa, beans, and potatoes—were Andean. In the Chaco Santiaguino (Handbook, 2: 655) and La Candelaria (Handbook, 2: 661), east of the Andes, flood-plain farming was practiced, but in the extreme deserts to the west irrigation was necessary. Irrigation and farm terracing, both Central Andean traits, extended to the Quebrada de Humahuaca in northwest Argentina (Handbook, 2: 619) and to the northern *Araucanians*.

On the coast, fish were taken in some quantity by means of hooks, drugs, weirs, spears, baskets, and harpoons, but habitation sites were limited to the few sheltered coves where there was fresh water. The central coast of Chile was far more favorable for occupation than the north coast.

Transportation was well developed. In the northern part of the area, pack llamas were driven over roads and balsa canoes were used along the coast; on the south coast, dugout and planked canoes were employed.

THE SOCIOPOLITICAL AND RELIGIOUS PATTERNS

Sociopolitical patterns.—Despite this fairly efficient technological basis, the population of the Atacameño deserts was very sparse and the people lived in widely separated, fortified villages, each consisting only of a few one-family stone houses. The village apparently was made up of related families under a patrilineal chief and it was

totemic; perhaps it might be regarded as a patrilineal lineage or sib. The *Atacameño* seemingly had no warrior class, though they carried on some warfare, and there is no evidence that they accorded chiefs special privileges or special burial. They lacked a temple-priest cult. In the area of the Puna de Atacama and the Quebrada de Humahuaca, evidence of temples and idols is also absent, and the archeological remains of villages and graves are basically like those of the *Atacameño*. Warfare in these areas was carried on for individual honors, and trophy skulls were kept.

The Argentine *Diaguita*, also living in an arid area, had towns similar to but probably somewhat larger than those of the *Atacameño*. The destruction of a house at the death of an occupant, however, is evidence of some settlement instability. The *Diaguita* may have been divided into lineages, but political consolidation probably extended beyond kin groups and incipient class structure is evident. Village chiefs, whose succession was patrilineal, sometimes contracted war alliances with other chiefs. Chiefs may have received special burial; in Chile, certain especially elaborate archeological graves contain the bodies of several persons. The *Diaguita* used large armies in their wars against the Spaniards. Their warriors achieved special status and may have formed a distinct class. They took and tortured captives and made head trophies, but they were not cannibals. The *Diaguita* had no temple cult and probably no class of priests, though special persons seem to have conducted agricultural rites.

The *Araucanians* had a much more fertile habitat and a much denser population than the *Atacameño* or *Diaguita*. The communities were not larger, however; instead, the population was simply divided into a greater number of comparative small communities, which were structured on a kinship basis in the Tropical Forest pattern. Each settlement consisted of a single house, which, reputedly for fear of witchcraft, was located some distance from its neighbor and was often palisaded or fortified. It was occupied by 3 to 8 individual families—15 to 40 persons—who apparently constituted a totemic, patrilocal, and probably patrilineal lineage, i. e., what is sometimes called a localized sib. The lineage had its own chief and apparently owned its farm land.

The lineage or settlement seems to have been comparatively independent, but several related lineages sometimes constituted a larger, less cohesive unit, and the several units of a district in turn may have been united into loose tribelets under chiefs with only slight power and nominal titles.

Araucanian society, though based on kinship and essentially democratic, had incipient classes: (1) wealthy men and chiefs, who, however, had no great privileges except that of polygyny and who did not receive special burial; (2) commoners; and (3) slaves and captives,

who did not constitute an important labor class. Wealth and military exploits were the means of acquiring status. Today, with an economy based on sheep, horses, and mules as well as on cultivation, the rank is more strictly economic: the wealthy, the commoners, and the poor peons who work on the estates.

The *Araucanians*, always outstanding warriors, not only took slaves, but they captured sacrificial victims. The victims were tortured and killed for cannibalistic but not ritualistic purposes. Their flesh was cooked and eaten, their hearts sucked by headmen, and their bones burned and the ashes drunk in chicha. Sometimes their skulls were made into trophy cups, their long bones into flutes, and their facial skin into masks.

Religion and shamanism.—There is little evidence of a true temple-idol-priest complex in the southern Andes, but central Andean influence is seen in public ceremonialism that was connected with tribal gods and that was held probably according to a ritual, agricultural calendar.

The *Diaguita* held agricultural ceremonies, and the *Atacameño* worshiped the sun and had fertility rites for their fields. The most important public *Araucanian* ceremony was conducted by priests who supplicated the Supreme Being and Creator with prayers and blood sacrifice of animals to send good crops and health. The *Araucanians* also had a cult connected with a deity responsible for natural catastrophes, and they appealed to various spirits, including the sun.

There is some question whether all these public ceremonies were entirely Andean in origin, but there is little doubt that many of the ritual elements were. The latter included: llama, and later sheep, sacrifice, and extraction of the heart blood; aspersions of blood and chicha; and four sacred directions.

Shamans obtained their supernatural power from deceased shamans and from the Supreme Being, to whom they prayed. The *Araucanian* shaman, like that of the northwest of North America, was often a transvestite and used a tambourine. He prognosticated, made rain, and cured disease. Disease was believed to be caused by the intrusion of foreign objects, by soul loss, and by magic.

Disposal of the dead.—Interment of the dead was the rule, but mummification and stone-cist graves, both Andean traits, were not uncommon. Urn burial occurs archeologically in *Araucanian* territory and in the area of La Candelaria and Chaco Santiaguense.

TECHNOLOGY AND MATERIAL CULTURE

Pottery.—Pottery was made by all southern Andean tribes but that of the *Araucanians* was mostly undecorated and crude whereas the *Diaguita* wares were bichrome and the *Atacameño* were trichrome.

Basketry.—The earliest basketry in archeological sites was coiled, but twining was introduced at a later period and a twilled weave was finally adopted by the *Diaguita* and *Araucanians*.

Weaving.—Weaving was done on the true loom with llama and alpaca wool as well as cotton. The *Diaguita* also used vicuña wool. Textiles had woven-in designs in color.

Metallurgy.—Copper and some gold and silver were smelted and worked by the *Atacameño* and *Diaguita*, the latter also making bronze and doing some casting. All three metals were used in native times by the *Araucanians*, but it is doubtful whether they were mined or smelted. To the east, metallurgy did not extend to the Chaco Santiagueño.

Weapons.—Andean weapons present among these tribes were slings, lances, bronze knuckle-dusters (*Diaguita*, *Atacameño*), stone-head clubs (*Atacameño*, *Humahuaca*, *Diaguita*), bronze axes, star-head copper clubs (*Diaguita*), shields, helmets, hide-cloak armor, and spear throwers (*Araucanians*, *Diaguita*). More characteristic, however, were bows and arrows (poisoned only in the Chaco Santiagueño area) and bolas (*Atacameño*, *Diaguita* doubtfully, *Araucanian*, post-Columbian). Lassos among the *Araucanians* were post-Columbian.

Dress.—Andean-type shirts, dresses, and mantles were common, especially after the *Inca* conquest.

Musical instruments.—The musical instruments were panpipes, end-flutes, rattles, trumpets, drums, bells, and ocarinas.

Stimulants.—All these tribes probably made chicha.

THE SUB-ANDEAN AND CIRCUM-CARIBBEAN TRIBES

The fourth volume of the Handbook describes the tribes whose distinctive sociopolitical and religious patterns were called Circum-Caribbean or Sub-Andean. These tribes were distributed from the Antilles across the northern coasts and northern Andes of South America and through Central America to the *Maya* frontier in Honduras. With the data of all volumes of the Handbook available for a more detailed comparison of the cultures, it is now evident that the Circum-Caribbean tribes of Volume 4 were somewhat interspersed with Tropical Forest peoples, while several tribes, especially those of the Ecuadorian and Colombian Andes belong with the Sub-Andean rather than Central Andean cultures. It is also clear that several tribes of eastern Bolivia, though included with the Tropical Forest peoples in Volume 3, really had the Sub-Andean patterns.

In the following classification, all tribes are called "Circum-Caribbean," but the designation of "Northern Andean" is used to distinguish certain tribes of Ecuador and Colombia from the other Circum-Caribbean peoples, because they had a considerable number of

specifically Central Andean culture elements (below). Classed as Circum-Caribbean are the following: The peoples of Central America south of the *Maya* frontier (Handbook, 4:26-28), of the northern Colombian Lowlands (4:329-338), of the Highlands of western Venezuela (4:469), of portions of northern Venezuela (4:22-23), and the *Arawak* of the Antilles (4:507-546). The additional name, "Sub-Andean," is applied to the *Chibcha* (Handbook, 2:887-909), the peoples of the southern Colombia Highlands (2:915-960), of Coast and Highlands tribes of Ecuador (2:785-821), and, in eastern Bolivia, the *Mojo* and *Bauré* (Handbook, 3:408-424), the *Paressí* (3:349-360), the *Manasí* (3:389-393), and the *Pawnaca* (3:396-397).

The subsistence of the Sub-Andean and Circum-Caribbean tribes differed considerably in the different areas. In general, it was based on fairly intensive farming, but on the sea coasts and in the Antilles, ocean resources were added to agricultural produce. The population was fairly dense, and villages were generally large and permanent.

The Circum-Caribbean and Sub-Andean village resembled some of the Tropical Forest communities in consisting of several houses, each with an extended family or lineage, but the bonds between kin groups were reinforced by stronger political controls and society was characteristically divided into somewhat fluid classes. A group of hereditary chiefs constituted the upper class. The nobility seems to have developed from a class of warriors who attained rank through the display of human trophies and the capture of women for wives and men for cannibalistic feasts and for occasional religious sacrifice. Tribal gods were represented by idols kept in special temples, and they were served by priests. Few tribes had a special hereditary class of priests, however; instead, shamans performed priestly functions, thus acquiring social prestige and political power. The temple rites tended to be oracular seances for the benefit of individuals rather than communal ceremonies for the public good.

Politically, many tribes had achieved multicomunity states, both through federation and conquest, but they never succeeded in consolidating them against revolt in a stable empire like that of the *Inca*. Political consolidation and economic exploitation of conquered masses of people had not advanced sufficiently to supersede a slave class made up of captured individuals.

Technologies and material culture, like those of the Tropical Forests, included pottery, loom weaving, domesticated cotton, woven basketry, dugout canoes, frame and thatch houses, palisaded villages, hammocks, and various other items adapted to tropical or subtropical regions. They also resembled those of the Andes in the use of metals, salt, the litter, guinea pigs, and other elements which were more numerous in Colombia and Ecuador than in Central America or the Antilles. (See

below.) Material goods were produced in sufficient quantity to allow local specialization and trade, especially in such items as metals, salt, cloth, pearls, and shell beads.

The Sub-Andean or Northern Andean tribes of Colombia differed from the Circum-Caribbean peoples of Venezuela and the Antilles in possessing such characteristically Andean culture traits as copper smelting and alloying of copper with gold; construction of roads, bridges, hilltop forts, and stone houses; working of salt; balance scales; quipus; maces; gold pincers; coca; war banners; the practice of brother-sister marriage; mountain worship; and wrapping of funeral bundles. Some of these features extend through Central America to México.

The *Chibcha* of Colombia were previously classed with the Central Andean tribes because it has always been supposed that they loomed above their neighbors in political and religious achievements. Now that the Conquest Period culture of the other Colombian tribes is better known, it is clear that the *Chibcha* were not very distinctive. They had no urban centers, and their political expansion was little if any advanced beyond that of several other Northern Andean tribes. Their temple cult was patterned on oracular performances for individual ends rather than public worship according to a ritual cycle, and their hereditary priesthood, though unusual, was not unique. Their technology and material culture was not outstanding, even by Sub-Andean standards.

Ecuador was incorporated into the *Inca* Empire about 1495, some 40 years before the Spanish Conquest, but in pre-*Inca* times it was definitely Sub-Andean. The effect of the *Inca* conquest was to make the native social stratification more rigid and more hereditary, to impose strong political controls which stemmed downward from the *Inca* Emperor, to establish a number of truly urban centers, and to introduce the *Quechua* language and such material items as new crops, more extensive irrigation works, llama herding, weaving with wool, increased building in stone, cultivation of coca, and new ceramic styles.

The Sub-Andean tribes of eastern Bolivia form enclaves within more backward peoples, as if they had been compressed within migrating groups of Tropical Forest and Marginal tribes. Their material culture, moreover, is much like that of the Tropical Forests, and it lacks metallurgy entirely. Even the *Arawakan* language, which all but the *Manasí* speak, has affiliation with the Amazon, Guianas, and Antilles rather than with the Highlands, and but for their class system and temple-idol cult, they should be classed with the Tropical Forest peoples.

Because the Sub-Andean tribes of eastern Bolivia are widely sep-

arated from those of the Circum-Caribbean and Northern Andean areas, they will be described separately.

In the following pages, the culture of these tribes is projected against that of the Tropical Forests and Central Andes. The historical significance of the many features found also in México is discussed subsequently.

THE CIRCUM-CARIBBEAN AND NORTHERN ANDEAN TRIBES THE ECOLOGICAL BASIS

Intensive farming and favorable environments helped support somewhat larger and more permanent villages than in the Tropical Forests. Whereas the Tropical Forest peoples cultivated migratory slash-and-burn plantations in the rain forests or the gallery forests, the Sub-Andean tribes of the Highland farmed in more open country, and even the lowland peoples frequently inhabited and cultivated savannas. Irrigation, an essential and characteristic practice in the Peruvian and Chilean deserts, though unnecessary in the rain forests, occurred sporadically: *Cañari*, the *Chibcha* or their neighbors, the *Timoteans* (who also had storage tanks), the tribes near Barquisimeto in Venezuela, *Cumanogoto*, the *Island Arawak*, and probably other tribes, the determination of which will depend upon archeology.

Some indication of the development of horticulture in the Northern Andes of Ecuador and Colombia is afforded by the number of crops: in the high altitudes, above 8,000 feet, potatoes, quinoa, ullucos (*Ullucus tuberosus*), and cubios (*Tropaeolum tuberosum*); middle altitudes, 5-8,000 feet, maize, arracacha (*Arracacia esculenta*), quinoa (*Chenopodium quinoa*), achira (*Canna edulis*), and auyama (*Cucurbita verrucosa*); and, somewhat in the middle altitudes but more in the lowlands, sweetpotatoes, maize, yuca or sweet (but not bitter) manioc, beans, peanuts, and a considerable number of fruits including pineapples, avocados, papayas, custard apples (*Annona squamosa*), chirimoyas (*Annona cherimolia*), and capuli (*Physalis peruviana*). Additional crops were peppers (aji), cotton and/or maguey and cabuya grown for fibers, altramuces (*Lupinus* sp.), and various grasses.

Most of these crops also occurred to the north, throughout Central America, but the number dropped off sharply in Venezuela, where bitter manioc appeared. Bitter manioc apparently spread in post-Colombian times to the Antillean *Arawak* and to Central America. The Antillean *Arawak* grew only potatoes, peanuts, beans, arrowroot, and a primitive variety of maize.

An indication of the relative importance of farming is that it was mainly woman's chore in the Tropical Forests but among many Circum-Caribbean and Sub-Andean tribes, as in the Central Andes, men also were cultivators.

In the lowland and coastal areas of both the continent and the Antilles, however, sea and river resources compensated for the smaller number of crops, and the population density was extremely great.

Llamas became important in Ecuador only after the *Inca* conquest and they never reached Colombia, but the guinea pig extended throughout the Northern Andes, probably to most of Colombia. The Muscovy duck and perhaps the dog were found in all Northern Andean and Circum-Caribbean tribes. The chroniclers mention a curious "mute dog" among the *Aburrá* of Columbia and the Antillean *Arawak*, but it may have been a tame bush dog (*Iticcyon venaticus*) or a fox. The domesticated turkey may have been kept by some Central American and Venezuelan tribes. The Antillean *Arawak* and perhaps the Columbian tribes had "domesticated" parrots. (See Gilmore, Handbook, vol. 6.)

SOCIOPOLITICAL AND RELIGIOUS PATTERNS

Settlement pattern and composition.—The population density was very great, being exceeded only by that of the Central Andes, but the settlement pattern, though imperfectly known, seems to have varied. Among the Highland tribes of Ecuador and southern Colombia, the dwellings—one-family pole-and-mud houses (Ecuador and probably *Chibcha*) or frame houses (*Páez*, *Móguex*, *Pijao*)—were somewhat scattered, and their occupants affiliated with one or another religious or political center. Compact, or nucleated, towns were exceptional. For defense, people retreated to hilltop forts. Examples of these dispersed settlements are the *Cara*, who had but few large towns; the *Puruhá*, whose villages consisted of only 10 to 12 houses; the *Palta*, who lived in isolated houses but assembled on occasion at a ceremonial center or at a chief's house; and the *Páez*, whose village consisted of several polygynous families living in houses clustered around the residence of the village chief.

In the lowland area and in the north Colombian and Venezuelan Highlands, the pattern was one of tightly nucleated towns or villages. The houses were carefully arranged around a central plaza which contained the chief's and nobles' houses or palaces and the temple, and the village was surrounded by a palisade (western Colombia, *Chibcha*, *Timoteans*, *Corgabo*, *Lache*, northern Venezuela, Panamá and *Talamanca* Division). Data on village size are few: the *Lache* village had up to 4,000 persons; the *Aruacay*, 1,000; the Antillean *Arawak*, 3,000; tribes of Panamá, 1,500; and tribes of the Caribbean Lowlands in Central America, 100 to 500.

A feature of the Circum-Caribbean village that relates it to the Tropical Forest village both in settlement pattern and social structure is the occurrence, in some cases at least and perhaps quite commonly, of the large, communal frame house which sheltered an ex-

tended family or lineage. The main difference is that the Tropical Forest community often consisted of only one such house, whereas the Circum-Caribbean village had many, so that there was no village exogamy and an individual's social and political affiliations and obligations extended beyond his kin to the many fellow villagers who were unrelated to him. Where the bilateral or conjugal family was the basis of society, as was evidently the case in Ecuador, sociopolitical affiliations also extended to nonkin.

Among most of the Circum-Caribbean tribes, the lineage seems to have been matrilineal and there are some cases of matrilineal clans, but evidence bearing directly on this point is scant. The Talamanca Division and the Antillean *Arawak* seem clearly to have had matrilineal extended families or lineages, and the *Bribri* and *Guaymí* had matrilineal clans. The *Mosquito*, the Southern and Western Colombian Sub-Andean tribes, and the *Chibcha* probably had matrilineal descent of chieftainship. The *Fincenú* even had female chiefs. Female warriors were also commonly reported in this area. A further suggestion of matrilineal descent in Colombia is the common occurrence of avuncular marriage, but this, like the brother-sister marriage practiced by *Inca* Emperors and by some Colombian chiefs, including the *Chibcha*, may have been merely a form of family endogamy designed to keep descent within the family. Kirchhoff believes that northern Perú also tended to be matrilineal (this volume, p. 295), though the Central Andes seem to have been very definitely patrilineal.

In the social classes, the system of warfare, and the priest-temple cult, these tribes found a new set of factors for governing interpersonal relations. In contrast to the Tropical Forests, the kinship basis was thrust into the background; status came to pertain to class membership rather than to age, and political control fell to the village chief instead of the kinship head. Individualism was greatly curtailed, special associations gave way to warrior groups (the only recorded men's secret society is that of the *Guaymí*), and shamans became functionaries of the temple cult as well as private practitioners.

The political pattern.—The Northern Andean and Circum-Caribbean political unit was a group of houses or lineages, and its head was a chief rather than family elder. The office usually was quite strictly hereditary, though in some tribes, for example in Ecuador, a chief was theoretically supposed to have exceptional qualities of leadership, and in some tribes (*Talamanca* Division), chiefs, especially war chiefs, were elected.³

³ A problem that would repay careful comparative study is the political power and social status of the priests or shamans, the hereditary secular chiefs, and the war leaders. The place of all three in the sociopolitical hierarchy would throw much light on the origin of the social classes.

The village usually constituted the state, but there was considerable tendency to tribal federation and even to the formation of empires or kingdoms. Though political consolidation of groups of villages fell far short of that of the Central Andes, the *Cara* are alleged to have dominated most of Ecuador in pre-*Inca* times. Apart from the *Cara* realm, all the *Puruhá* were under one chief and the *Cañari* were federated under a hierarchy of chiefs. On the coast, the seven *Puná* divisions had an over-all chief, but each *Esmeralda* town and probably each *Manta* town was independent. In Colombia, federations of sub-tribes to form the tribal political units had made some progress among the *Páez*, *Lile*, *Ancerma*, *Catio* and other Highland tribes, and among the *Quimbaya*, *Tolú*, *Cenú*, and *Mompox* of the North Lowlands. There were also incipient empires that were economic if not political. They were established by conquest and tribute was exacted from the conquered peoples. Among the *Chibcha* and their neighbors, at least two important realms were established (Handbook, 2: 887-897). Northern Venezuela is inadequately known, but considerable inter-village solidarity seems indicated for the *Timoteans*, *Caquetío*, *Cumanagoto*, and *Aruacay*. The *Arawak* of the *Antilles* and the *Guaymí*, the *Talamanca* Division, and the *Güetar* of Central America had also achieved loose states.

Social classes.—The class structure really stemmed downward from the chiefs, nobles, or lords, who constituted the most clearly defined class. Usually, several chiefs formed a privileged and more or less hereditary aristocracy of considerable power. Among the Tropical Forest tribes, the main privilege of the chief was that of practicing polygyny. The Northern Andean and Circum-Caribbean tribes accorded chiefs this privilege in greater degree, and the *Puná* and the peoples of the Unaré River in Venezuela are alleged to have had eunuchs to guard the chief's harem. A chief's wives or concubines were so bound to him that they were often interred with him at death. A chief or lord was also distinguished from commoners by occupying the largest and most central house of the village; by having his land cultivated and many other services performed by common folk; by bearing a special title; by wearing distinctive garments, insignia, and ornaments of precious metals, gems, and sometimes pearls; by sitting on a special stool, usually of carved wood; by being carried in a hammock or gold-adorned litter (*Chibcha* and other Colombia tribes, *Caquetío*, Unaré River in Venezuela, Antillean *Arawak*); by receiving obeisance from his subjects; by marrying his sister (*Chibcha*, *Carrapa*, *Picara*, *Paucura*) or his sister's daughters; and by elaborate disposal of his body at death. As many of the special funerary practices can be inferred from archeological data, they afford an excellent clue to the presence of a class system among prehistoric peoples.

Among the Northern Andean and Circum-Caribbean people, the chief's body was: Put in a tomb, seated on a stool (*Puruhá*) and accompanied by his wife (*Panzaleo*); buried with his wives (*Cañari*); buried with wives and slaves (*Chibcha*); buried with wives in a deep-shaft grave (*Pasto*, *Quillacinga*, *Caramanta*, and perhaps several Ecuadorian tribes); desiccated or cremated and buried (*Quimbaya*); mummified and kept or else buried (*Popayán*, *Güetar*); buried in a mound-enclosed vault (North Colombia Lowlands; Cauca-Atrato region); desiccated, placed in a hammock, later cremated, and the ashes drunk (*Caquetío*); desiccated and hung in a house (*Chiribichi*); desiccated, buried, and later roasted or reburied (*Piritú*); disemboweled, desiccated, and kept as an idol or buried with several wives (Antillean *Arawak*); desiccated and kept in a house or buried with wives and retainers (Panamá); or sewed in a mat and buried with retainers (*Mosquito*). Mound burial was probably more common than reported, but it seems not to have been characteristically Circum-Caribbean, and it was not a typical Andean trait. The custom of cremating the ashes and drinking them with chicha occurs among a few Venezuela tribes, but is more characteristic of the Tropical Forests, especially of the western Amazon.

Colombian tribes have been accredited with matrilineal succession of chieftainship, and in view of their strong matrilineal tendency (see above), this is perhaps expectable. The evidence for matrilineal succession is usually that a chief was succeeded by his sister's son, but where brother-sister marriage was practiced, the sister's son was also the chief's son. A chief's marriage with his sister's daughter would also give the superficial appearance of matrilineal succession. True matrilineal succession, therefore, can only be assured where, as in the case of the *Chibcha*, it is specifically stated that the title passed to a nephew rather than to a son.

The aristocratic or noble class was ill-defined and less fixed by heredity than that of the chiefs or lords. It seemingly consisted of lesser chiefs—certainly the chiefs of villages within a federation and perhaps the headmen of lineages within a village—and probably of famed warriors and wealthy persons who achieved their status. Shaman-priests are probably also to be counted in this social stratum. The nobles were neither so fixed by heredity, so absolute in power, nor so neatly subdivided as in the *Inca* Empire. Their privileges resembled those of the overlords in kind—a *Chibcha* noble might have 100 wives—but not in degree.

The commoners must often have been the kin of chiefs and nobles, but they lacked privileges, were servile before their betters, and had to supply cloth and other produce and labor to the upper classes. They probably had some potential upward mobility, however, and

could improve their lot through achieving war fame and capturing slaves.

The slaves were war captives, but little is known of their true place in society. Whether they formed an important group of drudge laborers, helping their captors to rise economically, is an open problem. It appears that male captives added to their captor's prestige mainly through affording them victims for sacrificial rites, cannibalistic feasts, and human trophies, and that female captives augmented their masters' polygynous households. Women may also have performed useful economic tasks, but farming in these cultures was the work of both sexes.

Warfare.—Circum-Caribbean and Northern Andean warfare was of vital importance to the sociopolitical patterns. It was highly developed, the armies numbering in the thousands and even including women. A few tribes, notably the *Chibcha*, conquered neighboring peoples and exacted tribute, but probably none wholly abandoned the pattern whereby individual warriors gained fame through the capture of slaves and sacrificial victims. The armies were not wholly subservient to the state, as in the Central Andes, where alien masses were incorporated into the empire and became commoners and soldiers.

Warfare afforded an individual four means of attaining rank: (1) The honor of taking war captives for cannibalistic feasts; (2) augmentation of his polygynous household with captive women; (3) the use of titles and the display of insignia and human trophies; and (4) probably the acquisition of some drudge slaves, male and female, who increased his wealth. A *Cuna* warrior received a title for killing 20 victims; *Mosquito* and *Sumo* warriors gained rank and insignia; among the *Bribri*, *Cabecár*, and *Terraba*, they were accorded special burial; and among the *Caracas*, they had a graded military class, with special insignia.

Cannibalism was strongly developed in Colombia (South Colombia Highlands, except the *Páez*; tribes east of the Cauca and of the Cauca-Atrato region), and in Venezuela (*Cumanagoto*, *Marcapana*, *Palenque*). Among the *Caramanta* and tribes east of the Cauca River, it amounted to a definite appetite for human flesh. The *Piritú*, though not cannibals, drank powdered human hearts in chicha. The *Chibcha* and certain Central American tribes sacrificed victims for ritual rather than for cannibalistic purposes, except that the Meso-American tribes were strongly cannibalistic, the *Sumo* were probably cannibals, and *Lenca* warriors ate their victims' hearts. The Antillean *Arawak* did not eat human flesh, though the *Island* and *Mainland Carib*, both of them Tropical Forest peoples, were fierce cannibals.

Correlated with cannibalism was the display of human trophies:

skulls (Panamá, South Colombia Highlands, *Chibcha*, Western Colombia, tribes east of the Cauca); the flayed skin stuffed with ashes or straw, and the hands, feet, and other parts of the body (South Colombia Highlands, Western Colombia, tribes east of the Cauca, *Manta*, doubtfully the *Corbago*). Except for the *Piritú*, who made flutes of human bone, a Tropical Forest trait, the *Timoteans*, the tribes of northern Venezuela, and the Antillean peoples used no trophies. In Central America, the *Sumo* used human teeth.

Warfare was an adjunct to religion in that it supplied victims for sacrificial rites among the tribes east of the Cauca River, the Cauca-Atrato area, the *Chibcha*, the *Cuna*, the *Güetar* and perhaps others of the *Talamanca* Division, and some of the Meso-American tribes. (See also below.)

A little-understood phenomenon is the occurrence of female warriors among many of the Colombian tribes, the St. Croix Island *Arawak*, and probably the *Cuna*. It seemingly had some connection with transvestitism. Male homosexuality, on the other hand, was common on the coast of Ecuador.

RELIGION AND SHAMANISM

The priest-temple-idol cult of the Central Andean peoples centered around the public worship of tribal gods, who were represented by idols kept in temples, and the worship was directed by a special class of priests. Few Circum-Caribbean and Northern Andean tribes had precisely this pattern. Some tribes lacked almost all features of it, whereas others had a cult that was transitional between the Tropical Forest and Central Andean patterns. The priest-temple-idol cult seems to have been entirely absent from Northern Venezuela and attenuated among the *Island Arawak*. In Central America, some tribes may never have had it, but post-Contact deculturation probably eradicated it among many others, so that its aboriginal presence can only be ascertained by further perusal of the early chroniclers. Certain archeological remains that appear to have been ceremonial centers and many fragmentary beliefs and customs that survive among the modern tribes strongly suggest that a temple-idol cult was formerly present.

In the Circum-Caribbean and Northern Andean tribes, the main religious functionary was the shaman. Like the Tropical Forest shaman, he cured disease, divined, and practiced magic with the aid of his personal spirit helper, but he also assumed the duties of intermediating between the community and its gods. That the same person filled the dual roles of priest and shaman seems clearly evident for many tribes (*Evégico*, *Panzaleo*, *Pijao*, North Colombia Lowlands, Northern and Northwestern Venezuela, and probably *Popayán* and *Ancerma*). In these tribes, membership in the priesthood was not

hereditary but depended upon the supernatural experience by which an individual obtained shamanistic power. Through his community services the shaman attained considerable status, and often he had political power equal to that of the chief or he was the chief (Northern Venezuela, *Caquetio*, *Island Arawak*). Thus, priests, like warriors, had a potential upward mobility, and they were not fixed in the social structure by heredity, as in the Central Andes. A few tribes, however, apparently had a special, hereditary class of priests (*Chibcha*, Meso-American tribes, *Bribri* and *Güetar*).

The religious rites were largely private, oracular séances and served the needs of individuals rather than of the community. Public ceremonials were not unknown—the *Chibcha* had a harvest and New Year's rite and an annual pilgrimage—but there was no fixed ritual cycle, associated either with a calendrical system or with agricultural phenomena. The oracular performances resembled the public séances held by Tropical Forest shamans but differed from them in that they dealt with tribal gods rather than with the shaman's spirit helper. The distinction, however, was not always clear-cut: shamanistic, fetish, and guardian-spirit concepts are interwoven with those of tribal gods, and spirits pertained to groups of different magnitudes: tribe, community, family, and individual. The *zemis* of the *Island Arawak*, for example, were household fetishes of different kinds, some of them derived from dream experiences; only the major *zemis*, which were kept in the chief's house and were worshiped in communal rites under the chief's direction, can be regarded as idols in a temple cult. Many tribes had household idols, which, though frequently worshiped with supplication and sacrifice, were family, not community deities.

There was great variation in the nature of the community deities, and they must represent the results of several streams of historical influence combined with diverse local patterning. Most common were celestial beings, especially the sun and moon (*Manta*, *Cañari*, *Chibcha*, *Coconuco*, *Carrapa*, *Catio*, northwestern and northern Venezuela, Caribbean Lowlands, *Lenca*). In the Tropical Forests, celestial deities frequently were myth characters and often creators and High Gods, but they were rarely the objects of community worship. The jaguar, another widespread South American supernatural being, attained cult status in some tribes (*Nutibara*, *Fincenú*). The worship of and myth of emergence from sacred places, especially mountains (*Cara*, *Cañari*, *Puruhá*, *Timoteans*), springs (*Manta*), a cave (Antillean *Arawak*), and rocks and trees (*Manta*, *Cañari*) are probably Andean-derived. The *Lenca* pilgrimages to sacred hills may be a vestige of the same idea, which extends to México. The *Chibcha* had shrines to lakes, rivers, mountains, and caves. The nature of many

deities, however, is unknown, though the idols representing them are described in detail.

Idols took many forms, such as clay (*Puruhá*), pottery, wood, stone, and thread (*Timoteans*), various materials including three-cornered stones (*Island Arawak*), and most elaborate of all, painted and gold-sheathed images (*Fincenú*, *Ancerma*).

Ancestor worship was probably very general, though the function of the ancestor of the common family is not reported. Deceased nobles and lords often became tribal deities. In the case of the *Chibcha* and Antillean *Arawak*, it is fairly clear that they were worshiped, and in many other tribes it is probable that their remains were mummified, as in the Central Andes, so as to preserve them as community or tribal fetishes or gods.

In the Tropical Forests, the men's club was usually the scene of religious activities, such as initiation rites, séances, and other ceremonies, and it, or a similar, early inter-American men's house, may have been the prototype of the temple. The Northern Andean and Circum-Caribbean peoples seemingly had no men's clubhouses, and, though minor rites might be practiced in family dwellings or before various lesser shrines and major ones sometimes in the chief's house, as among the Antillean *Arawak*, many tribes had a special structure in which the idols were kept and the priest-shaman conducted the oracular sessions and other rites. Offerings were made to idols only in family dwellings among the *Páez*, *Moguez*, *Caramanta*, *Pozo*, and *Arma*, and on the upper Cauca and in parts of Central America. There was a special temple among the *Pasto*, *Timoteans*, *Chibcha*, the Meso-American tribes, and the tribes of northwestern Venezuela and western Colombia. In Colombia and among the Meso-American groups, the temples were sometimes placed on mounds, but they seem rarely, if ever, to have been built of stone, as in the Central Andes. *Chibcha* temples, like houses and palaces, were of poles and mud. Among the tribes of northern Venezuela, the Antillean *Arawak* and probably the *Timoteans*, shamans performed oracular functions in a cave.

Human sacrifice was practiced by many though not all of these tribes—the *Timoteans* and *Island Arawak* had none. It was closely associated with warfare and, in some cases, with cannibalism. The victims seem everywhere to have been war captives, except among the *Manta* and the *Chibcha*, who also sacrificed fellow tribesmen. The purpose of sacrifice was extremely varied; to honor idols and sacred mountains (*Puruhá*); to insure war success (*Pozo*); to effect cures, to insure fertility by offering blood to fields, and to provide hearts to be worshiped as gods (*Huancavilca*); to appease the Sun, especially during droughts, by tying-up the victim, killing him with darts, and cutting his heart out (*Chibcha*, who also sacrificed animals); to obtain

ghosts to guard family idols (*Pijao*); to control the weather by offerings of hearts (*Caramanta*); to supplicate the sun for rain with the blood of young girls (northwestern Venezuela); to appease temple idols, the skins of the victims including animals being filled with straw and ashes (*Manta*); to honor burial feasts and each moon (*Güetar*); and to accomplish unspecified purposes (*Arma* and *Quimbaya*, who performed the rite on special platforms; *Picara*; *Ancerma*; and Meso-American tribes).

Ritual elements link the tribes with various areas. Taking salt (Ecuador) and coca (Ecuador and Colombia) are Andean; burning copal and the steam bath, are Mayan and Mexican; flagellation (*Puruhá*) is Tropical Forest; fasting (*Páez*), hair cutting (*Puruhá*, at girl's puberty; *Panzaleo*, as punishment; *Puná*, as a men's style), and use of tobacco are widespread.

The strictly shamanistic practices of the Northern Andean and Circum-Caribbean tribes are in the general South American pattern, representing beliefs that had considerable antiquity and persisted with great stability regardless of the cultural context they entered. Under the influence of a stimulant or narcotic (tobacco, *Datura*, coca in the Andes, or other drugs according to local species), the shaman used his supernatural power to cure disease by sucking out the object causing illness, and he divined, prognosticated, and worked various kinds of magic. There was also considerable herbal curing.

TECHNOLOGY AND MATERIAL CULTURE

The basic technologies in handling wood, stone, ceramics, fibers, and metals were like those of the Central Andes, but the manipulative skills were definitely inferior. Despite the quantities of wood available, its stylistic handling was stilted and no vigorous wood carver's art is reported. Stone was little used. In ceramics, where unlimited scope in molding, modeling, and painting would have been possible, the art was confined to narrow plastic traditions and to restricted painting. Few woven products have been preserved, but the textile art seems to have fallen far short of the craft potentialities realized in Perú, the products having a limited range of techniques and designs being painted on rather than woven in. Metallurgy achieved smelting and alloying of gold and copper, but smelting and alloying of other metals and probably many of the Central Andes metallurgical processes were not employed.

Metals and gems.—Copper and gold were smelted and alloyed among the *Pasto*, *Páez*, *Mogues*, *Pijao*, and the tribes of the upper Cauca, east of the Cauca, and in Panamá, but copper occurs only rarely again in Central America until the *Maya* frontier is reached. In Ecuador, the *Cañari* had gold, gilded copper, and silver, but only

gold and silver are ascribed to the *Esmeralda* and *Manta*. Metal objects, including gold-copper alloys, were traded as far as the Antilles, but smelting seems not to have been practiced by the *Chibcha*, the *Timoteans*, and the other tribes of northwestern and northern Venezuela, perhaps for environmental reasons, though some of them may, like the *Island Arawak*, have worked placer gold.

Several precious stones including emeralds were used from Ecuador to Panamá. Pearls were probably used by most Coastal tribes, including the *Manta*, *Cuna*, and the peoples of northwestern and northern Venezuela.

Ceramics.—Pottery reached a high development, mainly in the modeled, incised, and appliqué tradition, wares of this type being found throughout the Sub-Andean and Circum-Caribbean area. Three-color polychrome is mainly Central America and is probably of Meso-American origin.

Weaving.—True loom weaving characterized all these tribes, except perhaps a few in Central America and the Antillean *Arawak*, who netted their fabrics. All fabrics were of cotton and other vegetable fibers. Use of llama wool, except that obtained in trade, was post-*Inca* in Ecuador, and use of sheep wool was post-Columbian everywhere. The cotton was generally cultivated, but the *Talamanca* Division and the tribes of northern Venezuela and perhaps the *Island Arawak* used a wild species. Ornamentation of cloth was usually with painted rather than woven-in designs.

Basketry.—Basketry appears to have been general, and, to judge by a very few descriptions of the weave, was of the Tropical Forest twilled and woven varieties.

Bark cloth.—Though also a western Amazon trait, bark cloth was made in Ecuador (*Pasto*), western Colombia, and Central America.

Transportation.—Dugout canoes were used by all coastal and river tribes. The Antillean and Panamanian peoples built huge, highly ornamented craft. The *Manta* used large, sea-going balsas, as well as dugouts.

The carrying basket is ascribed to some tribes, but the netted carrying bag appears to have been more common (*Páez*, Central America, Northwestern Venezuela, *Island Arawak*).

Gourds.—The manufacture of decorated gourd containers was a characteristic feature, and in Colombia (*Pasto*, *Moguel*, *Páez*) the gourds were ornamented with a special "*Pasto* varnish."

Household furniture.—Like the Tropical Forest peoples, these tribes used hammocks and wooden stools. The platform bed is mainly Andean, but occurs sporadically: *Chibcha*, the Central American peoples, and the *Ancerma*. Among the *Island Arawak*, commoners used platform beds while chiefs used hammocks.

Stone metates and stone stools occurred especially in the Andes and Central America.

Weapons.—The bow was used by a few tribes in western Colombia (*Aburrá*, North Colombia Lowlands), the *Timoteans*, some peoples of northwest Venezuela, and most of the Central American tribes. As elsewhere in South America, it seems to have spread at the expense of the spear and spear thrower, with which it had a negative correlation. The spear thrower remained typically Highland in this area; it instead of the bow was found in Ecuador, southern Colombia, among the *Chibcha*, and in parts of western Colombia. It also survived among the Antillean *Arawak* and some peoples of the Venezuelan Andes.

The sling, another typically Andean weapon, was widespread in Ecuador and Colombia, and also survived among the *Aburrá* and on the Island of Trinidad. Stones were also thrown by hand among the *Pasto* and in the Antilles.

Clubs, especially of the flat or macana type, were general. Shields were used in southern Colombia and the Venezuelan Andes (map 5). Axes seem to have had a wide distribution, but they were generally of stone, being made of copper only in Ecuador.

The blowgun was common to the Sub-Andean and Central American tribes, but the latter shot pellets; the poisoned blowgun dart only reached the *Cuna* during historic times.

Esthetic and recreational features.—A variety of musical instruments are recorded, the most characteristic and widespread being shell trumpets, flutes, drums, and rattles.

Of stimulants and narcotics, chicha was probably general. Tobacco had a wide distribution (map 9), but it and other narcotics were consumed principally by shamans. Coca, too, was widespread, though its occurrence in Central America and the Antilles is questionable. Natively, it had ritual and shamanistic rather than popular use.

THE SUB-ANDEAN TRIBES OF EASTERN BOLIVIA

Considerable deculturation of much of eastern Bolivia during the historic period and inadequate data on aboriginal culture leaves in doubt just how many native tribes might be considered Sub-Andean. The *Mojo* (Handbook, 3: 408), *Bauré* (3: 409), *Manasí* (3: 388), *Paressí* (3: 349), and *Paunaca* (3: 396) were Sub-Andean from a sociological rather than material point of view. They formed enclaves among Tropical Forest and Marginal tribes, but, as the archeology of eastern Bolivia suggests considerable cultural development and some Andean linkage at approximately the Tiahuanaco Period of the Highlands, it is evident that the Sub-Andean culture had been partially swamped out and compressed by tribal movements bringing more primitive peoples into the area. All these tribes are *Arawakan*,

except the *Manasí*, and they probably moved into the area from the Tropical Forests to the north.

SOCIOPOLITICAL AND RELIGIOUS PATTERNS

Intensive cultivation of tropical root crops as well as of many domesticated plants derived from the Andes supported a stable though not very dense population. Villages, except among the *Mojo*, were large and comparatively permanent. *Mojo* villages averaged only 60 to 80 persons, though some were larger, but those of the *Paressí* had 10 to 30 large houses, each with 30 to 40 persons, a total of 300 to 1,200 persons per village. Villages were very compact and carefully planned, the chiefs' houses and ceremonial hall standing in the center. *Bauré* villages were palisaded.

Social classes.—All these tribes were socially stratified, some more than others. *Mojo* chiefs were probably hereditary, except that shamans often became chiefs. War captives, though allowed to marry into the tribe, were held in contempt and probably ranked below the commoners, but the classes were obviously ill-defined and quite fluid.

The other tribes had the characteristic Sub-Andean class pattern. Among the *Bauré* were: (1) a somewhat endogamous group of chiefs, who were furnished food by commoners; (2) commoners; and (3) a servile class, presumably war prisoners. The *Paressí* classes were: (1) hereditary chiefs, who lived in special houses and were furnished food and other necessities by their subjects; (2) nobles, who were heads of independent families (probably of the communal houses made up of many captive wives) and who controlled drudge slaves; (3) war captives who were serfs or slaves. The *Manasí* had: (1) powerful hereditary chiefs, whose special houses in the village plaza were used as temples and public meeting places, who received homage from their subjects, and who were accorded special burial in a stone- or wood-lined grave; (2) shamans and a distinct class of priests; (3) a nobility made up of headmen (of households?); and (4) commoners. The chiefs of these tribes were not carried in litters, did not receive elaborate burial, and lacked many of the other evidences of status of the Andean and Circum-Caribbean area.

The extent of the political group is unclear. In all tribes, each multilineage or multifamily village had a chief, and among the *Mojo*, two or three villages constituted a subtribe under a chief. The *Paressí* waged wars of conquest, but it appears that these wars incorporated alien groups into the *Paressí* class system rather than created incipient empires or even federations.

Warfare.—There was some warfare, especially among the *Paressí*. The capture of men for slaves and women for wives enhanced indi-

vidual prestige. The cannibalism-human trophy complex of the Circum-Caribbean peoples is not reported, but a kind of substitute is found in the *Mojo* jaguar cult. Jaguar hunting was the, or one of the, main avenues to fame, and jaguar trophies were kept along with other religious objects in the hall which served as men's clubhouse and temple. Human skulls were kept, but these may not have been enemy trophies.

The priest-temple cult.—These tribes had temples—in some cases, the chief's house; in others, the men's club—dedicated to cult worship, but in some cases the cult had vestiges of a men's secret society. Shamans, who gained power through individual experiences, served as oracles and priests, consulting and making offerings to the gods. Apparently only the *Bauré* and *Manasí* had a special, hereditary class of priests, but the deities were true tribal or village gods, not mere shaman's spirit helpers.

Among the *Paressí*, the men's club was the temple, and it was dedicated to the serpent god and perhaps other deities. It contained the sacred paraphernalia, including a trumpet, which represented the voice of the gods and was taboo to women—a feature of the Tropical Forest men's secret society. The *Mojo* also used their men's drinking hall, a special structure, as a temple. Taboo to women, it was the shrine of jaguar trophies and perhaps of sacred musical instruments. Here, special shamans consulted jaguar spirits and made offerings to the gods, among them certain celestial and nature deities. Both men and women were shamans, but the women may not have served as priestesses. *Manasí* priests were initiated into their class. They alone entered the sanctuary, which was in the chief's house, to consult and make offerings to the gods, who carried them away. The gods seem to have had some association with the thunder.

TECHNOLOGY AND MATERIAL CULTURE

The technology of these tribes was preponderantly of the Tropical Forest type, and it lacked such Sub-Andean elements as metallurgy. Cultural elements shared with the Tropical Forests include cotton hammocks, wooden benches, dugout canoes, bark cloth, twilled and hexagonal basketry, good painted pottery, cotton weaving on the vertical loom, Muscovy ducks, bows and arrows, blowguns, spears, rubber-ball game, hollow-log drums, and pole-and-thatch houses. More specifically Andean, perhaps, were: trade in salt; the *Mojo* causeways, which had lateral canals that served as canoe channels (it is quite possible that these antedate the *Mojo* by a long period); balsa rafts; primitive bridges; and slings. The *Mojo* also used bull boats, bolas, spear throwers, panpipes, trumpets, and clarinets.

THE CENTRAL ANDEAN PEOPLES

The high civilization found in the Central Andes at the time of the Spanish Conquest was that of the *Inca*, and it was coterminous with the area in which the *Inca* culture largely replaced that of the local tribes. One hundred years before the Spanish Conquest, the *Inca* occupied only the immediate vicinity of Cuzco, and the remainder of Perú, Ecuador, Bolivia and North Chile was divided into more than 100 independent states, some of them small tribes, others, groups of tribes forming kingdoms. Each had its own culture (Handbook, 2: maps, 3, 4, 5, 7). The *Chimu* Kingdom of the north coast of Perú was probably an empire, but it did not compare with the final *Inca* achievement. The *Inca* ultimately subjugated all peoples from the Colombian border of Central Chile and incorporated them in a single state or empire. They deeply affected the cultures of Perú and Bolivia, especially the sociopolitical patterns transcending the communities, but they only superficially influenced Ecuador and North Chile. The *Uru*, an insignificant enclave in the Río Desaguadero-Lake Poopo region of Bolivia (Handbook, 2: 575), remained quite primitive. The *Inca* culture, in other words, was imposed on the peoples of the continuous mountain mass of the Central Andes, while beyond, to the north and south, where the Cordillera begins to fork and breaks into smaller blocks, considerable cultural diversity persisted.

The Central Andean culture brought into sharp focus the patterns foreshadowed among the Sub-Andean and Circum Caribbean tribes. Intensive farming and herding supported the most dense population and the largest and most permanent communities in South America, and it yielded a surplus which released large portions of the population for manufacturing and for governmental, religious, and other non-survival assignments. Social classes became hereditary, endogamous castes, but the nobility rested on a large base of wealth-producing commoners rather than on a slave class. Warfare was an implement of imperialism instead of a source of slaves and ritual victims, as among the Sub-Andean and Circum-Caribbean peoples, and, though commoners could improve their condition through war exploits or by becoming special artisans, they could not rise to a higher caste. Religion was organized around a state temple cult and a hierarchy of state and local gods, and it was served by a graded priesthood, the priests being clearly differentiated from shamans, who were curers and magicians. The whole society was highly regimented through a code of laws and regulations which were enforced through a judicial system, and little was left to tribal custom with its less overt sanctions.

In agriculture, herding, road building, megalithic construction,

metallurgy, weaving, and in achievements in government, conquest, and class organization, the *Inca* surpassed anything in the New World. In intellectual attainments, however, and in esthetic achievements, except in pottery and textiles, they fell far short of the *Maya*. They had no system for writing words or sounds, and they recorded numbers only with knotted strings, or quipus. Their graded hierarchy of deities lacked the refinements of the Mexican pantheon, with its esthetic concomitants in ritual, architecture, and religious art. Their ceremonial calendar, based on a simple adjusted lunar count and ignoring even solstices and equinoxes, did not compare with the carefully calculated day counts and other means by which the *Maya* reckoned time.

The main difference between the *Inca* and the *Maya* is that the former intensified warfare and conquest to create an empire dominated by political and social organization, whereas the *Maya*, a comparatively peaceful people, developed their temple cult so that an esoteric priesthood became the leading class and life revolved around ceremonial centers. Highly developed religious art, architecture, and ceremonialism and a refined calendar marked the peak of the *Maya* achievement. Probably both trends were implicit in the Circum-Caribbean culture.

THE ECOLOGICAL BASIS

Andean farming was the most intensive in the Hemisphere. It employed vast irrigation works, terracing on steep hillsides, and use of fertilizers. More than 30 different species (Handbook, 2:5; 6: Sauer) and innumerable varieties of plants were cultivated; of potatoes alone there were scores of varieties. Llamas, alpacas, guinea pigs, and ducks were tended. Local specialization according to the environmental zones of the Andes (about one-third of the crops were Highland specialties) coupled with considerable exchange of produce gave a varied as well as abundant food supply. Wild foods were of minor importance, except for fish on the coast and fish and birds in the Lake Titicaca-Río Desaguadero region.

The efficiency of Andean agriculture is evidenced by the large population, which was the most dense in South America, and by the surplus production. The latter was insurance against want and released people for other work. Although the land was worked by commoners, only one-third of the produce went for their keep, the other two-thirds going respectively to the state and church, whose members did not engage directly in material production. Moreover, vast numbers of commoners were taken from their fields for military service, special craft production, household service, and labor on roads, bridges, buildings, and other public works. It is said that 30,000 persons were used

in the construction of the fort at Sacsahuaman and that the emperors sometimes had difficulty in inventing projects to keep people busy.

Extensive movement of produce and goods was made possible by the use of large, seagoing balsa rafts on the coast and of pack trains of llamas traveling over an excellent system of roads in the interior (Handbook, 2:229). This efficient transportation, an important factor in the development of urban centers, surpassed anything in the remainder of South America, where goods were moved in pre-Columbian times by human carrier, or by dugout canoes, as along the Amazonian waterways and on some of the coasts. The post-Columbian horse of the Pampas was probably of great efficiency, but the population and economic surplus of this area was insufficient to support large and elaborately structured aggregates of people.

SOCIOPOLITICAL AND RELIGIOUS PATTERNS

Settlement pattern and composition.—The combination of productive and transportation efficiency in the Central Andes gave considerable latitude to the settlement pattern: people could live either dispersed over the countryside or concentrated in urban centers. Archeology has shed very little light on settlement patterns, not for want of data but because of preoccupation with other problems. Chanchan on the north coast (Handbook, 2: pl. 51) is an extraordinarily large, compact, planned urban center, covering some 11 square miles. Possibly the restriction of settlement areas on the coast to the narrow confines of irrigated valleys, which are bounded by stark deserts, predisposed valley and coastal communities to strongly nucleated types. The Highlands, dependent on rainfall more than irrigation and given over to considerable herding, allowed more dispersed settlements.

The cultural factors which operated within the ecological limits to determine the size, composition, and distribution of communities are partially evident from ethnographic data. The unit settlement of the Highlands was a compound enclosing as many as six houses of stone, stone-and-adobe, or adobe, grouped around a court. Each house sheltered a biological family and the compound apparently accommodated an extended, patrilineal family (Handbook, 2:223). In pre-*Inca* times, the compounds were irregularly clustered around ceremonial and administrative centers to form unplanned villages, which rarely included more than 300 families, or about 1,500 persons. The villages, located near but not on arable land or grazing areas, were not fortified, refuge being sought in hilltop forts.

The *Inca* set out on a deliberate program of (1) creating new administrative and ceremonial centers and (2) moving villages away from the old hilltop forts to sites nearer the fields. The new villages and cities,

planned by government architects, were ideally laid out in squares, each with one to four extended-family, unit-type compounds or enclosures. Except for capital towns, however, the new settlements were little larger than formerly. Cuzco attained great size, but a considerable portion of the 100,000 inhabitants of the valley were scattered in small villages, separated from one another by farm lands.

The ayllu, a much-discussed social and territorial unit (Handbook, 2:253, 483, 539; this vol., p. 293), seems to have been essentially a land-owning group consisting of one or more villages and having a myth of common origin for its members. The land, at least under the *Inca*, was distributed annually to the constituent families. The ayllu cannot have been a clan as defined herein, because it consisted of many lineages or extended families which were probably related only through marriage, because marriage was endogamous within it, and because kinship terms reflect a bilateral kinship structure. The ayllu was patrilineal only in that office was so inherited; group affiliation in an endogamous group is necessarily bilateral. In pre-*Inca* times, the ayllu often seems to have constituted a politically independent tribelet, which was constantly at war with its neighbors, but some tribes consisted of several ayllus. Under the *Inca*, and perhaps before, the ayllus were grouped into twos (moieties) and sometimes threes for administrative purposes (Handbook, 2:255, 262). The *Aymara* moiety, however, is now and perhaps was in Conquest time an endogamous territorial division of a community within the ayllu (Handbook, 2:541).

These population groupings of different magnitudes—community, ayllu, moiety, and state—gave the individual a set of ever-larger spheres of interpersonal relations that extended far beyond his kin. Within this enlarged framework, his behavior was governed by a complex set of behavior patterns, which were adjusted to the vertically graded positions in the hereditary class system as well as to his kin. Many of these patterns were sanctioned by law. The graded system, which patterned military and religious as well as civil institutions, is reflected among the Sub-Andean and Circum-Caribbean tribes and it was foreshadowed among some pre-*Inca* Andean peoples, especially the *Chimu*, but it developed in full strength under the *Inca*, who imposed it on all the peoples from North Perú to southern Bolivia and who somewhat modified the peoples even farther to the north and south.

Classes and government.—Before the *Inca* conquest, the Andes surely had social classes with a hereditary tendency, but status, at least in the smaller groups, could also be achieved through warfare. The expansion of the *Inca* state created an elaborate and rigidly fixed society throughout the Empire. Starting in the vicinity of Cuzco with

a sociopolitical system probably not very different from that of their neighbors, the *Inca* expanded during the century before the Spanish Conquest and subjugated tribe after tribe, incorporating them into their own scheme. The *Inca*, especially the direct descendants of the Emperor, became the dominant group and highest caste, but the chiefs and nobles of conquered peoples were also given high status in the realm. Commoners generally remained such. Unlike the Sub-Andean and Circum-Caribbean peoples, among whom the status of the nobles seems to have depended partly upon a captive slave class, the *Inca* nobility was supported by the commoners, who tilled the third of the land set aside for the state and performed other services. In other words, conquest incorporated wealth-producing commoners within the realm rather than adding a captured slave class to the bottom of society, where there was really no place for it.

(1) The state was headed by the *Inca* Emperor, who had absolute authority in all civic affairs and was considered a divinity. The later emperors were so exalted that they married only their full sisters. The Emperor lived in a splendid palace, wore special garments and ornaments, bore a title, was carried in a litter, and sat on a stool atop a throne while attendants sheltered him with a parasol and fanned him. People paid him obeisance according to elaborate rules. He had many wives and servants. At his death, the members of his retinue were made drunk and killed by strangulation. The Emperor's body was mummified and placed in a shrine. Deceased *Aymara* chiefs and nobles were placed with their wives, llamas, and other property in stone towers (chullpas).

(2) The nobility consisted of (a) the descendants of past *Inca* emperors and special persons called "*Inca* by privilege" and (b) the local curacas or chiefs. The *Inca* nobles were divided into 11 ayllus, each the patrilineal lineage of an emperor, and they had the highest rank, held the most responsible positions, and were accorded many honors comparable to those claimed by the Emperor. The curacas were made up of local chiefs and nobles, who had held power before the *Inca* conquest, and of certain provincial rulers who, though appointed originally for their ability, were given permanent and hereditary status. The nobility became an elaborately graded hierarchy, the upper levels approaching the Emperor in status and the lower standing little above the commoners. There was little possibility of upward mobility because the system became fixed, but degradation of rank could occur as generations became more remote from high-ranking ancestors. This degradation was somewhat checked, however, not only by class endogamy but by intrafamily marriage, the Emperor taking his full sister as his principal wife and the nobles their half-sisters, although both took additional wives or concubines from the commoners. These mar-

riages in violation of incest regulations may have been a fairly old Andean custom, designed to protect a hereditary chief's class, for such marriages occurred north to Colombia.

(3) The commoners formed a fairly homogeneous group. They could improve their status only through warfare, in which they captured additional wives or for which they were rewarded with property and women, thus distinguishing them from the ordinarily monogamous commoners and giving them benefits nearly equal to those of the lower nobles (this vol., p. 298).

The commoners produced their own sustenance on the third of the land allotted them and cared for their own wants, but they were not permitted to make luxury goods. The extended family lived in its compound, preferably near its ancestor's mummy, but, unlike the nobles, the commoners rarely traced their forebears beyond the grandfather. Their daily life centered in the activities of the ayllu and community (Handbook, 2: 483).

The commoners were obliged to supply produce for the *Inca* church and state from the lands set aside for them, to do "mita" labor on public works, to serve in the army, to provide domestic help for the nobles, to supply the "Chosen Women" as wives or concubines for the nobles and as nuns for the convents, and to work as special artisans for the Emperor. In the absence of an important slave class, they performed all the basic and menial labor as well as a few of the more skilled tasks.

Warfare.—In pre-*Inca* times, most Andean warfare probably resembled that of the Sub-Andean and Circum-Caribbean tribes in that its purpose was plunder, revenge, and the taking of women, slaves, sacrificial victims, and human trophies. Cannibalism had probably been abandoned, if it had ever been practiced; it is accredited to only the *Aymara*, who roasted and ate their captives and drank their blood (Handbook, 2: 548).

The objective of *Inca* warfare was to incorporate conquered territories into the Empire. For purposes of exploitation, it was more expedient to leave the populations intact and exact labor and tribute from them than to bring them into the *Inca*'s already crowded lands as a slave population. The *Inca* governmental and class system was imposed upon the provinces, and, though whole populations might be moved from one region to another to prevent rebellion and to speed acculturation as far as possible, the pre-existing class structure was retained.

The early phases of the conquest quickly elevated the *Inca* proper to the upper nobility. As the conquest progressed, however, the commoners, who served in the great armies, profited little. Their role as the main wealth producers was fixed by law and their rank was fixed

by heredity, so that their main rewards consisted only of extra wives, a small amount of property, and a few favors.

After the Empire had become consolidated and the socioeconomic system functioned to supply the nobles with all the goods and services they desired, warfare was carried on more and more for the sheer power and glory of conquest and as a religious crusade to impose the *Inca* cult on the Andean peoples.

Human war trophies, apparently a very old Andean trait, were still displayed under the *Inca*—skull cups, ash- and straw-stuffed skins, leg-bone flutes, skin drum-heads, and tooth necklaces. Presumably these were evidences of individual prowess. There was no cannibalism, and animals rather than human beings were ritually sacrificed. The few human sacrificial victims were taken from the Empire population rather than from war captives; the sacrifice of war captives seems to have been restricted to victory celebrations.

The government and society.—The *Inca* Empire introduced to the Central Andes a state culture over and above the local or folk culture, and a system of control through state law which was probably only slightly evident among the Sub-Andean and Circum-Caribbean peoples and was entirely unknown among tribes whose sociopolitical unit was the kin group. This system resulted from the need to weld together a tremendous and farflung population and to make it serve the conquerors.

Under the *Inca*, the conjugal and extended family remained, but age became a basis for assignment to government tasks and the kin head was superseded politically by a government appointed ruler. The ayllu and moiety became administrative units. Men's clubs and other social groups apparently were unknown. Birth customs probably were not greatly altered, and girls were isolated at their first puberty, but boys' adolescence was celebrated by what appears, at least in the case of the nobles, to have been an initiation into a warrior status. At puberty, all boys received a breechclout and new name; in the case of noble youths, llamas were sacrificed and the boys were whipped, made to race, and given earplugs and weapons. Games were a characteristic feature of mourning ceremonies. Local worship, with its shrines, deities, spirits, magicians, and curers, was not abolished, but the state religion was forced on all communities.

In the greater part of their behavior, the commoners were regimented by government decree. They were counted, classified, graded, and assigned to groups ranging from 10 upward in a decimal system. They had fixed obligations to serve in the army, work for the noble class, and lend their services to innumerable government projects. They learned to speak *Quechua*, the *Inca* language. They were forbidden use of the distinctive garments, insignia, objects of precious

metals, and other privileges reserved for their betters. Even their marriage was regulated, and their very lives were at the disposal of the Emperor. A large portion of their behavior was governed by the legal code, which applied different principles to the different classes.

The Empire was kept under control by large armies and garrisons, a system of couriers who carried messages and supplies over the roads, large numbers of census-takers, overseers, and the like, and a judicial system.

Religion and the temple cult.—Prior to the *Inca* conquest, local religion had been based on belief in various tribal gods and the spirits of sacred places. There were shrines and perhaps true temples, and probably there was some kind of priesthood. The conquest imposed *Inca* religion, with its temples, idols, priesthood, and system of ritual on the local groups, but did not entirely destroy the local worship, which remained in the same general pattern as that of pre-*Inca* times. It served mainly to create a hierarchy of gods, to introduce more elaborate temples and more ambitious rituals, and to establish a large, graded priesthood.

Under the *Inca*, the deities and other objects of worship were ranked in order of importance: Viracocha, who was the Creator, Supreme Being, and culture hero, and whose servants were the other gods; the Sun, ancestor of the Emperor; Thunder, the weather god; the Moon; various stars; the Earth and Sea; huacas or shrines, such as tombs, buildings, and sacred places, especially springs, stones, and mountain peaks; and amulets and images, probably including those representing individual guardian spirits. These range from national gods down through ayllu and community deities to household and personal spirits. There were also various evil spirits.

The temples housed images of the gods, religious paraphernalia, priests, and the sacred women, but the ceremonies themselves were held out-of-doors.

The priesthood, headed by a close relative of the Emperor, at times may have rivaled the civil authorities in power (Handbook, 2: 298 ff). Under the high priest were the priests in charge of the various temples, their importance varying with that of the temple, and under them were a great number of assistants.

Ceremonies were held according to a ritual calendar (Handbook, 2: 471). Most common rites were agricultural, but some honored the various gods and others were held against sickness. Special rites were performed in times of crisis. Under the leadership of the priests and their assistants, the gods and supernatural powers were propitiated with blood sacrifice, prayers, fasting, and offerings of coca, sea shells, chicha, and other things. Sacrifice was usually of llamas and guinea

pigs, but in times of crisis children were strangled and their hearts offered to the god. Similar ritual was used in a smaller degree for the lesser spirits and shrines, to which anyone could make prayers, offerings, and the like.

Personal sins were believed responsible for sickness and misfortune, and a task of the priests was to confess sinners and impose penances.

In addition to the use of shrines and idols as oracles, divination was practiced by sorcerers, who communicated with their spirit helpers, made spirits speak from fire, examined lungs, seeds, or coca, observed the movements of birds or animals, and interpreted omens and dreams.

It would seem that shamans, that is curers and diviners, were distinct from priests. Among the *Inca*, curers obtained power from a vision of a spirit helper or through rapid recovery from disease. Disease was thought to be caused in several ways: punishment by the gods for sin, sorcery, winds, evil forces, and soul loss brought on by fright. The concept that sin causes disease represents a socialization of more primitive taboo transgression theories. The state was a socioreligious structure, with a divine Emperor at its head, and social offenses became also religious, disease-causing transgressions. (See Ackerknecht, this volume, p. 633 ff.) Disease was manifest as a foreign object in the body, displaced organs, or a toxic condition. The curer sacrificed to his visionary power, divined the cause of the illness, prayed and sacrificed to the Sun, heard confessions, prescribed the penance or procedure the patient should follow, and finally sucked out the object, rubbed the body, or used other appropriate means.

Modern *Aymara* curers do not have spirit helpers; they are persons who have been struck by lightning or who have learned their art. They practice both black and white magic.

Andean ritual elements included fasting, praying, offerings, blood sacrifice, washing (*Aymara*), use of chicha and coca, the concept of sacred directions, and three as the ceremonial number.

TECHNOLOGY AND MATERIAL CULTURE

The Central Andean peoples were distinguished from the remainder of South America by the number of processes they employed in manufactures and by the quantity and quality of their production rather than by essential technologies. They are also distinguished, especially from the Marginal and Tropical Forest peoples, by the specialization of production and consumption according to the social and political system. Simple goods and buildings were used by everyone, but most of the finer constructions and craft products were destined for the government, priesthood, and nobility.

Building arts.—The finest construction of the Central Andes was devoted to public buildings—palaces, temples, warehouses, and forts built of carefully dressed and skillfully laid stones. Construction employed lintels, small corbelled arches, columns, and occasionally two and three stories. Commoners' homes were small, one- or two-room, stone or adobe houses. The stone-faced agricultural terraces, the tremendous system of irrigation canals and ditches, and the canalized rivers were all community works. So were the paved and graded roads which ran throughout the Empire, crossing chasms on suspension bridges and swamps and rivers on pontoons.

Metallurgy.—The Central Andean peoples differed from those of the Sub-Andean and Circum-Caribbean area in that they not only smelted and alloyed gold, silver, and copper but they used platinum, mercury, lead, and arsenic and made various alloys, including bronze. They also employed a larger number of metallurgical processes (*Handbook*, 2: 246–248). The art products and ornaments of precious metals were intended mainly for the nobility, but, in contrast to other American Indians, there were many metal utility objects which anyone might use, for example, knives, axes, bolas weights, clubheads, mirrors, needles, bells, crowbars, and tweezers.

Weaving.—The people of the Central Andes had three types of true loom—horizontal, vertical, and belt—and they were the only aboriginal people in the New World to use wool on an important scale. In addition to domesticated cotton, they wove with llama, alpaca, vicuña, and bat wool. Their textiles are unequalled for their fineness, number of techniques, and variety of woven-in designs. The finer ornamental products were made by special artisans for the nobility.

Ceramics and containers.—The pottery art was no less outstanding than the textile craft, for the Central Andes have yielded wares of all known American techniques—molding, modeling, and other plastic treatment, negative painting in three colors and polychromes in as many as 11 colors. The *Inca* ware, reflecting standardized mass-production, fell short artistically of earlier ceramics, but technologically it equalled any of them.

The Central Andes is also notable for its carved and lacquered wooden cups and for its painted gourds.

Basketry.—Basketry is little known; apparently the Indians devoted the greater part of their attention to ceramics. Twined, twilled, and coiled baskets are all known archeologically and may have been in use at the Conquest. Coiling is still used by the *Aymara*. Mats were also woven.

Weapons.—Central Andean weapons seem to have changed little throughout the archeological history of the area, perhaps because they

were adapted to open country. Hunting weapons included slings, bolas, nets, snares, nooses, and clubs. The bow and blowgun were absent at the Conquest. In warfare, spears and spear throwers were used for preliminary attacks, but they were superseded in the Highlands by the sling and bolas. For close fighting, star-headed clubs of stone and metal, macanas, and copper or bronze axes were wielded. Protective devices included quilted-tunic armor, slat shields, and helmets. It is difficult to see why the bow was not used. Suitable bow wood cannot be had in the Andes, but presumably it, like many other products, could have been obtained in the Montaña.

Transportation.—Effective transportation was provided on the north coast by sea-going balsa wood rafts that carried 50 men and were propelled by sails and paddles, and on the south coast by inflated seal-skin rafts. In the interior, human carriers, who used square clothes slung on their backs, and llamas, which could bear 100 pounds, transported goods over the roads.

Household furniture.—Houses were equipped not only with cooking utensils and other articles, but they had wooden stools, those of the nobles being elaborately carved. People usually slept on the floor, though the platform bed may have been used in some regions. The hammock was unknown except as an occasional substitute for the litter in which to transport nobles.

Clothing.—The basic garments for men were a breechclout, sleeveless tunic, cloak, leather sandals, and head band, and for women a long dress, a mantle pinned at the chest, sandals, head band, and head cloth. Details of dress indicated class status. Bracelets and necklaces were worn, but lip and nose ornaments seem not to have been common; earplugs were used by the nobles. There was some head deformation but it seems to have been a trait of local custom, not a mark of class; it also occurred in early times.

ESTHETIC AND RECREATIONAL ACTIVITIES

In esthetic and recreational activities it appears that many of the refinements were associated mainly with the nobility and the church. Art manifestations were principally on containers, ornaments, and textiles, which embellished the persons or buildings of the noble and priestly classes. Chicha was a libation and a ceremonial drink as well as a beverage of popular consumption, coca was used especially by the nobles, priests, and diviners, and even tobacco and *Piptadenia columbrina*, another narcotic, were probably employed exclusively for religious purposes.

Musical instruments, though perhaps used by commoners for their own enjoyment, were employed mainly for military affairs and

religious festivals. These included end-flutes; bone flutes; plug flutes; panpipes; trumpets of shell, of wood, and of gourds; tambourines; two-headed drums; metal bells; and shell rattles.

Games were little developed, except dice games, which were the occasion for betting. A ball game and various athletic sports and contests were common, and children played with a number of toys, including tops, which were whipped.

Literature in the form of ballads, created by bards serving the nobles and priests, and secular plays contributed to the entertainment of the upper classes.

THE HISTORY OF SOUTH AMERICAN CULTURES

THEORETICAL CONSIDERATIONS

In the theoretical approach to problems of American culture history there are important differences between those who postulate that the principal Indian cultures originated in the Old World and those who hold that they were indigenous, New World products.

Theories deriving Indian cultures from outside the Western Hemisphere range from sensational accounts of the wholly imaginary "lost continents" of "Mu" in the Pacific and "Atlantis" in the Atlantic to more plausible treatises claiming that basic cultures and civilizations were carried by trans-Pacific migrations from China, Indonesia, India, or the Near East. Some writers claim only that individual culture elements reached the shores of America from overseas.

The "lost continent" theories do not merit consideration. Some of the other theories have perhaps received more attention than they deserve, but interest in them is kept alive by our still very insufficient knowledge of early American developmental stages, which challengingly has been used as proof that they are to be found outside America. Among the more influential and serious writers are Elliott Smith (1929) and Perry (1926), who derive New World civilizations from Egypt; Graebner, Schmidt, and their followers, who postulate that a series of *kulturkreise*, or cultural strata, each a more advanced cultural stage, were brought from the Old to the New World; and Rivet, who has attempted to show the trans-Pacific affiliation of certain Indian races, languages, and cultures. (See Schmidt, 1913; Rivet, 1943.)

These theories, especially the *kulturkreise*, assume: (1) that cultural classification based on Old World data are valid for America, and (2) that each culture consisted of a complex of political, social, and religious institutions, economic activities, and material elements, which diffused as a whole though the cultures interblended subsequently. A corollary though unstated assumption is that the American Indian

lacked the capacity to create what are assumed to be fundamental features of culture. They pay comparatively little attention to the relative chronology of Old and New World culture history, or else they assume dates for the latter which cannot be proved, and they ignore geographical and cultural barriers to culture diffusion.

American scholars find that these theories are too deductive and *a priori*.⁴ Concerned for two generations with intensive field work, they have preferred to classify Indian cultures in terms of New World, not Old World, data. Schmidt's primitive cultural stratum of Food Gatherers, for example, is meaningless for America because, as we have seen, the Marginal peoples are highly diversified and have little in common with one another. His farming, mother-right stage, when woman were cultivators, simply does not fit the Tropical Forest peoples, who were overwhelmingly patrilineal. The assumption that herding was associated with male dominance is not valid for the Andes, for it would be a complete misunderstanding of Andean culture to suppose that llama herding was at all comparable to Old World cattle raising and that it accounted for patrilineal tendencies in the Central Andes. Similarly, any theory that classifies New World civilizations with those of the Near East ignores the vital differences in culture patterns and element content.

The New World cultures not only differed from those of Africa and Asia, but they differed from one another in complex ways. Most particularly, the elements and patterns had very different distributions and, inferentially, different histories, so that the problem is not to trace total cultures through time and space but to ascertain the history of each element, element complex, and pattern. American anthropologists, therefore, deal with each on its own merits. Approaching the problem empirically, they study the manifestation of each cultural feature in specific situations before theorizing about its broader time-space relationships, and they prefer to reconstruct local segments of history before constructing world schemes. They have been unwilling to concede that the American Indian was wholly devoid of inventiveness, and they insist on examining the data of archeology for evidence both of local development and of the relative Old and New World chronology of features shared by both hemispheres. They ask for concrete proof of how migrations could have occurred and how geographical barriers were overcome, especially in the period before ocean travel developed.

Few American anthropologists deny the possibility of transoceanic influence on New World cultures, though most of them repudiate the theories that bring total cultures from overseas. It is conceded that

⁴For further methodological analyses see Cooper (1941), Lowie (1937), and Dixon (1928).

individual elements and even groups of elements may have been imported, though absolute proof is difficult to produce in any single case. At one extreme are features such as matrilineal descent, moon worship, moieties, megalithic monuments, agriculture, or writing, which are not really culture elements or patterns but classificatory labels. To derive phenomena with such labels in America from similarly tagged Old World phenomena would be to ignore the realities of culture. At the other extreme are certain American domesticated plants, whose identity and genetic connection with Old World species can be established beyond reasonable doubt. On botanical evidence, the pre-Columbian occurrence in American and the Old World of sweet potatoes (Maori varieties), an edible root (*Pachyrhizus*), calabashes (*Lagenaria*), north Peruvian cotton (*Gossypium*), plaintains (*Musa paradisiaca normalis*), and perhaps peanuts and coconuts indicates a diffusion from one hemisphere to the other. Between these extremes are a very large number of dubious cases, which involve subjective judgments and which have entailed considerable controversy. (See Dixon, 1928.) Whether these elements are attributed to a single world origin or to multiple origins depends upon one's opinion concerning their uniqueness and man's inventive capacity. Some elements, such as thatched, frame houses, penis covers, or crutch-shaped paddles, would seem simple enough to have been invented several times. Others, such as the blowgun, bark cloth, panpipes, chewing lime with a narcotic, star-headed clubs, trepanning, the venesection bow, and the ikat weaving technique, might be argued with some plausibility to have been invented only once, but it would still be necessary to prove that they were earlier in the Old World than in the New World and to show how they could have reached America. A good many of them might have been introduced to America by individual boatloads of voyagers in the course of settlement of the Polynesian Islands during the Christian Era. It is somewhat more doubtful that voyagers reached the shores of America in the earlier milleniums, when the American civilizations were taking form.

The history of these many moot items, however, is not very important to an understanding of the development of American cultures, for, as has been shown previously, they occurred largely irrespective of the main cultural patterns, all of which developed in distinctive American traditions. The patterns found among the Marginal peoples clearly stemmed from local adaptations beginning in early post-Glacial times, before trans-Pacific navigation was conceivably possible. Those of the remaining tribes, including the civilized peoples of México and Perú, acquired distinctive American patterns at a very early time. They developed along with the cultivation of the more than 100 native American crops on which they were based, and

these crops required centuries if not milleniums of plant breeding. None of them, except the few plants mentioned above, came from the Old World, and it can hardly be conceived that the idea of agriculture lay dormant in Indians during hundreds or thousands of years while they migrated from the agricultural areas of the Eastern Hemisphere across Siberia, Alaska, and southward into temperate America. Our present ignorance of the formative beginnings of these civilizations is not proof of their Eastern Hemisphere origin; it simply means that, until recently, attention has been accorded their elaborate manifestations, not their primitive origins.

American anthropologists, though in general accord in minimizing the importance of transoceanic influences on American Indian cultures, disagree somewhat on problems of the time and place of cultural origins within America and the directions of cultural diffusion. Their principal methodological tools have been the culture-area concept, the age-area hypothesis, and archeological sequences. All the tribes in a culturally homogenous area are held to have derived their culture from the same source, which is assumed to be the place with the greatest complexity of structure or with the greatest element content and which is known as the culture center. The age-area hypothesis holds that, other things equal, the oldest cultural features are those having widest distribution. Where a distribution is interrupted, it is assumed that where the features are absent they were superseded by other features. Archeology affords verifiable culture sequences, but these are usually in terms of culture elements rather than patterns, which are more difficult to abstract from its data.

For the New World as a whole, the following history is generally accepted by American anthropologists. As the oldest and more elaborate cultures based on farming are found in México and the Central Andes, it is believed that one of these areas was the center of origin of an early, ill-defined, Formative Period culture from which the main complexes and patterns found among all farming tribes were derived. During subsequent development, both México and Perú became specialized as secondary centers. What might be called tertiary centers developed in many places, such as the Southeastern and Southwestern United States, the Circum-Caribbean area, the Southern Andes, Northwest Argentina, and the Amazon Basin. When similar culture elements and patterns are found among farmers at the northern extremity of the farming area in North America and the southern extremity in South America but not among intervening tribes, the age-area hypothesis would hold that they are survivals of what was present during earlier periods in the culture centers and throughout the intervening areas. The hunting and gathering, or Marginal, peoples located beyond the agricultural tribes in both continents are as-

sumed to represent survivals of an earlier, preagricultural period that was also once pan-American.

The difficulty with this methodology is that sociopolitical patterns are treated as if they were comparable to culture elements and as if they occurred irrespective of the cultural ecology of each area. Certain ornaments, ritual details, items of material culture, social usages, and many other elements did occur in very different cultures and environments and their antiquity may be roughly estimated by the age-area hypothesis. The history of sociopolitical patterns, however, cannot be reconstructed by any such reasoning. Many of these patterns were the result of ecological adaptations to local environments as much as of historical influences and tradition. At one extreme, the sociopolitical structure of the group seems to have grown very directly out of the interaction of subsistence habits, settlement type, settlement composition, and socioeconomic activities. At the other, where the ecology afforded potentialities for a variety of patterns, the tribes were more susceptible to historical factors and the age-area method may be used within limits to ascertain the succession of sociopolitical patterns.

Applying the age-area hypothesis to American data, it can be shown with reasonable certainty that the Marginal tribes of both North and South America had certain elements, such as flint chipping, and certain institutions, such as crisis rites and shamanism, which were the heritage of a primordial period. Some of these retained a pan-American distribution; others survived only among the Marginal tribes and were superseded elsewhere. None of the dozens of different sociopolitical patterns of the Marginal peoples, however, was the survival of a primordial pattern and, therefore, antecedent to the Tropical Forest, Sub-Andean, or Central Andean patterns. Simple patterns, such as those based on kin groups, unquestionably preceded more complex ones, such as those including both kin groups and social classes, but the specific nature of the earlier forms cannot now be reconstructed with certainty by any method. Thus, in terms of sociopolitical patterns, the Marginal tribes represent neither a cultural type nor a cultural stage, but rather a class of tribes which had in common only the lack of the distinctive patterns of the remaining peoples. It is only in terms of certain technologies, material elements, and isolated ritual and social usages that they can be said to retain a primordial culture.

Where the combination of environmental potentialities and subsistence techniques allowed considerable latitude in the development of sociopolitical patterns, the age-area hypothesis is more useful in reconstructing the history of these patterns. Both the Andes and the Circum-Caribbean area, for example, had an efficient economy, a dense population, and large population centers. As certain patterns are found in both, it is reasonable to assume that the simpler forms

of the Circum-Caribbean area were derived from the more complex ones, like those of the Andes, and, though the former were somewhat adapted to local environments, they retained many features found in an earlier period of the Andes. This earlier period culture, in fact, probably extended from the Andes to the Mexican and *Mayan* areas.

The following pages will sketch the outlines of South American culture history from the earliest periods to the Spanish Conquest. Evidence on the antiquity of man in America is drawn principally from North America, where the record is better known. The earliest culture will be reconstructed from archeology and, so far as the age-area hypothesis is applicable, from ethnography. The development and spread of the cultures based on farming will then be traced. A subsequent section will outline post-Conquest changes among the Indians.

ORIGINS OF NEW WORLD CULTURES

RACIAL ORIGINS

Racially, the American Indian is predominantly Mongoloid. He is, therefore, most closely related to the peoples of Asia and was probably derived from Asia by way of Bering Strait. Some anthropologists have seen Mediterranean, Melanesian, and other non-Mongoloid strains in the Indian. This thesis cannot be disproved, but, unless it be postulated that these strains came from comparatively recent arrivals, it involves a genetic assumption that is difficult to defend: that several racial prototypes could intermix for hundreds if not thousands of years and the genes subsequently segregate out in the original combinations, a chance of one in thousands.

Though predominantly if not entirely Mongoloid, the Indian was extremely variable in head form, stature, and other characteristics. (See McCown, Stewart, and others in volume 6 of the *Handbook*.)

ANCIENT MAN

Many authors claim to have found evidence of ancient man in the Western Hemisphere, but few claims have withstood all criticism. The evidence is usually of three types: (1) typological, New World artifacts being accorded an antiquity comparable to their Old World homologues; (2) association of human remains with locally extinct fauna or flora; and (3) human remains dated in terms of geological, especially glacial, chronology.

Mere typological evidence has been generally rejected. Lithic types, which have been accredited with an antiquity comparable to their first appearance in the Old World sequence—usually a period of the Paleolithic but according to some, even the “Eolithic”—often survived in America in comparatively recent cultures. Some early American

periods have certain features in common with Eastern Hemisphere periods, but none of the Old World Paleolithic or Neolithic cultures are present in typical form in the New World, and there is no reason to suppose that America had a cultural sequence that corresponded precisely in either content or chronology to that of the Old World.

That human remains are associated with extinct fauna and flora is not per se evidence that they had great antiquity, for the extinct species may have survived locally until quite recently. Such associations have greater significance when the extinct species is evidence of important climatological change.

Attempts to place human remains in major geological periods have yielded no generally accepted results because, first, the time calibrations are too gross to be of value in cultural sequences, and second, the presence of human remains in older geological strata often has been found to be a later, secondary intrusion. Satisfactory field techniques have not always been utilized to demonstrate the contemporaneity of cultural and geological remains. The correlations of human remains with glacial periods, however, is susceptible to finer calibration, and certain correlations of cultural materials with periods of great climatological and physiographic changes in North America are now generally accepted. Though opinion concerning the absolute dates of these changes is not yet unanimous, the margin of disagreement is being steadily narrowed.

To judge by the correlations of human remains with glacial and postglacial phenomena in North America, man unquestionably was present in the New World at the close of or during the last phases of the final glacial period, which was certainly 15,000 and possibly 35,000 years ago. The Folsom bison hunters, represented by a lithic industry with some Upper Paleolithic features, are definitely known to have lived in the western United States during the final phase of the Wisconsin glaciation (Roberts, 1945). The oldest remains in Sandia Cave, New Mexico (Hibben, 1941) are probably earlier than Folsom. It is presumed that man reached America from Asia via the land bridge which linked Siberia and Alaska during the glacial period. (See also Sauer, 1944.) Ocean navigation was certainly not developed at this time; the Polynesian migrations in the Pacific occurred mainly during the Christian Era, especially the last millenium.

No remains of Central or South America have yet been shown to have antiquity comparable to those of North America. It should be remembered, however, that there has been comparatively little search for ancient man in South America and that, before the Folsom remains were found 20 years ago, orthodox opinion assigned the Indian a maximum antiquity in North America of only 5,000 to 8,000 years.

Central and South America, however, have remains of pre-agricul-

tural periods. The human footprints found in the volcanic mud in Nicaragua almost certainly antedate the earliest *Mayan* civilizations. The oldest remains of southern Patagonia and at Magellan Strait may plausibly be assigned an antiquity of 5,000 years—2,000 is the minimum—and correlations of these with glacial periods may require that their age be extended (Handbook, 1: 21–22). Human remains have been found associated with extinct fauna at various sites, but the fauna is rarely dated with any certainty. The Lagoa Santa or Confins man, however, seems to have been relatively primitive (Handbook, 1: 399), and the associated human and extinct sloth remains in Tierra del Fuego (Handbook, 1: 22) may have fairly great age.

None of these primitive South American remains resembles the oceanic cultures, but they are generally similar to preagricultural materials in North America. As there were no serious obstacles to man's gradual migration southward from North America, it is far simpler to derive the Indian of South America from this source than to postulate that he came by trans-Pacific migrations.

EARLY AMERICAN CULTURES

The data of both archeology and ethnology now leave no doubt that man first entered America from northeastern Asia, long before agriculture was known anywhere in the world, and that he possessed a Stone Age culture. He spread slowly throughout the hemisphere, from Alaska to Tierra del Fuego, adapting his mode of life to local environments with the aid of a simple technology and a limited number of material aids. Subsequent to the original migrations from Siberia and prior to the voyage of Columbus, Old World influence on the New World was very limited. Asiatic influence is evident on the northwest coast of British Columbia and Alaska; some elements have a circumpolar distribution around the subarctic, and a few traits, such as the bow, spread very rapidly and widely in America. Within the last two or three thousand years, other traits (p. 744) may have spread directly across the Pacific to the western shores of America.

Certainly no less than 3,000 years ago, and probably much more, the Indian began to bring native American plants under domestication. The cultivation of these plants spread widely, but did not reach the tribes in the northern and western parts of North America and in the southern and some of the eastern parts of South America. The peoples beyond the limits of agriculture remained Marginal, and they retained fairly primitive cultures, which, though in no case precisely like that of the first immigrants to America, were comparable to it in general simplicity. A comparison of these cultures together with the archeo-

logical evidences of early man gives a general picture of what once was probably pan-American in distribution. But it is impossible to reconstruct the original sociopolitical patterns, except in terms of absences, such as social classes. The early culture must be reconstructed largely in terms of isolated elements, and many of these, though very old, may date from anywhere in the thousands of years of prehorticultural times for which there is scarcely any archeological record.

Technology and material culture.—To judge by the Period I remains of the middens of Tierra del Fuego (Handbook, 1: 17 ff.) and by the Folsom finds, the earliest Indians knew the use of fire, pressure flaking of stone, and the process of cutting and grinding bone. They used some kind of projectiles—possibly but not certainly spears—which were tipped with stemless, chipped-stone points, and they worked skins with stone scrapers and bone awls.

On the other hand, the very considerable list of traits found among the Tropical Forest, Circum-Caribbean, and Andean tribes but not among the Marginal peoples, were almost certainly absent from the early American cultures. These include: domesticated crops; traits of food preparation, such as use of salt, babracots, metates, and mortars; technologies, such as ceramics, loom weaving, basketry, true tanning of skins, and metallurgy; and esthetic and recreational traits, such as musical instruments, chicha, tobacco, and other narcotics and stimulants; many elements of warfare, including cannibalism and human trophies; and temples, priests, idols, human sacrifice, and a ritual cycle.

Sociopolitical and religious patterns.—No special sociopolitical patterns can be shown to have characterized the early American cultures. We have seen that at the Conquest the Marginal tribes had a wide variety of patterns which reflected their ecological adaptations no less than their cultural heritage from the past. An effort to reconstruct the early patterns on a split-distribution basis by drawing parallels between the Marginal peoples of North and South America would be very misleading, for the similarities are quite superficial in most cases.

Among the *Shoshonean* tribes of the Great Basin of western North America and among the *Nambicuaras*, society was based on the individual family, but the socioeconomic activities of the two areas were very different and so were the larger, looser, multifamily aggregates that occasionally assembled. The patrilineal bands of the *Ona* (60 to 100 persons), each with a hunting territory for big game, were only superficially similar to the conjugal family groups of the *Algonquians* of Canada, each of which owned a territory for trapping fur-bearing animals. The large, caribou-hunting *Athabaskan* bands (200 to 250) of Canada had no parallel in South America, and the matrilineal vil-

lages of the Chaco have no precise North American counterpart. The main similarities between North and South American Marginal tribes, in fact, occur where the ecological adaptations were similar. For example, patrilineal bands were found in southern California and in Tierra del Fuego—both hunting areas but without large, migratory game herds. Mixed bands, i. e., multilineage or multifamily bands, of some size were found in the Great Plains of North America and Patagonia, and in both areas they were post-Columbian and were based on use of the horse for hunting and traveling.

The most that can be claimed for the earliest American tribes is that the people lived in small, generally somewhat migratory, groups which probably consisted of kin, that they lacked developed chieftainship and social classes, and that they were structured on the basis of sex, age, and socioeconomic activities.

There are, however, a number of socioreligious features which are virtually universal among primitive peoples and may be accredited with great antiquity: fasting by the mother of a newborn child; the isolation and fasting of a pubescent girl; earth burial; belief in a soul and a life after death; belief in supernatural beings; belief in witchcraft and omens; shamanism (but see p. 588); and shamanistic curing by massage, blowing, and sucking.

Some individual culture elements.—A considerable number of culture elements which were frequently found among the Marginal peoples of North and South America but less often among the farming tribes have been interpreted as possible ingredients of an early, preagricultural period. (See Nordenskiöld (1931), Krickeberg (1934), Ploetz and Métraux (1930), references cited by Cooper (1942), and Handbook, 2: 213-214.)

This application of the split-distribution method can be accepted only with certain qualifications. First, as Cooper (1942) points out, some of these elements may well have been invented independently in North and South America in response to environmental needs, so that their distribution is no evidence of their antiquity. Second, some elements may have developed as part of the agricultural complex and spread beyond its limits, later being superseded among many farming tribes. Such, for example, appears to have been the case with coiled basketry; and netting, though older than loom weaving, is not necessarily prefarming. Third, terms such as "archaic," "preagricultural," and "early" are altogether too general. Presumably, the preagricultural period in America lasted some 10,000 to 15,000 years, during which culture became considerably specialized in the different areas. Finally, few of the traits accredited with great antiquity in America occurred among all Marginal peoples. Rather, they seemed to cluster in certain tribes and groups of tribes, and it would be wholly unwar-

ranted to assume that they were once pan-American in distribution and have since inexplicably disappeared among many Marginal tribes. Certain games, for example, link the Gran Chaco and North America, but they are wholly absent in the Chilean Archipelago.

The most fruitful working hypothesis is that American culture history was very complex. To understand this history we need evidence of period and regional differences derived from detailed archeological sequences and ethnographic comparisons. A certain clustering of these early traits, for example, suggests that there were at least two streams of influence in South America, one carrying certain complexes down the Andes and the other entering the eastern part of the continent. Where archeological sequences are known, the culture appears to have arrived in a succession of waves. The following list shows the clustering of some of the elements that occurred also among North American Marginal tribes:

Araucanians.—Shaman as transvestite (also *Puelche*); use of tamborine; and pubescent girls race at dawn and carry fire wood (also West coast of North America).

Paraná River.—Finger mutilation as evidence of mourning; placing skewers through the skin; and fasting for a guardian spirit (all *Charrua* and especially Plains in North America).

Chaco.—Such myth motifs as the trickster, the vagina dentata, and the theft of fire, the last also Montaña, eastern Brazil (Métraux, 1939; see also Handbook, 1: 369); such games as ring-and-pin (also Montaña), dice (also Andean), "snow-snake" (*Abipón*, *Mocovi*, also eastern Brazil), gambling (also Andean), the musical bow (*Mataco*, *Toba*, *Lengua*, *Guaná*), and scalping (also Guianas).

Elements of a spotty distribution.—Some elements were fairly widespread among North and South American Marginal tribes, though not common to all of them: harpoons, spear throwers, nets, traps, fish-hooks, body painting, depilation, some kind of secret society, usually for men (*Ona*, *Chamacoco*, eastern Bolivian and Juruá-Puruá *Arawak*, Northwest Amazon, *Bororo*), and the belief that disease was caused by the intrusion of a foreign object into the body.

Features of more limited distribution include such elements as the ritual drinking tube (*Yahgan*), head-scratcher (*Yahgan*, *Tucanoans*, *Choco*, *Apinayé*, *Canella* and other *Ge*), steam bath (*Purí-Coroado*, *Botocudo*, *Araucanians*), fasting for a guardian spirit (*Charrua*), arrow swallowing (*Tehuelche*, *Tereno*), certain musical styles (*Yahgan*), deer-hoof rattles (Chaco, eastern Brazil), the soul-loss theory of disease (Andes, Chaco, eastern Brazil), the bull-roarer (*Sherente*, *Apinayé*, and *Mataco* as a toy; *Choco*, *Aymara*, *Bororo*, *Mashacalí*, Bolivia, Antilles, *Araucanians*, Central America, Guianas, *Apinayé*), hoop-and-pole game (*Sherente* and some Guiana, Amazon, Chaco, and Fuegian tribes), hockey (Chaco, *Tehuelche*, *Araucanians*, *Northern*

Cayapó), snow-snake (Chaco, eastern Brazil),⁵ boiling water with hot stones (*Aweicoma*, *Chono*, *Caingang*, *Shokleng*, *Botocudo*, *Purí*), heating water with hot stones (*Yahgan*, *Ona*); earth oven ("*Tapuya*," *Purí-Coroado*, *Caingang*, *Shokleng*, *Aweicoma*, Chaco, Argentina, *Northwestern Ge*, *Southern Cayapó*), and strike-a-light (*Archipelago*).

THE CULTURES BASED ON AGRICULTURE

ANDEAN CULTURE DEVELOPMENT⁶

On the basis of present evidence, American plant domestication appears to have been undertaken independently in many places and at different periods. Some species, such as maize, beans, and squash, attained wide pan-American distribution, but on the whole, each geographical area had, in addition to these plants, its characteristic complex of staple crops: the intermediate and low Andes, maize and other cereals; the high Andes, potatoes and quinoa; the Tropical Forests, root crops; and the Northern Andes, many fruits. The oldest evidence of agriculture is in the Central Andes, where it is conservatively assigned an antiquity of about 500 A. D. (*Handbook*, 2: 80). Some authorities believe that it should be dated 1,000 years or more earlier; in México the first evidence of farming is probably much later. Thirty-one domesticated plant species are known from the Early Periods of Perú, and probably most of them were cultivated during the preceding Chavín Periods. Several of these plants, such as lima beans, had already reached the limits of their genetic variability in the Early Periods, implying a long antecedent period of crop breeding. The domesticated llama and alpaca were also present in the Early Periods.

Botanical data suggest that some area to the east of the Andes may have been the locale of the first cultivation of the principal Andean crops, especially maize. Eastern Bolivia, with its milder and more varied climatic zones, is a likely place, and its archeology has rich remains that may include those of very early agricultural peoples. The archeology of eastern Bolivia, however, is scarcely known, and early sequences have not yet been established. It is necessary, therefore, to look to Perú with its long chronology, for a picture of the earliest cultures based on farming. The oldest Peruvian culture yet identified is that of the Chavín Periods, which may have covered a

⁵ For games and gambling, see Cooper, this volume, p. 503.

⁶ For the sequence of Andean Periods, see *Handbook*, vol. 2, p. 80. Research since 1945, especially the Virú Valley Project in Perú, has extended cultural sequences back to a preceramic, agricultural period, corrected the order of the periods as given in the *Handbook*, and thrown new light on the development of social, religious, and military patterns. (See Steward, 1947, 1948; Strong, 1947; Willey, ms.)

considerable time span. The remains of the Early Periods, which followed the Chavín, give a more complete picture of life at that time.

In the Early Periods, agriculture was already well established in Andean patterns, even extensive irrigation works being used. Farm productivity is evidenced not only by the fairly dense population but by the development of manufactures, the building arts, and religion and warfare, which took many persons from subsistence activities. Transportation was no less efficient than in later periods, for huge balsas were used for coastal travel and llamas packed goods over roads on the land. The precise settlement pattern has yet to be ascertained, but communities appear to have been small and dispersed, though closely spaced, and if they were nucleated at all, it was around religious centers, such as the classical Highland site, Chavín de Huántar. There was certainly no urban planning, as at later sites like Chanchan. Habitations of the Early Periods occur either as isolated dwellings or as loose clusters of dwellings forming small villages; any planning was of religious rather than civil centers.

Evidence of a stratified society, developed warfare, and a priest-temple cult is unmistakable in the ceramic decorations and representations (Handbook, 2:149-182), especially of the Mochica Period. These patterns must have resembled those of the later Andean peoples, and they appear to have been even more elaborate than those of the Sub-Andean and Circum-Caribbean areas.

Pottery designs show chiefs or nobles with characteristic evidences of rank: wearing special dress and insignia, sitting on thrones, being carried in litters, receiving obeisance from kneeling or bowing subordinates, being served by retainers, and meting out punishment. Considerable governmental control is implied by the construction of great mounds, buildings, irrigation systems, roads, and other public works, by the use of runners or messengers for communication, and by the public imposition of punishment.

Warfare was strongly developed. Armies wearing special dress and using all the principal weapons of later periods fought under powerful war chiefs. Captives were sacrificed, probably in religious rites, by being thrown from mountain peaks, or they were made into human trophies, especially head trophies. Cannibalism is not shown, but the quartering of prisoners is suggestive of the *Tupian* cannibalistic festivals.

The three-story stone edifice at Chavín de Huántar and the temple mounds of the Early Periods were almost certainly religious centers. The feline, serpent, and condor deities which are represented in so much of Peruvian art were undoubtedly among the principal tribal gods. Various other nature gods are also portrayed. An important class of priests officiated over public worship, but it is not known

whether they were also shamans and whether they had civil power. There was also ancestor worship, or a cult of the dead, as evidenced by the elaborate mummy burials. Shamanism was probably present with its familiar patterns, for shamans (possibly priests) are shown shaking rattles and, less often, massaging and sucking their patients.

On the technological and material side, the essential processes and complexes were thoroughly established by the Early Periods, and craft production seems to have been specialized. Painted, plastic, and modeled ceramics equalled those of later periods. All weaving techniques were known, and both cotton and wool were used. Basketry was made, evidently at first twined and twilled (Early Ancón and Supe), and later coiled; if correct, this sequence reverses the usual one elsewhere in America. The weapons used in Perú at the Spanish Conquest were already present: darts, spear throwers, clubs, and slings. The Mochica also had the blowgun and dart. The only industries to show technological advancement subsequent to the Early Periods are masonry and adobe construction and metallurgy. Chavín used only gold; Mochica added silver, platinum, and copper; only the Middle or Late Periods achieved bronze.

The Chavín culture, though the earliest known in Perú, is far from primitive or even formative, and it probably had the basic features of the succeeding Early Periods. Its subsistence was adequate, its technological and material culture contained virtually all essential elements and complexes of later periods of the Andes, its sociopolitical and religious patterns were clearly developed far beyond those of the Marginal or Tropical Forest peoples, and its esthetic attainments show considerable sophistication. Although its ultimate origins are still obscure, it is difficult to imagine that it was suddenly introduced full-blown from across the Pacific. Instead, it must have developed along with the indigenous American crops, on the cultivation of which it was based.

Cultures of the Chavín and Early Periods would seem to have contained the generalized patterns that were later to take more definite form. Undoubtedly there was some local variation, but the main features were more or less pan-Andean. If San Agustín has antiquity comparable to that of Chavín, a similar pattern probably existed in Colombia. In fact, it probably extended northward to México, and the basic *Mayan* pattern of a population loosely aggregated around great mound and temple sites seems to represent a special emphasis on the socioreligious components of this culture.

Subsequent Andean trends bring out the potentialities implicit in the early patterns. Population evidently increased considerably, for in the Middle Periods hillsides were terraced for farming purposes. After the Early Periods, civil and military affairs were developed on

a larger scale, a development no doubt partly explainable by the pressure of the growing population. An increasing proportion of construction was devoted to building forts, and, to judge by the pre-*Inca* or Late Periods, each of the kingdoms and many smaller states was constantly at war with its neighbors. Civil developments are evidenced by the construction of planned cities, especially on the North Coast of Perú, where large urban centers were laid out with protective walls, gardens, reservoirs, ceremonial mounds, cemeteries, streets, and blocks of houses.

Attainments in political organization and conquest, however, were probably not the same throughout the Central Andes. The Tiahuanaco culture, which widely affected local areas throughout Perú, may have represented a military conquest comparable to that of the *Inca*, which had the same cultural effect. Even the classical site of Tiahuanaco, however, seems to have been a predominantly religious center, and the Tiahuanaco influence may have been more religious than military. Subsequent to the Middle Periods, a considerable local autonomy seems to have returned. In the Highlands, the population remained dispersed, being only loosely nucleated around religious and administrative centers, and political control did not extend beyond the ayllu or group of ayllus which formed the independent state. Each Coastal valley, too, probably retained considerable independence. The North Coast, however, had a long tradition of centralized planning. This planning may have been only within the framework of independent city states, which began at sites like Chanchan, in the Middle Periods, but large irrigation systems evidence the extension of centralized control over considerable areas. By and large, it was in advance of the Highlands. The *Inca*, relative newcomers to empire building, were the first people who are known to have incorporated all of the Central Andes in a single state. The *Inca* (p. 734 ff.) developed government at the expense of other features of the earlier patterns. A somewhat fluid class structure, in which status could be attained by the capture of slaves and concubines, was replaced by a hereditary caste system, which was based on wealth-producing commoners and in which the slave class became unimportant. Even the priests came to form a stratified, hereditary class, and so did the very gods. All religious, economic, social, and military activities of state concern were fitted into a system of governmental control, though otherwise the folk culture was left intact (p. 737).

SPREAD OF THE ANDEAN PATTERNS

Outside the Central Andes, South American archeology has not yet disclosed with certainty the cultures that were contemporary with the Chavín and Early Periods of Perú. Ethnographic data and frag-

mentary archeological evidence, however, show that most of the basic technologies, many items of material culture, and even the sociopolitical patterns of the Central Andes attained a wide distribution.

The Marginal areas.—Eastern Brazil, the Pampas, Patagonia, and the Chilean Archipelago felt little influence from the Andes. None of these areas contains archeological evidence of anything more advanced than the Marginal tribes which occupied them at the Conquest, except those portions of eastern Brazil where a Tropical Forest type of culture attained limited distribution.

The Southern Andes.—The Southern Andes, to judge by the archeological sequences on the coast of North Chile, had a Marginal type of technology prior to the introduction of agriculture. The Early Period traits include: spear throwers, bolas, harpoons, fish-hooks, fire drills, and earth burial. With the introduction of agriculture came pottery, especially painted wares, coiled basketry, slings, wool and cotton weaving, sandals, dogs, and balsas (Handbook, 2: fig. 49). The southward diffusion from the Central Andes was intensified toward the end of pre-Conquest times by the development of overland trade, which was carried on by means of pack llamas traveling on roads. Under the *Inca* Empire, the Southern Andes received metallurgy, Central Andean type garments, and other traits, many of which reached the *Araucanians*.

The *Atamameño* and *Diaguita* had a rich Andean-type material culture, some of it probably acquired in Tiahuanaco times, but they retained a primitive sociopolitical pattern, because the forbidding deserts, even though exploited by Central Andean techniques, could not support sufficient populations to maintain a class system. The communities remained comparatively small and seem to have consisted mainly of lineages or kin groups. The *Araucanians* occupied a much more fertile area and had a denser population, but it was divided into a greater number of units rather than into larger, more close-knit ones. These units were of the Tropical Forest type, based on the lineage or kin; social classes had scarcely emerged and intervillage federation hardly begun.

The *Araucanian* culture has resemblances to areas other than the Andes. Frame, thatched houses, palisades, dugout canoes, urn burial, and cannibalism link it with the Tropical Forests. The association of transvestitism and the tamborine with shamanism are characteristic of Siberia and the northwest coast of America. The distribution of this peculiar shamanistic complex is difficult to explain; possibly transvestitism in Colombia, which had some ritual significance, is another survival of an early widespread complex.

The Northern Andes and Circum-Caribbean area.—At the Conquest, Ecuador, Colombia, and the area around the Caribbean Sea had

a pattern which contained all the essential features of the Chavín and Early Periods of the Central Andes, though these features were developed in differing degrees. An ample subsistence complex supported a very dense population, though crops adapted to the rain forests took the place of certain Andean species, and, along the coast, sea resources were added to farming. Society was stratified into chiefs or nobles, commoners, and slaves or war captives. There was a temple cult dedicated to tribal gods and served by priests or by shaman-priests. Warfare was strongly developed, and it afforded individuals a means of attaining rank. Technological processes included ceramics, woven basketry, and metallurgy in gold and tum-baga. In the Central Andes, metallurgy was the only technology to advance radically after the Early Periods; processes of the latter periods, such as copper smelting and bronze making, are not found among the Circum-Caribbean peoples.

The similarity in general patterns as well as in specific elements of the Andean Early Periods and the Circum-Caribbean culture of the Conquest indicates some connection between the two. This is not to say that the latter was derived entirely from the former. In a comparative analysis of the Circum-Caribbean culture in volume 4 (pp. 6-11), it was shown that the general patterns and traits of this culture extended from the Andes to México and Yucatán. It was postulated that an early, inter-American Formative Period culture underlay the more special later developments of all these areas and that this culture had the following features: Subsistence based on fairly intensive farming, especially of maize; dispersed settlements that tended to nucleate around religious centers; political units limited to the village or small state; a class tendency, status being accorded warriors and religious leaders, who probably exercised considerable civil power but whose rank was achieved rather than inherited; religious mounds, altars, idols, offeratories, and shrines; priests, probably becoming distinct from shamans; considerable warfare; captives taken as slaves and as sacrificial victims; children of slaves, however, probably free; human sacrifice; cannibalism, probably ritual; human trophies, especially skulls and flayed skins; ritual blood letting; celestial and animal deities, and incipient agricultural ritual; construction of causeways, aqueducts, canals, defensive works, and stone buildings; bark cloth; loom weaving of domesticated cotton and use of batik and tie-dyeing; painting, negative painting, incised, and plastic treatment of pottery; featherwork and feather mosaics; metates; armor; coiled and woven (?) baskets; cloth garments; sandals; and many other culture elements of a very wide inter-American distribution. This

Formative Period culture contained the essential features of the Circum-Caribbean culture.^{6a}

In many respects, however, the Circum-Caribbean culture is more closely linked with the Andes than with México or Yucatán. Central America and the Andes share such archeological traits as stone-cist and deep-shaft graves, stone stools, jaguar-form stools, Manabí-type stone slabs, and mace-head clubs, and such ethnographic features as the platform bed, burial of subjects with a chief, mummification, and tumbaga (the last north to Panamá). Colombia was even more closely linked with the Central Andes (p. 731) though it also shared very fundamental features with Central America. Thus, the *Chibchan* language extended northward to the *Ulua-Sumo-Mosquito* in Nicaragua, and both Colombia and Central America seem to have had a basically matrilineal society, in contrast to both México and the Central Andes.

There is also a number of traits which link Central America and the Tropical Forest tribes: manioc, Muscovy duck, babracot, pole-and-thatch house, palisaded villages, communal house, hammock, blowgun, and other elements especially adapted to tropical rain forests (p. 698).

A significant number of these Andean and Tropical Forest traits also reached the Antilles.⁷

Mexican influence is not wanting in Central America, but it appears to be relatively late, it is definitely concentrated among the Meso-American tribes who were scattered along the Pacific Coast, and it did not reach the Antilles (Handbook, 4: 199).

In the northern Andes and in Central America, it seems certain that a Formative Period culture preceded the Circum-Caribbean culture. The areal differentiation and interareal linkage of this postulated early culture are still to be determined. Possibly all of it came from South America; present data, however, cannot support such a contention. The data of both archeology and ethnology, on the other hand, show that apart from the general inter-American or Formative Period features, Central America is related to South America rather than to México or Yucatán; i. e., the flow of more specific features has been predominantly northward for a long time. For precisely how long this has been so remains to be determined. It is suggestive that litters were known in the Andean Early Periods and that deep-shaft graves in Colombia may have comparable antiquity.

There is reason to suspect that at some early period or periods, Andean culture may have spread with considerable vigor into tropical rain forest areas, where it subsequently diminished or disappeared.

^{6a} For modification of the following analysis, see Steward, 1947, 1949.

⁷ Some interesting features of the Circum-Caribbean culture were found among certain North American tribes, especially the *Natchez*, bordering the Gulf States. No historical connection of these cultures is postulated, but a comparison of them would be interesting.

This expansion was, perhaps, comparable to that of the Pueblo II Period in the Southwestern United States. A similar, though more restricted expansion, occurred during the Tiahuanaco and *Inca* Periods. Evidence of this early spread consists of such archeological remains as stone construction, irrigation works, terraces, roads and causeways, which are found in various places in eastern Bolivia, in the llanos of eastern Colombia, in the Venezuelan Andes, and in the Highlands and North Coast of Colombia. Though some of these remains may eventually be identified with the historic tribes, many of them seem to have fallen into disuse long before the Spanish Conquest. Central America, too, may well be found to have archeological remains evidencing a more complex culture than that found at the Conquest.

A highly speculative explanation of these remains is that an Andean-type culture expanded at an early period into sparsely populated areas well beyond the Andes. These far-flung remains, however, do not have a uniform style, comparable to that of the Pueblo II, Tiahuanaco, or *Inca* cultures, and the expansion must have been gradual. In the course of time, population pressure and warfare perhaps led to the development of tightly nucleated, palisaded villages in place of dispersed settlements, and to the abandonment of hilltop forts, roadways, stone construction, and the building of mounds, except for burial purposes. At the Conquest, the only great civilization outside Highland areas were those of the *Maya*, and these were in the dispersed settlement pattern. This unique *Maya* adaptation and survival presents an interesting problem.

The Tropical Forests.—A class-structured society was not characteristic of the Tropical Forest tribes, and, though the large villages found in many regions of exceptionally dense population would seem to have afforded a basis for it, social cleavage was predominantly in kinship groups. In many tribes, however, rank was accorded to warriors, who formed something of a class, and captives were kept as slaves. Among some tribes which acquired the horse after the Conquest, classes definitely emerged to form a pattern like that of the Sub-Andean tribes.

It is difficult to know whether the status accorded warriors was the result of purely local development or of diffusion, because stimulus-diffusion from the Andean and Sub-Andean cultures may have been far-reaching. An opportunity for such diffusion was provided by the proximity of some of the class-structured tribes to the Andes. The definite class-tendency of the *Atacameño*, *Diaguita*, and *Araucanians* can certainly be traced to the Andes. Even the *Comechingón* and *Huarpe* of the neighboring Pampas, despite their Marginal ecology, showed slight influence of this tendency. The *Abipón*, *Mocoví*, *Paya-*

guá, and *Mbayá* of the eastern Chaco, though definitely Tropical Forest in their technologies and material culture and separated from the Andes by the Marginal peoples of the western Chaco, may have been affected by the Andes at some time in the past. It is interesting that the *Mbayá* were one of the few non-Andean tribes to carry on wars of conquest. The *Tupí* and *Carib*, though extraordinarily warlike, were geographically remote from the Andes and the Caribbean area. They lacked true warrior classes, and captives were bracketed into kin groups rather than into a slave class. The *Arawak*, especially of the Guianas, possessed incipient social strata in that sons-in-law became a patriarch's retainers, though here, too, society was still on a kinship basis. It may be significant that some Guiana tribes, especially the *Arawak*, were matrilineal, like the Circum-Caribbean peoples, whereas most Tropical Forest tribes were strongly patrilineal.

The Tropical Forest tribes lacked a temple-idol cult. A few of them had public harvest ceremonies, and among many, the shaman performed oracular functions in the men's club. As a rule, however, the shaman apparently conferred with his own spirit-helper rather than with a tribal god. The gods, especially celestial ones, which became objects of tribal worship among the Circum-Caribbean and Andean peoples, were usually little more than mythological characters in the Tropical Forests. If there is a connection between Tropical Forest and Circum-Caribbean religion, it is probable that the latter, under Andean influence, built on a Tropical Forest pattern, assigning the shaman priestly functions, especially as an oracle for the tribal gods, which were represented by idols and kept in a special temple. The Tropical Forest men's house was the functional equivalent of the temple in being the scene of various rites, but it also had purely social purposes.

In technological and material culture, the Tropical Forest tribes had all Circum-Caribbean features except metallurgy; they used loom weaving, domesticated cotton, ceramics, and woven basketry. The resemblance of these material traits to those of the Circum-Caribbean area, however, is greatest in the Guianas and Amazon. In eastern Brazil and around the periphery of the Amazon, many of them disappear.

In addition to the technologies, which were perhaps ultimately Andean-derived, the Tropical Forests had many traits which they shared with the Circum-Caribbean peoples but hardly at all with the Andes. Many, but not all, of these are adapted to tropical rain forests: the tropical root crops; thatched, frame houses; scant garments; hammocks; dugout canoes; bark cloth; blowguns; arrow poisons; fish poisons; rubber-ball games; hollow-log drums; and use of tobacco.

The present thesis postulates that the Tropical Forest culture derived its essential technology from the Circum-Caribbean culture and that it also acquired certain rain forest traits, but it failed to borrow the Circum-Caribbean sociopolitical and religious patterns. In becoming adapted to fluvial, littoral, and rain forest areas, the technological complex spread via the main waterways. Specifically, it seems to have spread from Venezuela down the Atlantic Coast and up the Amazon and its tributaries (perhaps secondarily it spread up the Orinoco and down the Rio Negro), suffering successive losses as it reached the headwaters, where many of the tribes remained Marginal or Semi-Marginal. It is clearly evident, however, in the Peruvian Montaña (Handbook, 3: 535; 5: 697), and it may have broken through the watershed of the Amazon in eastern Bolivia to reach the upper Paraguay River, while another stream flowed down the coast of Brazil to the Paraná Delta. It followed the rain forest and major waterways but scarcely penetrated the plains and savannas, where the tribes remained Marginal.

Certain major linguistic groups with a wide distribution must have played an important role in the diffusion of the Tropical Forest culture. In the north, the *Arawak*, some of them with a Circum-Caribbean culture, and the *Carib* are outstanding. South of the Amazon, the *Tupí* were among the major culture carriers. On the other hand, the many small, linguistically isolated Marginal and Semi-Marginal groups clustered around the periphery of the Amazon Basin (map 18) are probably to be regarded as predecessors of the far-flung language groups. Pushed to the back country, they have long remained in isolation.

Some authors have ascribed to the *Arawak* the most important role in developing and spreading South American cultures. To these people has been attributed the diffusion of vertical looms, the cult complex with secret trumpets, elaborate ceramics, and other traits throughout the Tropical Forests. They have been accredited with the spread of Circum-Caribbean traits to the Antilles and to Central America and even the introduction of the early Chavín, Tiahuanaco, and other cultures to the Andes. In view of the very great differences in the cultures of local *Arawakan* tribes—for example, the Circum-Caribbean Island *Arawak* of the Antilles, the Sub-Andean *Arawak* of eastern Bolivia, the Semi-Marginal *Campa*, *Piro*, and others of eastern Peru, and the Tropical Forest *Arawakans* of the Guianas—it is necessary first to establish that there was such a thing as an *Arawakan* culture. In terms of their sociopolitical patterns, it is obvious that these tribes would have to be classed with their neighbors on a culture-area basis rather than with one another on a linguistic basis. The more important traits and technologies, such as ceramics, loom weaving, and

basketry, occurred among the *Quechua* and *Aymara* of the Andes, the *Chibcha*, *Carib*, and other tribes of the Circum-Caribbean area, and the *Carib*, *Panoans*, and dozens of smaller linguistic groups in the Tropical Forests; they were by no means peculiar to the *Arawakans*. The *Arawakans* certainly played an important role in diffusing these features in parts of the Tropical Forest areas, but even here their role has yet to be clarified. One has the impression that the importance attached to *Arawakan* migrations resulted in part, at least, from a certain methodological approach to the problem of culture classification: when cultures are described by linguistic groups rather than by areas, there is a concomitant tendency to assume a culture type for each language and then to reconstruct the history in terms of the migrations of language groups.

POST-COLUMBIAN CULTURE HISTORY

Comparatively little attention has been accorded the acculturation and assimilation of the Indian to European culture since the Spanish Conquest. The literature is probably adequate to sketch the broad outlines, but it has been utilized mainly to reconstruct aboriginal cultures and the data of the 400-year post-Contact Period have usually been compressed into a two-dimensional picture without historical perspective. What are virtually pioneering essays in identifying and describing some of the post-Columbian acculturational periods are, however, included in several Handbook articles. (See especially Kubler's on the Andes (2: 331), Murra's on Ecuador (2: 785), Steward's on the Montaña (3: 507), and Tschopik's on the *Aymara* (2: 501).)

The Europeans brought to South America a comparatively homogeneous culture. Though their initial penetration was variously made by the conquistador, missionary, or colonist, the prevailing purpose was to exploit the native peoples through a system of tribute, which soon amounted to mass labor. The success of this system and its effect on the Indian depended upon various local factors: the native population density; the incidence of disease on the natives; the accessibility of the area and its suitability for Iberian types of land use; and the predominating facet of European culture, such as proselytization, mining, or conquest.

The Marginal and Tropical Forest areas.—Along the coast and navigable interior waterways, the Indian was, as we have seen (p. 664 f.), quickly exterminated or absorbed. In the less accessible areas, he survived and retained a native culture, but he was strongly affected by the Whites even before direct contact with them. Throughout the Tropical Forest and Marginal areas, the course of pre- and post-Contact acculturation was similar in certain general respects.

Prior to the arrival of the Europeans in these areas, the Marginal and Tropical Forest Indians received certain vital trade goods which brought a temporary efflorescence of and even deep changes in their culture. New crops, domesticated animals, and steel tools, and, later, firearms and other items so improved exploitative activities and transportation facilities that villages of unprecedented size developed, tribal contacts were increased, and even aboriginal culture elements, such as the blowgun and bow, were diffused to many tribes which had previously lacked them. Communities became larger, requiring new political controls and involving new kinds of interpersonal contacts, and warfare was intensified by the pressures of dislocated tribes.

These changes were within the frame of the preexisting, native culture, but certain striking new sociopolitical patterns emerged from the new ecological patterns. In the case of the Pampean and Patagonian tribes, the effect was very similar to that among the Plains Indians of North America. The former were hunters of guanacos and rheas in grass and brush country; the latter, of bison on the prairies. In pre-horse times, both had lived in small bands. The Patagonian bands were probably patrilineal, like those of the *Ona*; some of the Plains tribes, however, had been farmers. In both areas, the horse and a limited amount of gear were introduced by the Spaniards, even before there was close contact between Indian and White. In both, the advantage of the horse as an adjunct of hunting and as a pack animal permitted the small aboriginal groups to amalgamate into bands, which, though unstable, were far larger than anything possible among foot Indians. Some of the North American farming tribes of the Plains gave up their fields to become horse nomads. The territory claimed by each of these bands became larger, though less well defined than previously, and the chief gained in power and prestige. The tribes bordering these areas were dislocated, causing intertribal strife and intensifying warfare. Cultural features less closely connected with the ecology were different in the Plains and the Pampas; for example, dress, house type, and ritual elements, but even some secondary traits, curiously, appeared in both. For example, people artificially lengthened their tresses with horse hair. (Regarding the influence of the horse on the Chaco, see volume 1 of the Handbook, pp. 202-203, 265-267, 304-308. For the *Goajiro*, see Volume 4, pp. 369-383.)

In some areas, the Indians were first contacted by missionaries. Though the orders adopted different policies, all of them settled the Indians in large villages and encouraged farming and various new crafts. This served to introduce considerable European culture, including Christianity, but disease rapidly reduced the populations in the mission villages. In many cases, the natives rebelled and returned

to aboriginal modes of life. Those who remained continuously in the missions have become very nearly extinct.

Where the European colonists established settlements, the Indians became attached, voluntarily or through enslavement, to them. Consumption of European food and goods, use of European dress, and completely new economic and social habits quickly eradicated native culture. The Indians who survived the frightful toll of epidemic diseases were gradually absorbed into the White and Negro populations.

Along the coasts in the tropical areas and in the Antilles, the population was comparatively great, and it might be thought that it could have served the Spaniards for mass labor, as in the Andes. Instead the Indians disappeared amazingly soon and their place was taken by Negro slaves. One factor in this decline was the devastating epidemics of European diseases, especially in the large settlements established by the Spaniards. Another was probably the violence of the European Contact on peoples far less prepared for virtual servitude than the Central Andean tribes who had been under the *Inca* Empire.

Each year, the surviving Marginal and Tropical Forest peoples decline further in numbers and become more assimilated to national culture.

The Sub-Andean and Circum-Caribbean tribes.—These peoples experienced the same general post-Contact trends as those of the Tropical Forests, except for a difference brought about by the distinctive features of their aboriginal culture. All the Indians of the Antilles and most of those elsewhere on the coasts and lowlands became extinct or were absorbed. In a few inaccessible regions, especially in the mountains, Indian cultures still survive, but they have experienced a phase of deculturation since the Conquest.

The older chronicles describe the class system, temple-idol cult, organized warfare, and esthetic craft production of Circum-Caribbean types. Modern ethnographies disclose only an unstratified society, a shamanistic religion, little or no warfare, and crude crafts. (See, for example, the *Cuna*, vol. 4, p. 257.) This step down to a lower level, very similar to that of the Tropical Forest tribes (4:2), was the direct result of the Conquest, which destroyed the higher levels of organization rather than the folk culture. The military basis for a class structure was removed when warfare, cannibalism, human sacrifice, trophy taking, and slavery were stopped, and the hereditary basis was lost when the chiefs and nobles were required to give over power to Europeans. In addition, the tribes were forced into submarginal lands which could not maintain the earlier sociopolitical patterns, and, during the Conquest, warfare and disease greatly reduced the popula-

tion. Organized native cult religion yielded to the Catholic Church, and only the more informal and private shamanistic and magical practices and folk superstitions survived. Technology and material culture deteriorated, partly because their finest expressions had been dedicated to the nobility, which was now gone, and partly because the new economy sank to lower levels of efficiency.

Stripped of their Circum-Caribbean and Sub-Andean features, these tribes were degraded to a generalized Tropical Forest level, and their patterns came to resemble those of the Amazon considerably more than those of their own ancestors. This phase of deculturation has considerable theoretical interest for it suggests that the Circum-Caribbean cultures were in reality Tropical Forest type cultures with an Andean overlay. This thesis, however, is antithetical to that suggested previously: that the Circum-Caribbean cultures might have represented a somewhat deculturated form of an earlier, fairly widespread Andean or Formative Period culture, which, by successive losses of social, religious, and esthetic features during both prehistoric and historic periods gradually converged toward the Tropical Forest pattern.

These observations are made to point up a problem rather than to offer hypotheses. To ascertain the prehistoric levels as well as the directions of cultural flow, archeology is badly needed, especially in Colombia, Ecuador, and Central America. To ascertain the precise nature of the native cultures at the time of the Conquest, the early chroniclers must be exploited to the fullest, for the ethnographies of the few tribes surviving today give a very incorrect picture of aboriginal cultures.

The Central Andes.—The Central Andes have differed from the remainder of South America in surviving the shock of Conquest and the attrition of European Contact without serious loss of population or culture. Epidemic diseases were not so devastating during the Conquest and Colonial Periods, and, though in the beginning the population may have been halved, the Indians have always outnumbered the Whites. The sheer numbers not only served to retard assimilation; they invalidated many European policies used elsewhere in South America. The minority of Europeans found it easier to exploit the vast labor supply through adapting their methods to preexisting native sociopolitical patterns than to completely uproot, recast, and finally replace the people. In the *Inca* Empire, they found a pattern more or less adaptable to their ends. They blanketed the higher officials into their class system and required the already subservient and much-exploited masses to supply tribute, taxes, labor (*mita*), and servants (*yanaconas*).

The effect of the Conquest on the Central Andes as a whole was similar to that on the Circum-Caribbean area. The more organized

and distinctive forms of the Central Andean pattern were replaced by European forms while the folk culture remained more intact. The *Inca* governmental system was replaced by the Spanish, with its fountainhead at first in Spain. The native social castes were leveled, except as the curacas found a place in the European scheme. The priest-temple-idol cult and the rites pertaining to it were formally replaced by Christianity, but shamanism, sorcery, and the more covert beliefs and practices organized around local shrines and deities, especially the mountain spirits, were able to survive. Craft production became more or less restricted to the plain, utility goods which had always been made for home consumption. The nobility and the *Inca* church, which had received most of the art products, were now gone. The Spaniards were more interested in tribute, and they mined their own gold, which they converted into money rather than into ornaments. Many European elements were adopted, for example, wheat, barley, cattle, horses, mechanical looms, methods of house construction, and types of dress, but these merely augmented an already rich native material culture. At first, the new produce enabled the Indian to pay his tribute but was not adopted into his culture.

There have been two main acculturational trends: (1) individuals and classes of individuals have been detached from the context of native life and assimilated quite rapidly into the national, Hispanic-American culture; (2) comparatively stable native communities have experienced more gradual acculturation.

Since the Conquest, large numbers of individuals have left their native communities to become a somewhat mobile, national proletariat that is integrated with the Europeans in a condition of economic and social interdependence. Their dislodgement from the native communities has several causes; work as servants to the Whites (the *yanacón*), mine labor, service in the army, the rewards of various new types of production in the urban centers, and the loss of their own lands. The last, and perhaps the most important, factor was brought about by: the early wars; new types of land use, especially cattle raising, which crowded out part of the dense farming population; complication of legal titles to land; increase of the native population; and the introduction of individualized land ownership and cash crops, which frequently led to land sales. Some of the detached persons became farm workers (*colonos*) on the haciendas and either were rapidly Hispanicized through the efforts of the landlords, or simply lost their Indian features without commensurate gain. Increasing numbers, however, have become urbanized and, though retaining some Indian culture, they wear European dress, work for wages, can read and write, and are geared to the national socioeconomic system through a national consciousness and a sense of solidarity.

The main reservoir of Indian life in the modern Andes is the community (*comunidad* or *ayllu*), which has remained relatively intact, except that during the Colonial Period its size was increased through the system of reductions (*reducciones*) to facilitate administrative control and religious proselytization of the Indian. In Perú and Bolivia, there are 3,000 such communities today, each with about 500 persons. Their patterns, despite a considerable content of Spanish elements, retain many of the principal characteristics of the native folk culture (Handbook, 2: 441): production mainly for local use, though always some surplus crops for sale; native methods of cultivation; exchange of labor with community members (*aine*); Indian garments, houses, and other material items; *Quechua* or *Aymara* speech, though much bilingualism; inability to read; local control of matters not of national concern; the extended family as the principal sociological unit; community solidarity expressed through village endogamy, local ceremonies, shrine (*huaca*) worship of local deities, especially those of mountains; and practice of shamanism and magic. Goods are exchanged at periodic markets, which may represent the resurgence of a pattern that was somewhat suppressed under the *Inca* control of production and distribution.

As the national economy expands, as education is extended, and as transportation brings the remoter rural areas into contact with the more advanced centers, the Indian communities are being gradually transformed. Almost all land is now individually owned, facilitating its loss through sale. The sale of land together with limited living space is breaking down the patrilineal, extended family, because new generations tend to bud off and become independent. This in turn weakens the community. Cash crops are introducing a money economy and greater use of manufactured goods. Special industries also augment community income and promote urbanization. Many non-Indian communities are incorporated and have producers' and consumers' cooperatives. Bilingualism is increasing. Formal education is introducing the national culture and a national consciousness.

Year by year, individuals who have been dislodged from their communities and even whole communities gradually pass from the category of Indian to that of national Ecuadorian, Peruvian, and Bolivian. The process of *mestizaje*, or assimilation to the national culture, seems likely to accelerate.

HISTORICAL SUMMARY

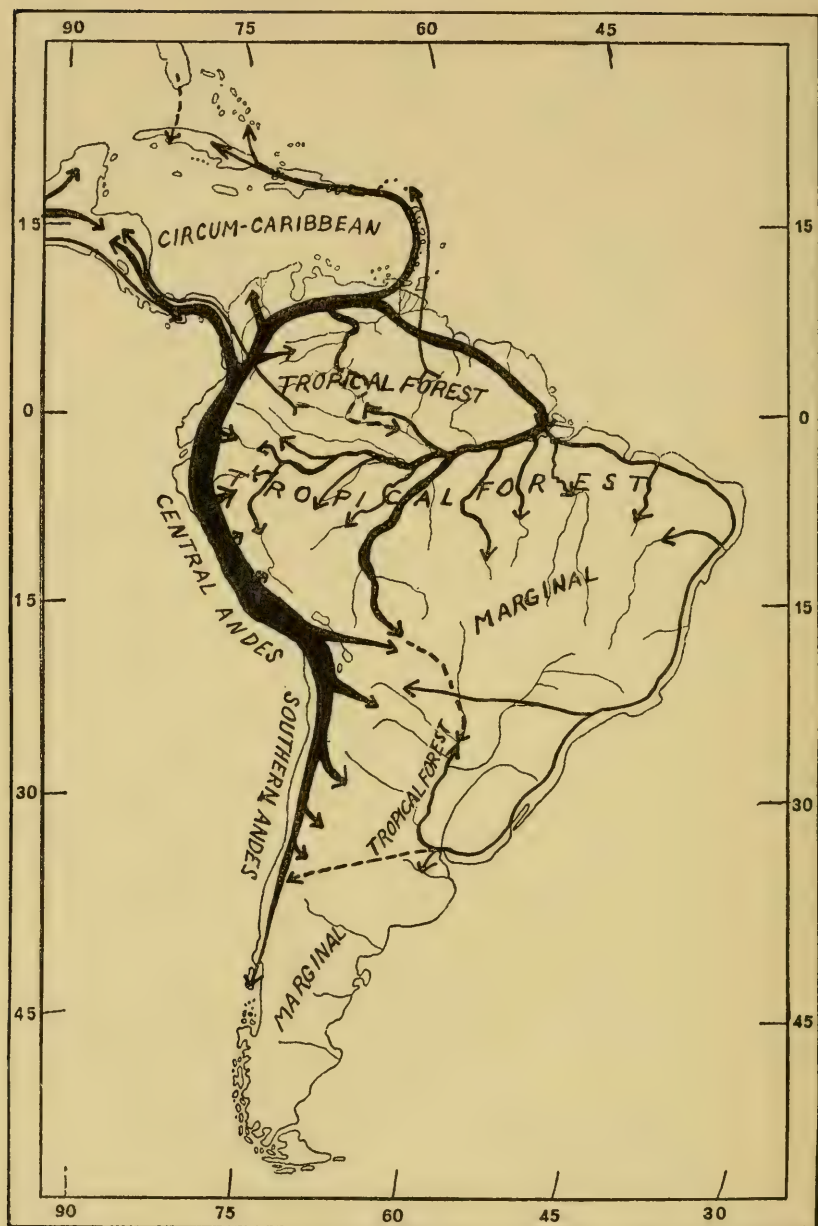
The first Indians reached America from Siberia some 15,000 to 20,000 years ago and eventually spread southward into South America, arriving in Tierra del Fuego not less than 3,000 to 5,000

years ago. Their original culture included fire making; flint chipping; spears; scrapers; skin dressing; birth, puberty, and death observances; shamanism; and perhaps various ritual elements. In the course of time, bows, harpoons, nets, traps, and other material items and various ritual and social elements were acquired in different areas. At no time, however, did the early hunters and gatherers have a uniform sociopolitical pattern. The extremely great local differences in environments, natural resources, and subsistence techniques imposed upon the population of each region the necessity of grouping itself and behaving in very different ways. Some were dispersed in conjugal family groups; others were nucleated in extended families, which were patrilineal or matrilineal, sedentary or nomadic, and permanent or temporary according to the socioeconomic activities required in each area. A few, where local abundance of food permitted unusually large population aggregates, were grouped into loose bands consisting of several conjugal families or of extended matrilineal or patrilineal families. The Marginal tribes encountered at the Spanish Conquest probably give an idea of the general variety of early ecological adaptations and sociopolitical types, but it is certain that all of these have changed over the years as new weapons and transportational facilities introduced new subsistence patterns and as the peoples migrated or were pushed into new environments.

The domestication of a considerable number of native American plants was begun several thousand years ago, certainly before trans-Pacific voyages could have taken place. By 500 A. D., and probably 1,000 years earlier, more than 30 of these were grown in the Central Andes, and some of them had reached the limits of their genetic variability. They became the basis of a large, stable population in the Andes, which, in its earliest known manifestations, the Chavín Periods,^{7a} had already attained mature esthetic, social, and political patterns, and, by the Early Periods possessed all essential technologies and refinements of material culture. It had a class-structured society, organized warfare and human-trophy taking, a priest-temple-idol cult, and excellent loom weaving, ceramics, metallurgy, basketry, architecture, and transportation.

It is postulated that a culture, called the Formative Period culture and possessing the general features of the Central Andean Early Periods, extended northward to México. The region of its ultimate origin is not known, but it might have been in South America. The archeological and ethnographic cultures of the Circum-Caribbean area represent a more specific formulation of the Formative Period culture, but the specific features link them most closely with the Andes and indicate a predominant northward flow of culture. It is pos-

^{7a} See footnote 6, p. 753.



MAP 22.—Main routes of historical spread of pre-Columbia South American cultures.

sible that the Circum-Caribbean cultures were more like those of the Central Andes at some early period than at the Conquest, by which time they may have lost certain elaborations.

The general cultural tradition represented by the Circum-Caribbean tribes spread southward from Venezuela, probably down the Coast of the Guianas to the Amazon, and up the main tributaries of the Amazon (map 22; compare with maps 2 and 3). The sociopolitical and religious patterns and metallurgy, however, were largely lost, and only ceramics, weaving, and basketry survived. Even some of the technological traits dropped out toward the headwaters of the Amazon. The tribes of the Northwest Amazon and some of those in the Montaña and the southern part of the Amazon Basin remained Semi-Marginal. In the more remote localities on the Amazon headwaters and between the major rivers, beyond the effects of this cultural diffusion, the tribes remained Marginal. The Marginal and Semi-Marginal peoples form an almost continuous area that extends like a great U around the Amazon Basin, from the Amazon-Orinoco watershed south through the Montaña and east across parts of eastern Bolivia and Mato Grosso to the Highlands of eastern Brazil. The U is broken mainly in the upper Madeira River region in eastern Bolivia, where Tropical Forest tribes pushed southward, meeting other Tropical Forest peoples who had followed the rain forests down the coast of Brazil and inland to Paraguay and beyond.

Thus, it may be postulated that a single great historical tradition originating from the Andes in an early period carried sociopolitical and religious patterns and a developed technology from the northern Andes into Central America and almost entirely around the Caribbean Sea where it became somewhat adapted to sea coasts. It almost completely failed to penetrate the Tropical Forests directly from the Andes. Instead, the technological traits were carried by seacoast and riparian peoples down the Guiana coast and up the Amazon, and they are absent beyond the areas of easy navigation.

In the southern Andes, the technological, material, and ritual patterns spread south to the *Araucanians*. Traces of Andean influence, perhaps even of sociopolitical patterns, may have reached the eastern Chaco, despite intervening Marginal tribes, and there blended with Tropical Forest influence from the coast of Brazil.

In its main outlines and element content, this development and flow of culture was indigenous to America. It is not impossible that a number of isolated elements reached America from the Pacific. It does not really matter, however, whether blowguns, lime chewing with a narcotic, bark cloth, a few domesticated plants, and even such things as frame, thatched houses were native to America. They were incorporated into various local patterns which they affected little if at all. The basic patterns, in fact, seem to have been established, perhaps in Perú, though quite possibly elsewhere, well before any trans-Pacific voyages could have been made.

After the Spanish Conquest, the Marginal tribes rapidly disappeared, except in the more inaccessible areas. Lacking a vigorous culture before the Conquest and vulnerable to European contacts afterward, the tribes of the Brazilian coast, of southern Argentina, of the Chilean Archipelago are virtually extinct, and they have left little cultural or racial imprint on the modern population. The greatest survival of Indians with a native, but not precisely pre-Columbian, culture is around the periphery of the Amazon Basin. It is solely a question of time, however, before these, like the other Marginal peoples, lose their Indian characteristics.

The Tropical Forest tribes disappeared as cultural entities along the coasts and navigable rivers, partly because they were absorbed or overwhelmed by Europeans, partly because they were not adaptable to slave labor under the European pattern, so that Negro slaves were imported to replace them. They have, however, left a strong racial strain and even many culture elements in the modern population.

The Circum-Caribbean tribes have become virtually extinct in the Antilles, but many survive in the portions of Venezuela, Colombia, and Central America which are unfavorable for White settlement. The loss of the class system, the war complex, the religious cult, and the technological refinements has deculturated them to a Tropical Forest level.

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ABBREVIATIONS

Åbo Akad. -----	Åbo Akademi. Åbo, Finland.
Acta Amer. -----	Acta Americana. Revista de la Sociedad Interamericana de Antropología y Geografía. Review of the Inter-American Society of Anthropology and Geography.
Agric. Journ. British Guiana.	Agricultural Journal of British Guiana. Georgetown, British Guiana.
Amer. Anthropol. -----	American Anthropologist.
Amer. Antiq. -----	American Antiquity.
Amer. Indíg. -----	América Indígena. México, D. F.
Amer. Journ. Sci. -----	American Journal of Science. New Haven, Connecticut.
An. Bibl. -----	Anales de la Biblioteca Nacional. Buenos Aires, Argentina.
An. Inst. Etnogr. Amer. ----	Anales del Instituto de Etnografía Americana. Mendoza, Argentina.
An. Mus. La Plata -----	Anales del Museo de La Plata. La Plata, Argentina.
An. Mus. Nac. Buenos Aires.	Anales del Museo Nacional de Buenos Aires. Buenos Aires, Argentina.
An. Real Acad. Cienc. Méd. Fís. Nat. Habana.	Anales de la Real Academia Ciencias, Médicas, Físicas y Naturales de la Habana. Havana, Cuba.
An. Univ. Chile.-----	Anales de la Universidad de Chile. Santiago de Chile.
Ann. Med. Hist. -----	Annals of Medical History. New York, N. Y.
Ann. Rep. Bur. Amer. Ethnol.---	Annual Report of the Bureau of American Ethnology, Smithsonian Institution, Washington, D. C.
Ann. Rep. Smithsonian Inst.---	Annual Report of the Smithsonian Institution. Washington, D. C.
Anthropol. Pap. Amer. Mus. Nat. Hist.	Anthropological Papers, American Museum of Natural History. New York, N. Y.
Anthropol. Publ. Univ. Penn- sylvania Mus.	Anthropological Publications, The University Museum, University of Pennsylvania, Philadelphia, Pa.
Anthropol. Ser. Catholic Univ. Amer.	Anthropological Series, Catholic University of America. Washington, D. C.
Anthropos -----	Anthropos, Ephemeris Internationalis Ethnologica et Linguistica. Salzburg and Vienna, Austria.
Archiv. Anthropol. -----	Archiv für Anthropologie. Brunswick (Braunschweig), Germany.
Archiv. Anthropol. Völker. Kol. Kultur.	Archiv für Anthropologie und Völkerforschung und Kolonialen Kulturwandel. Brunswick (Braunschweig), Germany.

Archiv. Gesch. Med. -----	Archiv für Geschichte der Medizin. Leipzig, Germany.
Archiv. Mus. Nac. Rio de Janeiro.	Archivos do Museu Nacional. Rio de Janeiro, Brazil.
Art and Archaeology-----	Art and Archaeology; the Arts throughout the Ages. Archaeological Institute of America. Baltimore, Md.; Washington, D. C.
Baessler-Archiv-----	Baessler-Archiv, Beiträge zur Völkerkunde. Leipzig and Berlin, Germany.
Bibl. Pedag. Brasil-----	Bibliotheca Pedagógica Brasileira. São Paulo, Brazil.
Bol. Acad. Nac. Hist. Buenos Aires.	Boletín de la Academia Nacional de la Historia. Buenos Aires, Argentina.
Bol. Acad. Nac. Hist. Quito.	Boletín de la Academia Nacional de Historia. Quito, Ecuador.
Bol. Estud. Hist. Pasto-----	Boletín de Estudios Históricos. Pasto, Columbia.
Bol. Filol. Montevideo-----	Boletín de Filología, Sección de Filología y Fonética Experimental, Instituto de Estudios Superiores, Universidad de Montevideo. Montevideo, Uruguay.
Bol. Hist. Antig. -----	Boletín de Historia y Antigüedades, órgano de la Academia Colombiana de Historia. Bogotá, Colombia.
Bol. Inst. Geogr. Argentino--	Boletín del Instituto Geográfico Argentino. Buenos Aires, Argentina.
Bol. Min. Agric. -----	Boletim do Ministerio da Agricultura. Rio de Janeiro, Brazil.
Bol. Mus. Nac. Chile-----	Boletín de Museo Nacional de Chile. Santiago de Chile.
Bol. Mus. Nac. Rio de Janeiro.	Boletim do Museu Nacional. Rio de Janeiro, Brazil.
Bol. Real Acad. Hist. -----	Boletín de la Real Academia de la Historia. Madrid, Spain.
Bol. Soc. Geogr. La Paz----	Boletín de la Sociedad Geográfica de La Paz. La Paz, Bolivia.
Bol. Soc. Geogr. Lima-----	Boletín de la Sociedad Geográfica de Lima. Lima, Perú.
Bot. Mus. Leaflets, Harvard Univ.	Botanical Museum Leaflets, Harvard University. Cambridge, Mass.
Bull. Amer. Ethnol. Soc. ---	Bulletin, American Ethnological Society. New York, N. Y.
Bull. Bur. Amer. Ethnol. ---	Bulletin, Bureau of American Ethnology, Smithsonian Institution. Washington, D. C.
Bull. Bur. Ethnol. -----	Bulletin, Bureau of Ethnology, Smithsonian Institution. Washington, D. C.
Bull. Free Mus. Sci. Art, Univ. Pennsylvania.	Bulletin of the Free Museum of Science and Art, University of Pennsylvania. Philadelphia, Pa.
Bull. Mém. Soc. Anthropol. Paris.	Bulletins et Mémoires de la Société d'Anthropologie de Paris. Paris, France.
Bull. Pan Amer. Union-----	Bulletin of the Pan American Union. Washington, D. C.
Chaski -----	Chaski, órgano de la Asociación Peruano de Arqueología. Lima, Perú.

- Col. Doc. Inéd. Hist. Hispano-América. Colección de Documentos Inéditos para la Historia de Hispano-América. Madrid, Spain.
- Col. Hist. Chile----- Colección de Historiadores de Chile y de Documentos Relativos á la Historia Nacional. Santiago de Chile.
- Col. Libr. Doc. Ref. Hist. Perú. Colección de Libros y Documentos Referentes á la Historia del Perú. Lima, Perú.
- Coll. Ling. Soc. Ling. Paris. Collection Linguistique, Société de Linguistique de Paris. Paris, France.
- Comp. Ethnogr. Stud. ----- Comparative Ethnographical Studies. Gothenburg (Göteborg), Sweden.
- Congr. Int. Amer. ----- Congreso Internacional de Americanistas; International Congress of Americanists; etc.
- Contrib. Mus. Amer. Ind. --- Contributions of the Museum of the American Indian, Heye Foundation. New York, N. Y.
- Ethnologica ----- Ethnologica. Leipzig, Germany.
- Ethnos ----- Ethnos. Mexico City, México (to 1925); Stockholm, Sweden (1936-).
- Etnol. Stud. ----- Ethnologiska Studier; Ethnological Studies. Gothenburg (Göteborg), Sweden.
- Etud. Ethnogr. Comp. ----- Etudes d'Ethnographie Comparée. Paris, France.
- Field Mus. Nat. Hist. Anthropol. Mem. Field Museum of Natural History, Anthropological Memoirs. Chicago, Ill.
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- Gesell. Naturf. Freunde, Berlin Sitz. Gesellschaft Naturforschender Freunde, Berlin Sitzungsberichte. Berlin, Germany.
- Globus ----- Globus. Brunswick (Braunschweig), Germany.
- Göteborgs Kungl. Vet. Vitt. Sam. Handl. Göteborgs Kungliga Vetenskaps- och Vitterhets-Samhälles Handlingar. Gothenburg (Göteborg), Sweden.
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